UNIVERSITY OF GEZIRA

FACULTY OF MEDICINE

PEDIATRICS LECTURES' NOTES



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Taking History

A. Demographic Data:

1. Name:

الاسم يكون رباعي عشان ما يكون في اختلاط في الفحوصات والعلاج.

2. Age:

في الأطفال التقسيم كالاتي :

1. Early neonate: 1___28 day
2. Late neonate: 28___56 day
3. Infant (2 months ___ 1 year)
4. From 1 ___ 5 years
5. More than 5 years

أي فئة عمرية مشاكلها بتختلف من التانية .

لو الطفل ولدوه وما صرخ حتجيني طوالي بي birth asphyxia, comatose, convulsion)

- لو ولدتو بي داية عدتها وسخانة حتجيني بي tetanus immediately
- الطفل دا حيجي TB لكن لو ولدوه كويس وبعد اسبوع سلمت عليه خالتو وكان عندها cough (Incubation period of TB) بعد ستة اسابيع بي

- Young infant 2 month ___ 1 year

Mortality increase in day 1 then decrease after 1 week then it decreases with age.

- كلما الطفل كبر ال mortality بتقل.
- Neonatal mortality in Sudan 42 / 1000 due to complication of pregnancy and delivery.

- Diarrheal disease common in under 5 years due to nutritional problems.

- 70% of children mortality is due to: RTI (upper respiratory tract infection), diarrhea, anemia and malnutrition.

- After 6 months there is haemoglobinopathies due to replacement of HgF with adult haemoglobic and problems in HgA

- Diarrheal disease Due to Nutrition, while viral infection due to he is unimmunized.

3. Residence: السكن

Important in epidemiology of the disease.

To know why pt of emergency came late

- أعرف لشنو اتأخر (زلط ...اسعاف) وهل عندهم خدمات ولا لا.
- مثلا لو كان في مشكلة صحية في منطقة معينة دا معناه انه في خلل في الخدمات المقدمة في المنطقة دي.

4. Schooling:

- الافضل للطفل انو يمشي من عمر ٣ سنوات عشان يكون عنده social interaction عشان ما يحصل suppression
وما يحصل neuronal cell لل
cognitive function لل delay

5. Informant: -(غالبا أمو) الزول البديك التاريخ المرضي بتاع الطفل

B. Complain: -

- طفلك دا عنده شنو والبيوجعو شنو ؟
- طفلك دا عنده حمى ، استفراغ ؟
- لازم تفهم الام عشان تحدد ليك الشكوى.

- The Respiratory system start in Nose...Mouth... which open in laryngx.
- The nose has Cillia which be fired any thing more than 10 Microbe.
- The protective mechanisms of the Respiratory system start with Cillia.
- After cillia there are goblet cells that produce mucus.
- بتلصق في ال ١٠ _ ٥ من اي حاجة mucous. دا
- Then mucus descends to larynx and produce cough.
  والكحة دي بتحصل عشان تطلع المخاط.
- لوالطفل ما بقدر يكح ...يا اما مناخيرو قفلت او سالت .....ولما الانف تقفل ف ما اللي هي فاتحة Eustachian tube بيقدر يرضع ولو المخاط اتلصق في الحلق بيقفل ال في ال pharynx
  واي تجمع الافرازات بيحصل فيه over added bacterial infection
  وبتحصل ال otitis media.
- This otitis media is second to upper respiratory tract infection URTI. المخاط حيمشي في القصبة الهوائية والطفل حيكح كحة جافة والكحة دي حتستمر لحدي ما يستفرغ والكحة دي كلها عشان يحاول يطلع المخاط.
  ثم ال trachea المخاط حينزل في ال
  وبعدين ال bronchi
  ثم ال lobe

- الطفل كان عندو نزلة وانفو بتسيل ...بعد شوية نفسو زاد ...بعد شوية ال اشتغلت accessory muscles
  (scalene muscle & diaphragm)
  بعد كدا بيحصل ... grunting
  ودا بيعني انو عاوز ياخد expiration
  عشان يحصل more gas exchange
  وبتحصل accumulation of co2 and cyanosis.

- More than 5 there is Tonsillitis which doesn't occur before 5years.

الفرق بين ال Tonsilitis والنزلة:

- النزلة بتكون مسحوبة بي كحة بسبب المخاط
- اما ال Tonsilitis ما بيكون في كحة.
- Age of Tonsillitis = 5___15 years …. & this is age of rheumatic fever.

- Acute laryngo trachio Bronchitis occur under 3 years, pt come with cough and stridor due to obstruction of upper air way.
- Acute epiglotitis: more than 3 years, pt come with High grade fever, stridor, open mouth وسيلان لعاب

C. systemic review: -

- Cough is first sign of cardiopulmonary problems.

- Difficult breathing without cough occur in acidosis, such as DK الطفل شايل نفسه من امس مافي كحة، ما في سيلان ، ببول كتير ، بيشرب كتير، بياكل بس ضعيف دا سكري ___.
- acute kidney injury __ الطفل من امس بيستفرغ وما ببول دا

- cardiopulmonary لو ما كان Difficult breathing لو جاك زول بي
  حيكون: renal failure, DM, metabolic error of metabolites لو كان صغير .
- Difficult breathing زول عندو
  orthopnea)) لمن يرقد
  heart failure ... congenital or rheumatic heart disease دي
- abdominal pain والurination ، وفي الحالة دي بسأل من ال
- urination وما بيحصل renal perfusion بيقل ال HF
  liver والدم قاعد في ال
  pain , عشان كدا بيكون في

عشان كدا في ال HF
بلقى الrenal والGIT بيتأثروا

D. past history: -

- زمان الطفل عمل حادث ونقلو ليه دم بدون ما يفحصوه وهسي الطفل عنده يرقان

- Dark urine, normal stool = Hepatitis بسبب الدم النقلوه ليه زمان دا. segment accumulation ويحصل acini ثم ال

Then super added infection then pneumonia.

- Lung compliance:

Mucus + fluid + inflammatory cell___ all this decrease lung compliance and there is decrease in exchange air then increase Respiratory rate.

- Increase Respiratory rate is the first sign of lung disease.
- lung compliance decrease due to accumulation of inflammatory exudate in the acini or due to constriction asthma and COPD.
- The lung has surfactant which prevents collapse.
- There is 20% residual volumeفي دايما بكون

- In case of asthma and Bronchiolitis there is decrease expiration so the residual volume increase
  exchange (قلت المساحة البحصل فيها)
  decrease in site of exchange then lung compliance decrease and increase in RR.

- In heart failure there is pulmonary oedema __ fluid in acini __ decrease lung compliance.

- Broncho pneumonia, obstruction and fibrosis are decrease lung compliance and first sign is increase RR.

- Not any caugh is pneumonia so cough may be URTI.

- To differentiate between URTI and lung disease must calculate RR.

• Count of respiration in children is varies between different age group.

- 1__ 60 days __ >= 60
- 2month___ year >= 50
- 1 year ___ 5 years >= 40

• Under 5 fever may be otitis media or pneumonia

E. Development, nutrition, vaccination: -

➢ <u>Developmental</u>.

• بتقول ليك الولد دا عمرو ٣ سنة لكن ما زي اخوانه لا بيمشي لا بيتكلم. طوالي هنا بتسألها الحمل دا كان كيف ، قالت ليك الحمد كان كويس وقالو لي تجي تولدي في المستشفى وما مشيت والولادة كانت صعبة واول ما ولدت الطفل ما صرخ وجاتو تشنجات .....دا Grade 2 birth asphyxia

• Pt with asphyxia hasn't increase in head circumfrance so his growth is affected and there is delay.

• <u>There are 4 arches of development</u>: -

1. Gross motor:

A. Neck support
B. Trunk support عشان يقعد

C. يحبى
D. يمشي

- At 3 months _____the grasp reflex disappears and neck support and all primitive reflexes disappear

- 5__ 6 months _____sit with support
- 7 months _____sit with out suppor

- ٩ شهور_____ يحبي
- سنة_____ يقيف
- سنة ونص_____يمشي
- سنتين _____ينزل من السلم
- سنتين ونص _____يجري
- ٣ سنوات_____ يسوق العجلة.

2. Fine motor: -

- بيخمش في الاول
- ٩ شهور بيمسك الحاجات
بعد كدا بيلبس الجزمة.

3. Speech: -

- 4 months _____start to pupil
- 6 monthلغة الشفاه_____
- 2 years _____ جملة من كلمتين

4. Social: -

- Start from 2 months ____with his mother skin to skin contact, breast feeding
- till 8 month ما بيمشي للغرباء_____.

+++++ PROGRESS REPORT +++++

It's important to remember that every kid develops skills at different times. It is perfectly normal for a baby to master some skills quickly and take a bit longer to acquire others. This is a rough outline of what to watch out for. If you have any concerns about your baby's progress, speak to your doctor.

	AT THE END OF **THREE MONTHS,** MOST INFANTS CAN...	AT THE END OF **EIGHT MONTHS,** MOST BABIES CAN...	BETWEEN **12 TO 14 MONTHS,** MOST KIDS CAN...
GROSS MOTOR	+ Roll from front to back + Control their head and neck + Raise their head and chest while lying on their front + Stretch out and kick their legs + Push down with their legs when their feet are on a hard surface	+ Roll both ways (front to back and back to front) + Sit on their own + Support their weight on their legs + Control their upper body and arms	+ Reach a sitting position + Crawl or scoot + Start crawling from a sitting position + Pull up to a standing position + Cruise, holding onto furniture + Stand briefly without support + Walk holding an adult's hand and take a few solo steps + Start to climb stairs with help
FINE MOTOR	+ Open and shut their hands + Bring their hands to their mouth + Bring their hands together + Bat a hanging object	+ Hold and shake an object + Move an object from one hand to the other + Use their hands to explore an object	+ Finger-feed using a pincer grip + Place objects into a container (and pull them out again) + Poke an object with an index finger + Push a toy + Begin to drink from an open cup + Start to use a spoon + Scribble with a crayon
SOCIAL/ LANGUAGE	+ Smile when you smile and on their own + Make facial expressions + Copy some body movements and facial expressions	+ Reach for a person they know + Smile at themselves in a mirror + Respond when other people express emotion + Mimic sounds	+ Be shy around strangers + Copy during play + Test limits + Put out an arm or leg to help when being dressed + Take off socks + Come when called + Say "mama" or "dada" and at least one other word + Stop an action if you say "no"
COGNITIVE	+ Watch faces closely + Follow moving objects + Recognize objects and people they know	+ Track a moving object and find one that is partially hidden + Struggle to get objects that are out of reach + Watch a falling object	+ Explore objects in different ways (shaking, banging, throwing, dropping) + Name familiar objects + Respond to music + Begin to explore cause and effect

SOURCE: **CANADIAN PAEDIATRIC SOCIETY**

➢ Nutrition: -

- Start from first hour
- Exclusively 6 months.
- Brain growth related to breastfeeding,
- head circumfrance increase (2) cm / month in first 3 months,
- then (1) cm/ month in second 3 months,
- Then (,5) cm / month in last 6 months.
- الرأس في السنة الأولى بيزيد ١٢ سم ، ٩ سم في الستة شهور الأولى، و3 سم في الستة التانية.
- بما انه بيزيد ٩ سم في الستة شهور الاولى هنا بتجي اهمية ال exclusive breastfeeding.
- After 6 month pt need to complementary food (small frequent) 2 × 1. لو انا بأكل ٣ وجبات هو مفروض ٦ وجبات

➢ Vaccination: -

1. At birth__ BCG and …… (0) oral polio.الصفرية
2. 6 weeks ٢ في الرجل و٢ في الفم___
first oral polio واحدة في الفم بمبية
والتانية (الروتا) لونها زي عصير القونقليز.

بديوها نقطتين oral polio الروتا بيدوها كلها لكن ال
بيدوه ال pentaو ال pneumo vax
3. 10 weeks_____ اسابيع 6 نفس ال
4. 14 weeks __ penta and pneumo vax, injectable polio بنسحب الروتا وبندي
5. 9 months ___ measles and Meningitis (2m).
6. 18 months ___ measles.

Haemolytic Anemia

Iron, Folic acid, B12 and vitC => all of these responsible from development of RBC.

- **Physiolog of RBCs: -**

- Normal RBCs have certain shapes.
- Diameter of RBCs 6-8,5 mm.
- Isn't sphirical it's Biconcave disk. this shape to circulate.
- mature cells aren't nucleated

Any cell contains: -

- Cell membrane
- Cytoplasm
- **Nucleus**: very important thing in nucleus is mitochondria which is responsible from metabolisms of cell (oxidative phosphorylation).

Reticulocyte is nucleated

- mature cell isn't nucleated so it doesn't depend on oxiditive phosphorylation, it depends on pentose phosphate shunt in oxidation
- protein is form skeleton of RBCs.
- Band protein 3 is connect between 2 layer.

- importance of double layer membrane:

بتمطى مع الخلية أثناء سيرانها في الدورة الدموية فبالتالي يحافظ على هيكل الخلية

- **composition**: -

- cell membrame
- Hb in the cytoplasm
- Cytoplasmic enzyme.

- **what is Hb?**

Haem + globin.
Globin is a group of amino acid
- chromosome 11, 16 is responsible from globin.
- Any defect in sequences of amino acid either decrease of number (< 150) which give Thalassemia.

sickle cell disease is change shape in amino acids in the sequence of polypeptide.

- According to polypeptide change there are **different Type of Hb:** - **Alpha** is constant in All Type of Hb the other one is changed.

Other chain is different as follow: -
- adult Hb is beta chain.
- A2 Hb is alpha chain.
- Hb F fetal Hb is gamma chain. And has high affinity to oxygen.

- the blood comes from the placenta of higher concentration goes to the head.
- then the blood from the head and upper limbs descend to the atrium and it's less oxidized it reaches 60 this go to the lower limbs.
- the body works to compensate the hypoxia by raising the RBG and this explains why the fetus has high PCV and Hb (22) and the physiologic jaundice that happens when this increased amount is destroyed.

- Hb H has 4 betas and it's incompatible with life.

- skeleton of RBCs is formed by 50% proteins, 40% fat, 10% carbohydrates.

- **Normal age of RBCs** is **120** days, macrophages take the old RBCs to the spleen which recycles them and form new ones.

- when RBC have abnormal shape or content it dies before 120 days. (premature destruction) so anemia appear when compensatory mechanisms didn't compensate This destruction.
- When the rate of destruction higher than the rate of production this results in anemia.

- fetus when start as yolksac, this yolk sac act as bone marrow and produce Blood cells after formation of lymphoid tissue (LNS, liver and spleen) it become responsible from production of cells, after that marrow convert to bone (long and Flat bone) this bone act as marrow.

    - After birth liver and spleen act as storage...

- long bone replaced by fat.
- Flat bone consist marrow.
- when Haemolysis occur, marrow start to Act so flat bone become widening (the forehead protrudes, the scapula widens but we don't see it, maxilla and the pelvis also widens but we can't see them because they are covered with muscles.

- These are features of marrow hyperplasia as compensatory mechanisms of anemia.
- When the marrow couldn't compensate the shortage the storage starts to act and extra medullary heamopoiesis occurs that's why liver and spleen enlarged.
- Hemolysis occurs in the spleen when recycling occurs and the polypeptide chains go to the liver as amino acids to form proteins including albumin.
- Heme give biliverdin and by biliverdin reductase give bilirubin (not water soluble) and transported to the liver by albumin (conjugated bilirubin- water soluble) excreted as urobilinigen which doesn't

change color of urine.

- unconjugated go to the biliary system to the gallbladder to help in fat digestion.
- decreased RBCs result in pallor.
- bilirubin means jaundice (unconjugated hyperbilirubinemia)
- stercobilinogen gives brown color of stool.

- **The <u>defect</u> which lead to <u>premature destruction</u> of the <u>RBCs</u> either**: -

1- **defect in cell membrane**, defect in it skeleton so macrophage can't identify it and destruct it, and this what happens in **spherocytosis** and **liptocytosis**.

2- **enzyme deficiency**:

-G6PD and pyruvate kinase
- G6PD in pontose phosphate shunt it give reduced glutathione give 2 NADH
and 2 ATP (ATPase give energy while NADH go to reduce glutathione).
as result of metabolism H_2O_2 is produced and convert to $2H_2O$.
if G6PD is deficient the glutathione is oxidized and H_2O_2 is present and RBCs destruct within the vessels and appear as Hburia This occur in G6PDD and malaria.

3- **heamoglobinopathies:**

- short chain (small cells) and decreased Hb polymerization => microcytic hypochromic anemia.

or sickled in sickle cell disease.

4- **infection** :
parasite produce pyrogene which destruct RBC
Type of severe malaria is malaria with severe anemia.
Bacteria => septicemia.

5- **Antibodies** :
Autoimmune Haemolytic Anemia

6- **Drugs**.

cell membrane defect

congenital spherocytosis is commonest.
- defect in protein cause weak membrane=> Fluid come bn 2 layers and give spherical shape => destruction of RBCs
- spherocytosis isn't severe anemia (Hb bn 6-9)
- pt with spherocytosis unless there is Aplastic crisis can live normal life.

- **clinical picture: -**

Black American, europian, chinese.

- decreased enzyme activity occur in black american in mature RBCs, while the chinese and europian Type the maturity occur both in reticulocyte and RBCs so there is spontaneous haemolysis.

- In Amirican no haemolysis unless there is risk which increase needs of G6PD such in stressful condition like infection Foods or drugs (Asprin and sulpha containing drugs).

- stressful condition increase the need for glutathione which if not reduced there is accumulation of H2O2 and Haemolysis.
- cell in G6PD D has normal shape so not related to macrophage.
- Hb in urine give black urine so called black water.

- Artisunate consist Sulpha, septrin.
if you give Aspirin to pt with viral infection there is reye's syndrome and pt may die with Liver failure.

- **diagnosis**:

enzyme Assay.

- **managment**:

- avoid eating peanut.
- avoid Asprin to fever.
- don't give sulpha containing drugs.
✓ if pt come to you with black urine you must:

- send him to hospital if this Hb doesn't wash it stay in tubules and cause Tubular necrosis and give Acute renal failure.

So when he come to you must: -
- do false alkaline diuresis
- give NaCL and normal saline + Lasix
- monitor the urine

Acquired Haemolytic anemia

There are 2 Type
1- worm Antibodies
2- cold Antibodies.
- warm Antibodies:
occur when the pt have high temperture.

- cold Antibodies:
  Occur when the patient is cold
- warm by IgG
- cold by IgM.
- There is hemangioma which destruct RBCs
- there is Aneurysm.
- essential hypertension have maintained flow which destruct the RBCs.

hemoglobinopathies

- There are 2 major problems either affecting alpha chain or beta chain which are botg polypeptide chains in the Hb.
- alpha chain disease is incompatble with life.
- Beta chain is either deletion or exchange of amino acid in the sequences
- The common problem is exchange of amino acids.
- The most common disease due to exchange of amino acid is sickle cell disease.

deletion is B thatashemia

sickle cell Anaemia

- AR disease - when the pt carry one gene is homozygous and is called sickle cell Trait
- the defect in sickle cell disease is amino acid exchange in the valine istead of glutamic acid.

glutamic acid is replaced by valine at position 6.

- **The pathogenesis:**

The cell is in a shape of net when it exposed to low O2 tension it transformed to longitudinal crystals so RBCs shape change and become sickled or crescent shape and the macrophages took it to the spleen so they processed and destructed.
to compensate the hemolyse cell there is Anaemia
The compensatory mechanism is bone marrow hyperplasia and extra medullary haemopoisis.

- **Presentation**:

1- hyper haemolytic crisis: -
Any sickle cell it will be destroyed but not massive but when there is infection as risk factor or any stressful condition which lead to destruction of the sickling of the cell (taken by spleen and destroied)
There is failure of compensation The child become anaemic and there is called hyper haemolytic.

In hemoglobinopathies there's chronic anemia.
When there's hyperhemolytic crisis the child will be:
- palor and jaundice
- anemia may lead to HF
If Hb is 5 without HF--> simple packed cell transfusion.
When there's HF you have to be careful to avoid overload, so it is better of interrupts the blood because the heart can't tolerate more volume. So
given in 2 sessions at least 4 hours apart.
sickler can tolerate the anaemia.
O2 dissociation means release of O2 from нь (1Hb carry 4 O2).
What causes which make O2 dissociation chain go to the right to release O2.
- Acidosis
- hypoxia.
- 2,3 diphosphoglycerate

- fever

There is high level of 2,3 diphosphoglycerate in SCD so the O2 is easly release so children with SC Anemia they yolerate anaemia.
sickler pt may have Hb2 but come to hospital with his leg due to high level of 2,3 diphospheglycerate.

2- <u>vasocclusive crisis</u>: -
- the vessels is been occluded.
- Biconcave RBC can pass in the microcirculation but when it become sickel shape the membrane become rigid when it passes in microcirculation
It stuck and aggregate platelet come and there is thrombosis => No blood supply => ischemia, depending on the site there is a **clinical presentation:**
Brain => CVA so patient may present with convulsion, cranial nerve involvement
Bone the commonest presentation is dactylitis on small bone of the hand and feet.

when occluded vessels to the internal organ like liver, spleen, heart, Kidney
pt can't present with appear symptoms unless the affect the heart.
- this is only cause of ischemic heart disease in pediatrics.
* it can occur in ling by 2 mechanisms: -
1- infarction in the acini
2- infarction in a large blood vessels and give Acute chest syndrome.
it occurs in spleen frequently and
end by autosplenectomy => the spleen become fibrosed and can't do its job.
The spleen become enlarged due to extra medullary haemopoiesis.

here the spleen became fibrosed it can affect the penis and give sustained erection (periapisin) which is an emergency.

3- Sequestration crisis: -

spleen is common site for sequestration.

occur of spleen is either in the infant or child with high HbF.

Those didn't have splenectomy

Young infants didnt get vasocclusive crisis so their spleen present as part of extra medullary heamopoiesis.

People with HbF not like sicklers have high O2 affinity and less release.

4. A plastic crisis: -

- Aplasia of bone marrow
- It's not specific to SCD, it can occur in all haemolytic anaemia.
- it is following viral infection, common virus which cause Aplastic Anaemia is

parvovirus B19.

- you should have differentiated bn Anaemia caused by Haemolytic crisis and Aplastic crisis:

pallor, jaundice => hemolytic crisis.

pallor without junduce => A plastic crisis.

- but you can find bleeds skin and mucous membrane due to platelet deficiency.
- pt may come with high grade fever due to decreased WBC. So you measure temperature 6 hourly.

If pt come to you with pallor, without jaundice you have to search about skin rash and measures his temperture 6 hourly.

- grading of pain: -

Just complain but he moves this pain doesn't affect activity. Grade 1 mild 1-3

moderate pain 3-7

Sever pain > 7 (7-10).

✓ moderate: -
Pain affect the activity.
✓ severe pain: crying, anxious, agony, awakening from sleep.
- if pain affect the activity give second line ibuprofen.

- if pt can't sleep give opiates.

- Start with Acetaminophen, if pain not relief and increase give ibuprofen, if increased but moderate give single dose of opiate.

- second step in managment is rehydration Start oral if it become moderate give IV Fluids.

- if it become severe pain give bolus of morphine and give interrupted dose, give IV fluid => maintenance is 5% glucose with Normal Saline.
  - Acute splenic sequestration: -

- child come either with anemia or shock.
shock is poor prognosis.
pt with shock come with:
- pallor.
- slow rapid pulse.
- enlarge spleen.
- shock need to immediate transfusion.
  - the follow up by:

- Pallor.
- vital signs.
- spleen.

  - Aplastic crisis: -

hyperhaemolytic, sequstration and Aplastic crisis => for blood transfusion (simple Transfusion)
CVA => For exchange Transfusion. priapism => For exchange Transfusion

- ✓ when you do exchange there is dramatic response.
- severity of SCD now is identified and

has caused by non globin chain.
where you find sickle cell disease?
في غرب السودان الكواهلة والمسيرية والبقارة والهباتية والتعايشة

- **investigations**: -
-

1. CBC:
When you find high WBC go see the retics cuz they're nucleated RBCs and means there's hemolysis

2- Sickling test -> use reducing Agent 3- Electrophoresis
you find sickle cell in pt who have
sickle cell more than 90%

Hemoglobinopathies doesn't appear till 6 months of age after the replacement of HbF by HbA.

electrophoresis is good in prognosis => it gives percentage of HbA, SS and HbF.
The more HbF less severity.
Sickle cell may have other Hb (C, E, thalassemia) the more the other Hb lessen the severity of the disease.

- **managment**: -

give folic acid which help in maturation of cell.
Increased Hbf Lower the crisis.
- pt with frequent or severe SCD like CVA give drugs which increase Hbf (hydroxy urea)
Hydroxyurea Also it is cytotoxic drug.

- thalassemia- Mediterranean anemia

The problem in the chain is deletion and according to it's severity depends the presentation: -

Thalassemia minor --> 25
Thalassemia intermediate--> 50
Thalassemia major--> 90 (it is severe and commonest) the cell is microcytic hypochromic.
The cell is not functioning and when the baby is 6 months will be white.
The bone marrow tries to compensate that's why they present with wide forehead, protruded teeth and enlarged liver and spleen.

- alpha Thalassemia is incompatible with life.
- moderate thalassemia is similar to

IDA -> pt have pallor but not severe anemia (Hb 6-9).
- pt have hepatosplenomegaly. prominent maxilla.
- both thalashaemia and IDA are
microcytic hypochromic.
* if pt come to you with moderate anaemia, prominent forehead and maxilla, hepatosplenomegaly => for electrophoresis if normal search about IDA.

10% of children with palpable spleen
they have IDA.

- **investigations**:

1- CBC:
↑WBC due to retices.
decreased HbA
- microcytic hypochromic
2- electrophoresis

- **Treatment**:

✓ regular Transfusion
If you left the bone marrow activated the forehead will widen and teeth will protrude and bone will very thin and pathological fracture will happen.
✓ bood Transfusion either hyper or super
1- Hyper transfusion:
- make Hb is 10.
- every 3 weeks.

2- super transfusion:
make Hb 12.
- every 2 weeks.
Patients with Thalashaemia dont get Transfusion from relatives avoiding Antigenesity. And patient may die by this reaction.

✓ Do Bone Marrow Transplant if available.
- enhance washing of Pyrogenes to prevent occurrence of fever. (So wash and filter).

✓ Thalassemia must be treated in thalassemia center which consist of multi disciplinary team due to it affect different area.
✓ the problem of frequanct transfusion is Iron overload:
1 mL of blood give 1 mg of iron

this iron deposite in the pitutary, heart, liver & pancreas and skin.
- in pitutary and pancreas so need to endocrinologist due to anemia cause stunted growth, iron cause disturbance in hormone and DM.
- deposit of iron in the heart cause cardiomyopathy so need to cardiologist
- deposit of iron in the liver cause haemochromatosis so you need gastrologist.

- Protrusion of teeth need dentist.
- bone fracture need orthopedic surgeon.
- need haematologist: because of every 2 wks Transfusion and specialist in blood bank to do the wash and filter.
- need psychologist
  - ✓ multi disciplinary team consist: -
- endocrinologist.
- cardiologist
- gastroenterologist
- dentist
- orthopedic surgeon
- haematologist
- psychologist.
  - opioid in need can't cause addiction.
  - kawahla also have G6PD defeciency.

There are 2 poly peptide chain in the Hb
1- alpha chain disease, incomatable with life.
2- Beta chain disease either deleting or exchange of an amino acid in the sequence
- common disease due to amino acid change is SCD
- common disease due to amino acid deletion is beta thalassemia.

SCD = autosomal ressive, so it need 2 genes for the disease to appear. (homozygous) ss.
- one gene => carrier, Called sickle cell trait heterozygous SC/SA

Defect => exchange of glutamic acid by valine in Position 6 chromosome 11.
when the cell is exposed to low O2 tension the normal mesh shape of the cell change into longitidenal crystals.
the longitudinal crystals cause problems not like the mesh shape that pass through the vessels with no problem.
the longstidanal crystals change the RBCs shape

from biconcave into sickle cell or crescent. macrophage don't recognize the abnormal cell=> spleen (processing).

• When the body fail to compensate for the handytise cell the anemia appear.
the compensatory mechanism is by: -

1- bone marrow hyper plasia
2- extra medullary haemopoiesis stressful condition.

Abnormal sickle cell taken by the macrophage => Destroy in the splean + failure of compensation => hemolytic anemia (hyper hemolytic).
So there is chronic anemia in haemoglobinopathy and a hyperhemolytic Crisis.

- In **crisis** the child will be: -

1- **pale** and **Jaundiced**
2- **severe anemia** may precipitace **HF**.
- sever anemia **without** HF => simple packed cell transfusion.

When there is **HF** you have to be careful because you may overload the child so better to give the blood in 2 sessions in at least 4hrs.
sickler tolerate the anemia.
Hb O_2 dissociation curve when shift to right means release of O_2 from the Hb.
- One Hb carry 4 O_2
- Causes by acidosis, hypoxia, 2.3 phospoglycerate.
- SCD patient have high level of 2,3 diphosphoglycerate so the O2 is release easily and thus sickler can tolerate anemia

#vaso-oclussive crisis: -
the vessels are occlused by the sickle cell. (Stuck)
so the blood doesn't reach => ischemia.
- C/P depending on the site.

If ocurrs in the **brain => CVA**, may present with convulsion, cranial nerve involvement, weakness.

If occurs in the **bone** cause **dactylitis** a small bone of the hand and feet- the commonest presentation early
can affect the mternal organ liver, kidney, spleen, heart = don't come with specific symptom.

- The only cause of **IHD in pediatric is SCD**.

- In the lung - Acute chest syndrome by causing Infarction of the acini or it cause Infarction in the whole Vessels.

- In the spleen + end by auto splenectomy early it enlarges due to extra medullary heamopoiesis.

- In the penis => sustained erection (periapism).

sequestration Crisis & (pooling of the blood):
Present with severe anemia (Anemic HF).
what provoke it is unkown.
- commonest site for sequestration is the spleen. (spleen found in Infant and child with Hby) => spleen present as a part of the extra medullary heamopoiesis.
- late sever presentation. SCD.

#Aplastic crisis =
A plasia of the BM.
Not specific for SCD, it can occur in all types of heamolytic anemia.
- Commonly following viral infection (commonest is paro viraus19)
- Present with sudden severe anemia without Jaundice.

--> anemia cause heamolytic crisis --> pallor and jaundice.
--> anemia cause by a plastic crisis --> sudden severe pale without jaundice
but you may find bleeding in the skin and mucous membrane due to plts decreased. (petechae, skin rash).

• A plasia of the BM affect the whole cell including the WBC => high grade fever.

grading of the pain

grade 1: mild pain Just pain, don't affect the activity or any thing.
grade 2: moderate pain affects the movement
grade 3: severe
Management of vaso-occlusive crisis:
- this classification important in Mx of pain:
1- the drug of choice in mild/ moderate pain is acetaminophen
- 2rd line is ibuprofen
2- Fluids: If mild Just encourage pt to drinks.
Severe = admit the pt and give IV fluid.
- Single dose of opiate can be given in moderate pain that affect the sleep.
Doesn't cause addiction.
- sever agonizing pain => admit to give: 1. bolus of morphine then interrupted dose.
2. give IV Fluids.
* Mx of sequestration: -
-mild sequestration Crisis => anemia
- severe => shock (poor prognosis) present with:
1- sudden sever pallor
2- small rapid pulse
3- increasing spleen in size.
 - for immediate BT.
 so, the follow up by=> pallor, vital signs,
spleen size
* Mx of aplastic Crisis: -
for Blood Transfuscon
 simple transfusion in:
 1- Aplastic
2- sequestration crisis.

* Indication of exchange transfusion:

CVA " dramatic response".
the severity of the disease (not the pain) determine by the non globin gene.
- if a child present with CVA you'll find another one in the family also have it => non globin gene.
.....

"Bagara" also have G6PD defeciency- normocytic normochromic anemia.
Hemoglobinopathies appear after 6 months of age after the replacement of HbF by Hb A.
Sickling test:
reducing agent 2% Na metabisulphate
reduce O2=> sickling
or Fixation by wax.

Hb electrophoresis show the prognosis because it shows the percentage of A, ss
The more HbF less the severity of the disease.
* Drugs used for Rx:
1/ Folic acid => help in mutation of the cell.
2/ hydroxy urea => increase HbF (high affinity to O2) => cytotoxic drug.
3/ some people give benzathine penicillin in repeated crisis => less infection.

thalassemia: -

- not common in sudan
- severe disease
- there's deletion in the chain
- occurance according to the chain deletion:
25=> thalassemia minor

50=> thalassemia intermediate
90=> thalassemia major (most common most severe)
Microcytic hypochromic anemia.

C/P: -
1. Pallor
2. Frontal bossy
3. Prominent maxilla
4. Hepatosplenomegaly
=> compensated by medullary and extra medullary haemopoiesis.
• alpha Thalashaemia => incompatible with life.
• thalassemia intermediate looks like IDA => Hb 6-9 And both are microcytic hypochromic anemia with moderate anaemia and hepatosplenomegaly.
* Diagnosis: -
1. CBC = total WBC is high (retics)
Hb is low
Plt is low
Microcytic hypochromic anemia
2. Electrophoresis: confirm the diagnosis.
* Managment: -
..... if you left the bone marrow activated the forehead will widen and teeth will protrude and bone will very thin and pathological fractures happens.

Blood transfusion either:
1. Hyper => Hb is 10 and transfusion every 3wks.
2. Super => Hb is 12 and transfusion every 2wks.
Wash and filter cells from ag to prevent reactions.
- usually treat in Thalashaemia center multidisciplinary team:
The problem of frequent BT is iron overload (1ml of blood= 1mg of iron).

The iron deposit in:
1. The pituitary => stunted growth and affect all hormones (endocrinologist)
2. Pancreas => DM
3. Heart => cardiomyopathy (cardiologist)
4. Liver => hemosiderosis (gastroenterologist)
5. Teeth => dentist
6. Bone => orthopedic surgeon
7. BT => heamatologist
8. Psychologists.

* note: thalassemia
Thalassemia means the sea
It's two types alpha and beta.
Failure to synthesize either alpha or beta. => impaired production of adult Hb.
• alpha thalassemia (4types): -
- one gene deletion = silent carrier
- two gene deletion = alpha Thalashaemia
- three gene deletion = Hb H disease
- four gene deletion = hydrops fetalis.
* what is the most common cause of anemia?
Destruction of RBCs in the bone marrow "ineffective erythropoiesis "rather than destruction of RBCs in the reticuloendothelial system (hemolysis).
- In thalassemia there's decrease in beta beta chain production => compensated by increase of gamma chain (HbF) and beta chain (HbA).

Presentation: -
1. Features of anemia => pallor, fatigue, palpitations and hemic murmur.
2. Features of hemolysis => jaundice, dark urine, dark stool.

3. Features of increase medullary erythropoiesis inside bone marrow => enlarged head, frontal bossing, prominent maxilla, depression of the nasal bridge, exposed upper teeth.
4. Features of extra medullary heamopoiesis => hepatosplenomegaly.
5. Endocrine involvement => stunted growth, delayed puberty and DM.

complication of blood transfusion:

1. Iron overload.
2. Circulatory overload.
3. Transient infection
4. Mismatch blood transfusion => hemolysis => hemoglobinuria and AKI.
5. Allergic reaction.

causes of death:

1. Heart failure
2. Liver failure
3. Increased susceptibility to infection.

* x-ray skull => show widening of diploid space ... (hair standing on end appearance)

Heamolytic Anemia:

RBCs are biconcave in shape to pass easily through the microcirculation.
- mature RBCs aren't nucleated.
We have different types of Hb:
- alpha is constant in all types of Hb but beta changes from one to other.
1. Infant HbF = alpha and gamma
2. Adult HbA = alpha and beta
3. HbA2 (2%) = alpha and delta
When compensatory mechanisms fail destruction > synthesis => anemia appear.
Why spleen and liver enlarge in anemia? Extra medullary heamopoiesis.

Causes of premature destruction?
1. Cell membrane problem so macrophages didn't recognize the cells (kyposis), spherocytosis

2. Enzyme deficiency: G6PD and pyruvate kinase.
* G6PD: -
Reduce glutathione, no G6PD => glutathione Increased => h2o2 =>destruct RBCs => hemoglobinuria, occur in 2 conditions " severe malaria, G6PD defeciency ".
3. Hemoglobinopathies (SCD, thalassemia)
4. Antibodies autoimmune anemia.
5. Bacteria

* Congenital spherocytosis: -
Autosomal dominant inheritance, people thought the problem is permeability of the membrane when you give fluids it enters and the cell become spherical (osmolarity test).

the problem is in the 2-layer membrane the fluid inters between them, and two things happen:

1. Don't pass through the microcirculation.
2. Macrophages destroy it.
- usually mild disease
- anemia 6-9 g/dl
- patient has normal life unless develop Aplastic crisis.
* when you find **pallor + jaundice** => **hemolysis** => **spleen enlarged** but **not clinically palpable.**
Severity of anemia detect by **pallor**.
Don't try BT unless the patient is in failure.

diagnosis: -
1. CBC: retics high due to hemolysis.
2. Fragility test.
3. Hb electrophoresis
4. Autohemolysis test
Any cell if put in 27° for 48 hrs you find 5% autohemolysed. If more than 5% => spherocytosis.

management:-
- ✓ Splenectomy.

G6PD: inherited sex linked, females are carrier and males affected.
Risk: stressful conditions
1. Drugs = aspirin, sulpha containing drugs, artisunate, septrin.
2. Food (favism).
3. Infection => increase need for glutathione => means it's not reduced => accumulation of h2o2 => destruction of RBCs.
The cells are normal in shape.

Not continuous hemolysis => no chronic h.-anaemia.

Diagnosis: **enzyme assays**
Management = only prevention
If present with hematuria => must be washed => tubular necrosis.
Wash by NaCL + lasix.

acquired:
Cold autoimmune heamolytic anemia, IgM, when temperature is low
Warm autoimmune heamolytic anemia "associated with infection ".

GENETICS

*genetic disorder responsible from one third of the total hospital admition.
*History: -
-pt with genetic disorder have:
1. …of similar condition.
2. repetitive hospital admition
3. H of suddun deaths
4. leading points toward some sore of genetic disorder

*Examination:
-do examination from head to foot search for genetic disorder.
* what is the different between family pedigree and genetic pedigree?
1. family pedigree have one generation, parent and sibles while genetic pedigree has more than one generation.
2.في ال family pedigree لمن اعمل سهم affected عملت م بكون genetic study و م
عرفت هل الباقيين affected لكن في ال ,ولا لا genetic pedigree لمن اقول دا affected
معناها هنا انا عملت ال study والعيان دا carry ……

. family pedigree has suspesion this pts affected, while in genetic pedigree I do genetic study then I say this pt affected or not.

patterns of **inheritance**:

1. Autosomal dominant.
2. Autosomal recessive.
3. x-linked dominant.
4. X-linked recessive.

1.Autosomal dominant: AD

- dominant gene, doesnot need to supportive gene from other partener to manifest in disorder. just one gene cause manifestation.
- there are sex chromosome and autosome chromosome.
- AD due to immune mutation.
- in AD all generation are affected (male and female).
- some time Autosomal Dominant inheritance disorder may behave as autosomal recessive disorders in rare condition, when disease have variable which mean the pt genetically affected but the manifestation doesnot appear in pt so the pt is not affected morphologicaly. on genetic pedigree it similar to AR disorder.

2.Autosomal recessive:"AR" condition: -

- in which there is skip generation (not affected).
- there is pseudodominant disorders which cause by percentage of carrier staus in general population.

- مثلا لو اتخيلنا كل العالم دا هو...,والcarrier rate for sickle cell ال ,عالي chance لل sickle cell anaemia في كل ال generation ممكن تلقي عالي لدرجة انك م skip generation, ال pattern ببقي بشبه ال AD.

3.x-linked recessive: - XLR

-it is due to immune mutation.

- abnormal gene carried on x chromosome y havenot genetic materials.
- if x of the mother is recessive the male will be affected while female will be carrier.

-in XL in the second generation just one gender affected (male), and female not affected, parent doesnot affected, and 3rd generation not affected.

* لمن القي كل ال generation فيها بس ال male هو ال affected طوالي بفكر في ال x linked disorder.

4.X-linked dominant (XLD): -

-لمن القي كل ال generation هم affected دا dominant disorder, طوالي بسال نفسي
هل دا autosomal ولا x linked بعاين لأي male, لو ال male ادي affected male دا
autosomal dominant, ولو ال male ادي م affected male دا xld.

- all enzymatic and metabolic disorders are AR disorders.
- haemophilia is x-linked.
- most of blood disorder are AR disorder like sickle cell, thalathemia .

poly genic inheritance: -

-هي هل احدد بقدر م autosomal, ولا x linked ذي ال DM ولا HTN.

- **Clinical examination** and dysmorphology:

-start with general look to know dysmorphic feature or abnormal facial or body manifestation.

dysmorphic features of down syndrome: -

- epicanthic folds:
  above the upper eyelids.
- broad flat nasal bridge.
- upslanting palpebral fissure.
- brushfield spot: iris spots
- hypotonic open-mouthed appearance.
- portruding tongue.
- one palmur crease
- Asymmetry body appearance occur in Russell-silver syndrome.

- very important to look for dysmorphism in genetic diseases.

- there are 22 pair of autosome chromosome and 1 pair of sex chromosome.
- some time during cell division there is nondisjunction in autosome chromosome so give trisomy.
- genetic material in cell called unbalance genetic material.
- normal genetic material in the cell called balance genetic material.
- trisomy in chromosome21 down syndrome.
- trisomy 18 called Edwards syndrome.
- trisomy 13 called patau syndrome.

klinefelter syndrome: -

-this is morphologicaly male but have feature of female (gynecomastia, raised body, no fascial hair, increase in diameter of the pelvic, voice).
-feature of morphology start later in 2nd decade.

-the most important exchange between the chromosomes is reciprocat translocation which occur between Acro sentromere chromosome.
Acro= peripheral
-any chromosome has centromere and long arm "q" and short arm P.
-some chromosome has just one arm and centromere"14, 15, 16, 21" this chromosome have Acro centromere.
-sometime during cell division, there is union between 2 chromosomes"14, 21" وحتكون في cell و شالت cell نقص ودا اسمو robertsonion translocation, ال cell الشالت حتزيد ال total genetic material و ال حقتها cell النقصت حتنقص ال total genetic material ودا اسمو حقتها. unbalance robertsonion translocation, ودا كتير في ال down syndrome.

- mechanism of down syndrome to happen:

1.Non dysjunctional: -

-common mechanism
-in old women who get pregnant above 40 years.

2.Translocation.

protein-energy malnutrition

د/ جورج ظريف

- congenital heart disease, chronic renal disease causes secondary malnutrition.

- **primary malnutrition**: -

- all proplems in nutritional supply.

- **secondary malnutrition**: -

-catabolism more than anabolism, No proplems in nutritional supply.

- malnutrition is responsible for more than one third of hospital admittion, they have long stay and serious killing complicaton.
- PEM is a serious proplem.
- recurrent illnesses cause primary malnutrition, if not treated it become secondary.
- chronic illnesses cause secondary malnutrition.

- **Risk Factors: -**

- late supplementation = exclusively breast fed more than 6 months.
- poor quality or quantity = misunderstand to importance of certain nutrition.
- short spacing and large family are active feeding.
- Rota virus prevent against most causes of diarrhea " rota virus infection"
- metabolic diseases cause secondary malnutrition.
- abrupt weaning in early infancy give kwashikor.

- **PEM**: -

- proteins are mainly affected.
- proteins are nucleus of polypeptide enzyme in the body, so proteins deficiency means polypeptides and enzymes deficiency -> hormonal deficincy and cell membrane defect, so all body functions depends on proteins.
- when protein is deficent, physiology of body is affected.
- Liver cells synthesis the albumen, other proteins prouced by other resources.
- there is an effect on immunoglobulin and protective immune system, so develop ulceration and dermatitis.

- loss of gastric acidity cause infection.
- reduce capacity cause vomiting and diarrhea.

Classification of PEM: -

- there are many systems to classify PEM:
- screening system
- clinical system.

clinical systems: -

A) wellcome classification: -

* is clinical classification depend on wt / age pluse presence or absence of edema.
* الجدول يقرأ بالطريقة الآتية: -
- if weight for age below or above 60% for expected with or without edema that is Marasmus.

- advantage of wellcome classification:

1 - rapid.

- <u>disadvantages</u>: -

1) no method to measure age accuratly.
2) does not consider the chronicity of the disease which consider by height.

3) Any pt above 80% outside the malnutrition.

B) M.U.A.C classification: -
- mid upper arm circumference.
- mid way b/n acromian and the olecranon.
- it is one of screening methods.
- up to 5 years old.
- it is quick and more reliable classification.

C) WHO classification :-
- it depends on standard deviasion.
- if hte pt below -2 standard devation, he is considered as malnutrition case.
- it depends on wt for height: -
- after calculate wt for height must be know how far from midline, if below 2 standard deviations from midline this is hazard malnutrition, if below than -3 standard deviation this is severe malnutritionand deal with him as marasmus or kwashikor in management - deal with hazard malnutrition as undernutrition.
- WHO classification is more accurate, due to this 2 measurement (wt and ht) done by ourself, and no need for Age.
- underweight is mild from PEM, while marasmus / kwashikor and marasmic kwashikor are sever from PEM.

clinical types of PEM: -

1) <u>kwashikor</u>: -
- discovered by a nurce who noticed the infants who abrupt weaning undergoes generalized body swelling and skin change.

* occational signs of malnutrition:
1) moon face 2) Hair changes.

3) skin depigmentation. 4) Anemia.
- this signs occurs in marasmus and kwashikor except moon face which is characterise kwashiorkor.

- cardinal features of malnutrition: -

1) psychomotor changes: -
- pt with marasmus is irritable while pt with kwashiorkor is apathic (not interest surrounding).
- malnuitied pt either irratable or apathic but not lethargic which mean pt with metabolic problems have level of uncontiousness.
2) weight loss, by measure the weight.
3) wasted muscles.
4) presence of edema in kwashiorkor (edema is abscent in marasmus).
* Any malnurished pt have this four cardinal features.
* firstly you search about cardinal feature then look for occasional features, becuace occasional features without cardinal features does not indicate malnutrition.
* clinical manifestations: -
- massive edema involve all over thr body (face, upper limb and lower limb) pitting edema bilateraly

2) marasmus: -
- wasting is more apparent in marasmus than kwashiorkor.
- wt loss more apparent in marasmus due to presence of oedema in kwashiorkor.
- marasmus pt is starved person but still can utilize own resource so undergo to sever wasting.
- utilize own resources means still his glucogenesis active and lipolysis also it active ~~> then there is loss of subcutanous fat and loss of stored glycogen in muscles and liver ~~> sever wasting.

- treatment of marasmus pt is very easy, not like kwashiorkor pt who can not utilize his own resources ~~> shut down of lipolysis, glucogenesis and glycolysis.
* protein of animal milk cause gastroentritis.
* alert but miserable ~~> marasmus.
* alert but apathetic ~~> kwashiorkor.

- old than age, skin involvement, Total muscles wasting but still thrive (قادر يبكي و يرفع كراعو) -> Marasmus.

- enlarged liver is bad prosnostic sign in kwashiorkor.

- ***Managment:*** -

- depend on clinical assessment, then investigation and complication.
* transition stage: -
with threputic milk and RUTF.
- RUTF: is formula which developed by WHO, it contains minerals and vitamins, it is brown in colour.
- RUTF = Ready to Use Theraputic Food.

- **How to assess malnurished pt?**

1) History: -
elevate risk factor of malnutrition ~~> recurrent gastroenteritis, prsistant diarrhea, congenital proplems, PMH operation, extending Family, poverty.

2) examination: -
general look:
- look irritable or apathetic, emaciated or not, buffy face or not.
- assess pallor and cyanosis and jundice.
- assess height.
- examine baby for the cardinal features of malnutrition.

- Full examination from head to toe of malnourished baby.

clinical signs:
1) visible sever wasting on ~~> ribs, inner part of the arm, inner aspecrt of the thight, gluteial region and interscapular area.
- sever gluteial muscle wasting called boggy pant sign.
2) nutritional bilateral pitting edema.
- compress dorsum of the feet for one minute: -
    + ~~> if involve dorsum of the feet just.
    ++ ~~> if involve the distal part of the upper and lower limbs.
    +++ ~~> if it generalizes.
- this division (+, ++, +++) is important in follow up.
* after detect the cardinal features give RUTF for pt and do appetite test.
* dermatosis is occasional sign.
* just discolouration of skin consider as dermatosis.
* dermatosis is important in follow up.

***examination from head to toe: -**
* hair: -
- if hair distribution is affected we called it spared ~~> ماشي على alopecia.
- if hair colour is affected ~~> whitich or yellowish ~~> ماشي على PEM
- Do puluckable sign:
تمسك شعرة لو طلعت معاك عادي دي بتكون easy puluckable.
- Flag sign of the hair which means different colour in one hair which indicate staging of affection and recovery.

* <u>eye</u> examination: -
- it is very imortant in malnourished pt due to vitamin A deficency is reversible stages, treatable, common with malnourished pt.

- must be discover vitamin A deficency early to Treat it, or give prophylactic vitamin A for every pt with malnutrition to prevent vitamin A deficiency.

* **signs of vitamin A deficiency**:
1) cloudness.
2) eye discharge ~~> any eye discharge in malnuorished pt consider vitamin A deficiency.
3) even cloudness of sclera and Bitote's spot.
* when we found signs of vitamin A deficiency starts theraputic dose of vitamin A.
* look for nasal discharge.

* oral cavity examination: -
- look for cholitis inf ection of lips and chelosis (inflammation of lips) ~~> both indicate riboflavin vitamin B2 deficiency.
- look for angular stomatitis of riboflavin deficiency.
- look for glossitis (stomach tongue) which may be painless in iron deficiency anemia or painfull in vitamin B12 deficeincy.
- look for chemical infection (oral thrush) which indicate low immune status.
- look for delay dental ruption.
تأخير بزوغ الأسنان الناتج من calcium and vitamin D deficiency.
- look for dental carries which indicate flouride deficiency.
- look for gengival bleeding which indicate vitamin C deficency.
- all minerals and viramins are malnutrition.

* <u>neck</u> examination: -
- For thyroid enlargment becuase thyroid tissue may be affected, becuase thyroid hormone is a protein in nature.
- look for lymphadenopathy

* chest examination: -
- look for chest infection witch may be sever and one from death cauces in malnutrion.
- auscultate for complication.
* examine pericardium for cardiomyopathy due to selenium deficency, which lead to poor contractility of heart and heart failure.
- Anemia from malnutrition, Anemia may lead to Anemic heart failure.
* abdominal examination: -
- examine for hepatomegaly.
- examine the genetalia and lower limbs for dermatosis / muscle and joint swelling and infection.

* **Examination include**: -
1) cardinal feature.
2) systemic examination (hair to toe).
3) complications.
* complications of malnutrition need for rapid management.
* complication of malnutrition consier as emergency.
* management of malnourished pt:
- admittion.
- give him formula 75: -
- 75 means each 100 ml contains 75 kal.
- density in 75 is low than in 100.
- start with 75 because there is Atrophy in GIT so there is low absorptive surface area, so high density meals cause osmotic diarrhea, so we give 75 and wait till regeneration of absorptive surface in 4 days then give 100.
- you give 75 in First 4 days of admittion then convert to 100.
- give antibiotic even if no infection, give it as prophylaxis due to he has immune disturbance.
- give minerals and vitamins whithout iron which is good media for growth of bacterial.

- unless there is strong indication for iron supplementation, usually after treatment of Anemia.
- give zink.
- treat complication.
* Rewarming Techniques: -
- kangroo technique: -
- skin to skin technique.
* signs of dehydration in malnutrition is called irreliable sign.
* it pt with malnutrition treated as case of normal baby with sever dehydration willl be overlooded with fluid ~~> iatrogenic death.
* Assessment of dehydration in malnourished baby as:
- pulse rate - pulse volume
- urine output. - history (vomiting).
* centile chart: -
~~> below -3 standard deviation ~~> for admittion.
~~> b/n -1 and -2 ~~> No admittion unless there is edema.
below 3- standard deviation ~~> severe PEM ~~ for admittion.
- if weight for length not below -3 standard deviation but there is wasting and visible edema ~~> sever PEM
و حأنتظرو لحدي م يحصل ليه below sd -3 و يبقى edema لأنو بعد شوية حيفقد ال improvment و يطلع above 1ل-

Gastroenteritis

Patient mainly presents with vomiting or diarrhea.
Major causes of gastroenteritis are microorganisms (bacterial, viral and protozoal).
Usually due to food poisoning.
Infection>>> for example:
1) Campylobacter jejuni usually causes gastroenteritis through improtect.....food.
2)Bacillus group can cause gastroenteritis in unpasteurized milk or cheese of unpasteurized milk.
Home canning food is contaminated with C. botulism.
Commercial food is also contaminated with C. botulism.
3)Botulism in the first comes with symptoms of gastroenteritis, later on within few days gives serious complications (diplopia and descending paralysis)
4)C. perfringens poisoning occurs in under cooked meat.

VIRAL INFECTIONS:
-Usually transmitted by files and by food worker.
-The contamination with spores causes the distribution of the viruses (rota virus and noro virus) which cause diarrhoeal outbreaks usually due to food workers.

-All sources of gastroenteritis are microorganisms transmitted by food or food workers.
-Those microorganisms have tge ability to produce toxins inside/outside the host so cause gastroenteritis through other mechanisms.

PATHOGENESIS:
1)Those produced toxins>>++cAMP and lead to Secretary diarrhea.

2)Those microorganisms destruct the microvilli within lumen surface (the absorpative surface)>>> Destruction of the absorpative surface>>
- - of absorption>> osmotic diarrhea.
3)The third mechanism is adhesion of this microorganisms to GIT surfaces >>> Acting like a barrier between absorpative surface and GIT cells.

These are the major pathways of microorganisms to cause non inflammatory diarrhea, No invasion of microorganisms to circulation.
-The microorganisms present in the GIT lumen and cause diarrhea through:
- TOXINS.
- Destruction of microvilli.
- Adhesion and acting like a barrier between cells and lumen.

INFLAMMATORY DIARRHEA:
There is invasion of microorganisms to the circulation>>>leakage of RBCs & WBCs into GIT lumen>>so the presence of RBCs in the stool indicates inflammatory diarrhea.
-Viral infection is very rare to cause invasion/usually through destruction of microvilli.
_Bacterial infection usually causes inflammatory diarrhea.

Onset of manifestations(vomiting/diarrhea) reflects the possible mechanisms of gastroenteritis:
canned food vomiting after 6 hours>>>>
This vomiting is due to performed toxins.
_It's unlikely of microorganisms to cause rapid rapid destruction of cAMP and give rapid manifestations unless there is performed toxins outside the host.
_Rapid onset of manifestations is definitely due to toxins.
_Antibiotics in rapid onset bacterial infection have no role.

Salmonella usually produces diarrhea 48-72 hours after getting infectious disease by salmonella>>>this due to destruction of cAMP and adhesion but rarely due to invasion of the GIT cells.
so it's very important to know the onset of gastroenteritis symptoms after taking food.
@@@@@@@@@@

Complications of gastroenteritis:
1)Fluids loss which leads to dehydration.
Alot of complications ARE if not corrected rapidly lead to necrosis? Which gives raise to chronic renal disease (which is irreversible).
_Acute renal injury due to dehydration is reversible.
_Hyperviscosity is one of the complications of dehydration (hyperviscosity syndrome).
_Hyperviscosity means slow flow>>>Thrombosis and ischemic pain in end arterial organs.
2)Electrolytes disturbances:
_in lower GIT there's the absorption of k >>>in diarrheal diseases there's hypokalemia and chloride deficiency.
_Hypokalemia leads to hypotonia and twitching of the muscles and cardiac arrhythmias.
3)Metabolic acidosis:
_Due to vomiting.
_As with tetanus.
Patient's have normal calcium levels but due to metabolic acidosis this calcium becomes inactive>>>tetanus isn't due to true hypocalcemia.
4)High grade fever due to fluids loss which is responsible for distribution of temperature.
5)Dissemination of infection gives:
_pericarditis.
_phlebitis.

_pneumonia.
_pyelonephritis.
_osteomylitis.
_all body organs could be affected by the bacteria.
6) after gastroenteritis due to campylobacter jejuni there's an ascending paralysis of GBS and may develop reactive arthritis.
7)May develop HUS in E. COLI infection (e. coli have certain species produce) bloody diarrhea.and later on patient presents within hemolysis (jaundice).
HUS is characterized by:
_prehepatic jaundice.
_features of hemolytic anemia.
_low platelets.
_acute renal failure.

Prognosis of gastroenteritis:
The gastroenteritis recovery is within few days (maximum 7 days) because most of gastroenteritis is due to viral infectionsinfection and non inflammatory processes.

- Gastroenteritis that persists for more than 14 days>>is called persistent diarrhea.

- Persistent Diarrhea:

_Is an acute gastroenteritis that persists more than 14 days.
_It's a risk factor for malnutrition, it causes losses and electrolytes disturbances.
_Long complications of this persistent diarrhea are serious complications.

- so it's very important to know if the patient has a persistent diarrhea or an acute gastroenteritis.

- In history you ask about:

_onset.
_Types of food taken by the patient.
_GIT manifestations.
_Search about complications.
_Ask for urine output which is decreased in severe dehydration.

Assessment:
_You assess patient mainly for dehydration.
_General look.

* **Categorization of dehydration** by the WHO:
1)No dehydration.
2)Some dehydration.
3)Severe dehydration.
Those 3 groups are according to 4 parameters:
1)Mood (key).
2)Sunken eyes.
3)Skin pinch.
4)Thrust of the patient.
*Assess General look to know mood which either:
_Normal>>no dehydration.
_Irritable>>some dehydration.
_Lethargic patient>>severe dehydration.

***Assess sunken eyes**:
NO>>no dehydration.
YES>>some and severe dehydration.
*Give the patient a cup of water to assess thursty:
_Able to drink and irritable>>>some dehydration.
_Unable to drink and lethargic>>>severe dehydration.

*Test the decreased elasticity of the skin due to less perfusion with fluids.
*In malnutrition due to loss of subcutaneous tissue, the pinched skin comes back very slowly.
_Do skin pinching and release the skin:
_Back in less than 2 seconds>>Normal.
_Back in 2seconds>>some dehydration.
_Back in more than 2 seconds>>severe dehydration.
*The management is different according to the degree of dehydration.

*Start the examination by the General look:
This patient is ill or well.
Conscious or not>>patient might be comatosed due to dehydration and electrolytes disturbances and hypoglycemia.
_Examin pallor, jaundice, and cyanosis:
Dehydrated patient's have pallor.
_Pulse rate is thready rapid pulse in severe dehydration.
_Respiratory Rate RR is rapid and fast due to the metabolic acidosis due to accumulation of lactic acid in dehydration.
_High temperature due to dehydration and microorganisms.

*Assessment of the patient **from head to toe**:
>>Sunken eyes.
>>Dry mucus membranes.
>>Absence of tears (crying without tears).
>>crypitation due to infection.
>>skin pinch
>> Complications might be found:
_Ascending paralysis.
_Skin infections.
_Arthritis.

MANAGEMENT:
1)Some dehydration:
Use oral rehydration solution.
2)Severe dehydration:
I.V fluids.

Correction of the dehydration status:
A) No dehydration:
_Counseling.
_Increase the intake, to decrease vomiting and diarrhea>>>Give non osmotic meals (sugar)/encourage breastfeeding.
_ORS after each motion.
_Minerals like zinc which improves the GIT cells' replacement (Zinc causes rapid regeneration of microvili).
_Increase the enteral feeding.
_Counsel the mother and tell her when to retain back@@@
If patient is lethargic and unable to drink>>>patient may convert from NO dehydration to SEVERE dehydration.
If patient develops bloody diarrhea >>>it needs Antibiotics.

B) Some dehydration:
_Oral rehydration solution at hospital level within 4 hours give 75ml/kg.
_Observe patient before and after rehydration (Sunken eyes, Thrust)
_If patient becomes no dehydration discharge it and convert to plan A (deal with him as no dehydration).
_At hospital is plan B.

C)Severe dehydration:
_plan C.
_Give I.V fluids>>>100ml/kg.

Ringer lactate is recommended because it has k and cation.
Or Normal saline.
_Give any type of fluids that must be present in the iv circulation and doesn't leak from circulation to the interstitium.
_Assessment before and after rehydration to know to which degree of dehydration it has converted to (plan B or plan A).

*Plan A>>home management and counseling.
*plan B>>at hospital(75ml/kg).
*plan C>>at hospital(100ml/kg).

*Other models of management:
_Antibiotics>>>recommended in patients with bloody diarrhea.
_Antibiotics are useless, it may destroy the normal flora and cause secondary bacterial infections.

ANTIEMETICS:
Vomiting reflects the presence of toxins outside the host.
No role of antiemetics in gastroenteritis because they give chance to organisms to stay longer.

*Don't Give antibiotics and antiemetics to patients with gastroenteritis.

*<u>How to treat patients with gastroenteritis?</u>
_Oral rehydration solution.
_Increase the enteral feeding.
_Minerals in form of zinc.
_Don't give antibiotics unless there's bloody diarrhea.

Juvenile rheumatoid arthritis

Manifesting in soft tissue => pain or tenderness in soft tissue.
*The definition subtype depends on first 6 months involve joint (number of joints)
*There is relation b/n specific HLA subtypes with RA.
*This HLA subtypes involve individual at risk and other protected.
فيهم ناس at risk و فيهم ناس protected ,,,
*Hotness due to hyperviscousity and infilteration of inflammatory cells.
* At the end of the day there is pain due to massive using of this inflammatory joints.

Types of arthritis: -
- mechanical or traumatic arthritis
- inflammatory arthritis
* RA is inflammatory disease.

Traumatic arthritis: -
Associated with pain more than morning stiffness.
الزول لمن يصحى الصباح م بكون عندو اي symptoms , لكن أثناء اليوم بعد ال massive using of traumatic joint بحصل ليه pain .
Inflammatory arthritis: - associated with morning stiffness.
بعد يصحى بكون معسم لكن التعسيمة دي بتروح شوية شوية و بجي ال pain .
*It is very important to differentiate b/n inflammatory and traumatic arthritis.
* any inflammation involves joint in which there is morning stiffness, morning stiffness doesn't present in RA only.
يعني الزول ما يربط ال stiffness بال RA بس .
*Morning stiffness associated with inflammatory joint disease what ever the underlying cause (CTD, RA, etc).

*Inflammatory joint disease destruction chronic disability.
Traumatic disease there is tearing in ligament regeneration of ligament … vessels … bone … and no chronic disability.
5ال تلقى انك مهم م cardinal feature ال بتاعة inflammation ال في joint.

Oligoarthritis involve big joints of lower extremities, not involve small joint … tarsal and meta tarsal bone.
*If the hip joint is presenting arthritis, it isn't RA keep to other cause of arthritis.
*On X ray there is total destruction of hip joint and ^ bone density and loss of space
 x poly arthritis: -
Tarsal, metatarsal, carpal, metacarpal and interphalyngeal joint may involve even vertebral column, cervical spine and atlanto axial joint, tempromandibular joint may be affected.
-Disease more than 20 joints may be affected.
-may cause chronic disability
-If involve atlanto axial joint may lead to cut of spinal cord and quadriplegia.
-hypotonic down syndrome at risk to develop subluxation of atlanto axial joint so develop quadriplegia.

*Systemic onset arthritis: -
Unless you think about it, you never reach to diagnosis; because it has common presentation with other common disease such as malignancy, leukemia, kalazar and other CTD.
-Rash at time of presentstion it doesn't appear, but it evokes after warm bath transient rash.
*Koebner phenomenon: -
-skin at the site of trauma become reddish.

-it is associated with autoimmune disease, joint disease.
-Not pathogenomic of systemic disease onset.
Diagnosis: -
-Inflammatory joint disease means morning stiffness more than pain.
-Laboratory investigation to know and suspected prognosis but not to confirm the diagnosis which confirmed by criteria.
-This laboratory finding occur in other CTD so it is not specific.

-Negative serotype doesn't exclude juvenile RA.
-The presence of rheumatoid factor carries poor prognosis and chronic disability.

*Iridocytilitis: -
-serious complication of oligoarthritis.
-inflammation of the eyes.
-lead to blindness.
-female < 5 years.
-positive ANA.
=Oligoarthritis with positive ANA need to regular ophthalmological examination 6-8 monthly.
-It is painless condition, have no symptoms, so it is serious.
*D D: -
#Juvenile ankylosing spondylitis: -
Complication of polyarthritis.
It is common in polyarticular type in male >8 years.

#Inflammatory bowel disease presented with arthritis.
-Crohn's disease, Ulcerative colitis.
#psoriatic arthritis: -

-have koebnar phenomenon.
-have skin involvement and arthritis.
#Reactive arthritis: -
-Usually follow upper respiratory tract infection.
-it is one from important DD of Rheumatic arthritis.
-affect big joint of lower limbs and response to anti inflammatory drug (NSAID).
-Any patient suspected as reactive arthritis put the pt in 6 months benzothine penicillin
1/ if there is recurrence consider it as rheumatic fever and continue.
2/ if no recurrence stop benzathine penicillin and follow up for 6 months and consider reactive arthritis.

- **Management**: -

There are three lines of choice:
1/ NSAID
2/ STEROIDS
3/ CYTOTOXIC DRUGS

There are other types of management rather than drugs:
Occupation, exercise.
Advise pt to go for hypotherapy which is very important to pt of RA.

- **Prognosis**: -

-ANA and RF aren't important for diagnosis.
-Erosions on X ray.
-Un remitting course
-Only 20% of polyarticular typr are carrying poor prognosis.
-Any erosion carries poor prognosis.

*If case start as oligoarticular, then by the time there is >5 involved joints polyarticular □ don't change subtype and still this is oligo □ oligo have risk for iridocyclitis.

Thyroid disorders

D – George Zareef

- The commonest 2nd endocrine disorder after DM
- T3 is more potent than T 4

1. Thyroid is important for growth and development (somatic + neurological growth)
2. Its catabolic hormones increase glycogenolysis _lipolysis and FFA oxidation
3. Thermogenesis
4. Potentiate action of catecholamines --> penetrate cell membrane --> intranuclear receptors --> potentiate action of mRNA

- Thyroid hormones act directly on intracellular receptors.

Hypothyroidism

- The commonest pediatric thyroid disorder
- Congenital or acquired (juvenile)
- Can cause irreversible brain damage or retardation --> cerebral palsy.
- Commonest cause is iodine deficiency
- Decreased iodine --> decrease T3 and T4 and inhibit negative feed back so TSH increase which is hyper trophic hormone leading to thyroid enlargement and goiter Formation
- If there is decrease iodine in intrauterine life the baby will develop neurological hypothyroidism or neurological cretinism which manifested by: -

1-developmental delay
2_Mental retardation
3_Spastic Paraplegia
4_Squint

Neurologically also can develop myxedema

- **characterized** by: -

1_developmental delay
2_ Dwarfism

CP: Congenital or acquired

- Usually don't appear at birth
- Always at the 6th week of neonatal life
- Pt has coarse facial features:

1. Dry hair
2. Dry skin
3. Protruded myxedematous swollen tongue (affect swollowing and crying)
4. Short full fatty neck
5. Umbilical hernia
6. Prolonged physiological jaundice_ whenever u counter pt with it investigate for hypothyroidism
7. Large fontanelle
8. Hypotonia and hypothermia
9. Constipation

#Hypothyroidism can cause Deafness
#The 1st deffrential diagnosis is mucopoly saccaridosis

- **Causes**:

- TRH deficiency
- TSH deficiency or mutation in TSH receptor
- Dyshormonogenesis

When the problem within pituitary (Panhypopituitarism) it is known as 2ndary Hypothyroidism

- **Investigations**:

- Free T4 is decreased
- TSH is increased in 1ry and decreased in 2ndary hypothyroidism

Rx : Thyroxine for life

Secondary juvenile :

Normal baby till todler then develop deceleration of growth
Occur in the first 2 Years
Commonest cause is hashimotos thyroiditis + Radiation (Head and neck).

- **Investigations**:

- T4 is decreased /TSH is increased
- Antibodies: Anti thyroglobin and anti microsomal antibodies (for hashimotos)

- **treatment** : Thyroxine For life

Hashimotos Thyroiditis

- Autoimmune disease
- Inflammation of thyroid tissue and infiltration of inflammatory cells, lead to unfunctional Thyroid Gland.
- Earlier euthyroid later hypothyroidism
- 10% of Hashimoto will be thyrotoxicosis.

- **Treatment** : Thyroxine for life

Hyperthyroidism

- **Causes**:

1. Graves disease
2. Subacute Thyroiditis
3. Factitious
4. TSH secreting pituitary tumor

- **treatment**: Propythiouracil /methimazole

In refractory cases: Radioactive Iodine

1- Graves disease:

- Eye signs: Lid retraction and lid lag
- Peak age 11 and 15 yo
- F:M ratio 5:1
- Antibodies act as TSH stimulating thyroid Tissues through Thyroid Receptor
- In pregnancy these Abs cross placenta and go to thyroid tissue of the baby so baby develop thyrotoxicosis (Causes of neonatal Thyrotoxicosis?)

CP:

- IUGR
- FTT
- Microcephaly
- Frontal bosing
- (In heart): Systolic HTN + wide pulse Pressure + Tachycardia + HF +Arrhythmias
- Diarrhea+ vomiting + flushing (catecholamine features)
- Jaundice + thrombocytopenia (resembles TORCH)
- Exophthalmas + Goiter

Pt undergoes spontaneous recovery within 6 Months (BCZ Abs are cleared).

- **Treatment**: as thyroid storm + Digoxin (If pt has HF and features of toxicity)

2- Thyroid Storm:

_ Occur in baby of mother with Graves

_ occur at any age

_ Normal child then acute onset of:

- Hyperthermia
- Tachycardia
- Irritability

_ If untreated: Delirium –Coma and death

_ Thy′ storm act as severe Infection.

- **RX**: Propanolol + k Iodide+ Radiation therapy as long term Rx (Technitium is radioactive Substance)

Radioactive iodine: Increase in hyperthyroidism and hypo due to dys hormonogenesis

3- Thyroxine binding globulin deficiency

Good Uptake + decrease T3.

Congenital Heart diseases

-Heart is a pump, control of circulation all over the body and maintaining Blood pressure.
-The Heart has two part (right and left side), they are working separately, the muscles are same but the function totally different, the right side has certain function, the left side has different function, simultaneously both sides are working together.
-so there is no direct connection between Rt and Lt side, totally separated, no internal connection, the connection around circulation.
- The Lt and Rt atrium are just continuation of the great vessels regarding the structure and function.
-The Rt atrium is continuation of SVC and IVC.
-The actual pump are Rt &Lt ventricles; the muscles of Rt&Lt are ejecting blood into circulation.
-CHD is common problem, 8per 1000.
-congenital mean the child born with this defect.
-CHD have certain association with other problems for example some congenital problem may be predisposing factor for CHD like congenital infection like rubella and torch in general.
-Torch infection is association with CHD (rubella and PDA).
-certain maternal diseases like DM is ass with CHD, also genetic diseases like trisomy (down's syndrome).
40-50% of pt with down syndrome are showing some part of CHD.
Trisomy 21,18,13 all these diseases are common ass with CHD.
Classification: -
According to the shunt: -

- Left to right shunt ->oxygenated blood goes to the lung (congestion).
- Right to left shunt ->deoxygenated blood goes to circulation (cyanosis).

-LT to Rt shunt ->congestion.
-Rt to left ->cyanosis.

-congestion and cyanosis are main problem which caused by CHD, may occur in some pt

- Obstructive stenotic lesions: - any valve stenosis PS, AS, MS ,coarctation of aorta.

- **Clinical classification: -**

1. Cyanotic CHD.
2. Acyanotic CHD.

Acyanotic CHD

- **Ventricular septal defect(VSD): -**

The most important disease in CHD in the world (22-23%).
*interventricular septum: -
-90% of it is muscular part.
-10% is membranous part.
(90% of VSD in the membranous part, 10% in the muscular part).

Hemodynamics: -
Usually the shunt is left to Rt because: -

- pressure in the Lt side more than Rt side.
- Contractility in the Lt side start before contractility in the Rt side in a fraction of second, so this will create pressure gradient from the Lt to the Rt that make shunt from Lt to Rt --> the blood goes to the lung and again goes to the left side in the large amount --> dilatation of the left side, the Lt ventricle dilated in VSD.

-bidirectional flow is serious problem.

- **Clinical features:-**

-History.
-small VSD have no symptoms, but some inelegant mothers may feel thrill, but usually it's discovered accidentally (during routine examination you find murmur but from history it silent). small VSD has no symptoms no complain and no complication.
-Moderate to large VSD --> the problem is noticed by the mother and family.
-The growth is affected (delayed growth).
-Difficult feeding in the infant reflect decrease exercise tolerance. (Breast feeding needs effort and energy by the infant).
-Repeated Chest infection, repeated admission to hospital and repeated congestion.
*when you examine the child, you find low weight and wasting. During infancy the weight affected more than height, after that even the height will be affected because it takes along time, so the child will be wasted and stunned.
*Bulging pericordium may be present or absent.
*visible pulsation.
*systolic thrill at left lower sternal border.
-displaced apex.
-VSD, TR, MR --> murmur of all these are auscultated in lower left sternal border.
-other murmur auscultated in upper LSB.
*thrill is palpable murmur (you may find murmur without thrill but you will never find thrill without murmur).
The characteristics murmur of this disease is pan systolic murmur in LLSB, usually this murmur is harsh, loud and heard all over the pericardium with maximum intensity at LLBS.
*you may find other murmur: -

- The amount of blood through the pulmonary artery more than usual --> volume over load through pulmonary artery lead to ejection systolic murmur at the pulmonary area.
- A lot of blood go from LA to LV through mitral valve --> diastolic murmur at mitral area.

*the classical murmur is pan systolic murmur other murmurs aren't important.

*you may find other signs which in the heart by other lesion or other signs outside the CVS, other signs are explain by other associated problem.

- **Investigation**: -

 ECG: -the classical presentation of ECG is:

-left axis deviation.

-feature of LV hypertrophy

 Chest X ray: -

-signs in ECG and chest X ray depend on the size of lesion.

-small defect --> normal ECG and chest X-ray.

- moderate and large defect --> LV dilatation.

- on X-ray you find:

- 1) dilated chamber or LV.

2)the heart shadow should be larger than usual.

-usually the whitish part compromise around 50% in adult and 60% in young infant

*more than 60% mean cardiomegaly.

3)Congestion.

4)you find also patchy (more whitish) at the area of distribution --> whitish more than blackish (normal lung is blackish in X-ray)

-in moderate to large defect or any causes of congestion you find whitish more than blackish in area of vascularity this is called **plethoric** or congested lung.

-X-ray is anteroposterior.

- Echo cardiography: -

-it's diagnostic.
-the definite diagnosis by echo.
*benefits of echo: -
-estimate site of defect.
-localize the defect.
-give idea about hemodynamic.
-detect complication.
-estimate the PA pressure and it association with pulmonary HTN or not.
-measure chamber dilatation.

*Natural History: -
1)if the lesion left without treatment,the possibility of **spontaneous closure** can occur in majority of cases (small and moderate defect).
-some defects will not close at all.
-small and moderate defect are the same in spontaneous closure.
-moderate and large defect are the same in clinical presentation.
-the timing of spontaneous closure is very difficult.
-you can't anticipate which defect is going to close.
-not every small defect will close.
-not every moderate defect will not close.
-the muscular defect closes more rapidly than membraneous defect.
*closure needs time (years).
-Majority of the defect will close before age 7-8 years (some defect take more than this period).

2)**congestive heart failure** is definite presentation in Heart defect.
-it occurs early during infancy, if the defect is large, appear after 6-8 weeks, before that time child is asymptomatic even the defect is large --> dominant Heart during intrauterine life is Rt side, pressure in Rt side in the fetal circulation is higher. after delivery the circulation is

changed and pressure in the Lt side more than Rt side but this needs time doesn't occur immediately (need few weeks, differs from infant to infant) during this time pressure in Rt&Lt side may be equal, then gradually the pressure in Lt side become high and Rt side become low.
(في الفترة دي ال في اس دي بكون ضائع في النص)
-the shunt become more Prominent after age of 6 weeks.

3)Eisenmenger syndrome: -
-it is catastrophic complication.
-it's n't treatable, you can't help the child.
-it's n't treatable unless you do Heart lung transplantation.
Pulmonary vascular obstructive disease: -
-this is complication of large defect
-this problem need long time (years)
-during this period changes in the lung occur gradually: -
- In the beginning there are repeated chest infections, then these changes affect the elasticity of the small blood vessels of the lung, these vessels become stiff and could be obstructed, the flow in this small vessels will be obliterated, pressure increases inside the lung and there is back pressure of the blood to the Heart gradually.

This lead to increased pressure in the Rt ventricle, then pressure on the Rt ventricle exceed pressure on left ventricle this will cause clinical **ES**.
Clinical presentation of Eisenmenger syndrome:

Syndrome will occur as we mentioned above, this reverse shunt could be, Rt to Lt shunt rather than left to Rt shunt.

#honey moon period: -
-symptoms are relived, congestion is improved, the pt become Asymptomatic for long period may reach 2-4 years, after that the child become cyanosed and all complication of cyanosis will occur in pt and the child will die after that time.

Infective endocarditis: -
Usually it's one of the complication of small septal defects.
-it needx pressure so it occurs with small defect.
-**IE** to occur needs abnormal endothelium.
-no **IE** in large defect because there is no pressure.

- **Management**: -

-management of small defect just conservative and you should follow the pt and do counseling.
-if the child became febrile he must be reported to clinic to make sure if this is infective endocarditis or not.
-if the VSD is moderate and the child is asymptomatic, do consulting, conservative treatment and follow up
-if the VSD is moderate size and pt symptomatic give Antifailure drugs.
-Medical treatment is given for symptomatic pt.

Antifailure drugs: -
-Lasix.
-Captopril
-spironolactone.
-give combination of Antifailure drug and follow up the pt.
-when the pt improved stop drugs.

#if the defect is large, it needs intervention: -
* intervention in VSD either surgical or by catheterization
Muscular VSD is favorable to catheterization.
-surgical intervention is suitable choice of intervention in majority of pt with VSD, and it is open Heart surgery.
-in open Heart surgery you put the pt in bypass machine (you cause cardioplegia of the Heart and stop the heart from the pumping).

-open means mid sternetomy.
-you take Heart outside and cause cardioplegia, so Heart stop working, then introduce heart in machine which act the working and lung and heart together, then close the defect by patch and suture this patch, then you should be sure about closing the defect, then pt is taken to ICU and put him in ventilator for hours.

Atrial Septal Defect

-it is common problem that discovered accidently.
-affect female more than male 2:1 .
#Types of ASD :-

- Ostium secondum
- Osmium Premium.
- Sinus venous: -

At the roof of the atrial valve at the upper part
-usually isolated.
-premium not isolated.
-SV usually may be associated with abnormal pulmonary venous drainage.
-ostium secondum at middle so it may be large or small .
-50% of OS is part of atrioventricular septal defect .
-AVSD also called AV canal .

#AVSD :-
Consist of OP ASD and VSD which usually at the inlet side .
-common AV valve with 5leaflet (mitral+tricuspid).
-this common valve will defective with cleft that may cause regurgitation of different degree of severity, sever &moderate.
-pt with down syndrome commonly have

atrioventricular septal defect(Treatment of it usually is surgical)

clinical presentation :-

-c/p of secondum ASD : -

usually Asymptomatic if the pt have
isolated Secondum ASD but this with exception especially if there is large ASD the pt will develop congestion

- on examination there is wildly split

Fixed S^2.

- S^2 have two component (pulmonary and aortic component).
- this splitting isn't audible but you can heard it during expiration
- but in this pt it is fixed and wildly split (the timing between 2 component is wide)
- fixed mean it heard during inspiration and expiration.
- you may heard a murmur which isn't Feature of this disease.
-in ASD there is no pan systolic murmur because pressure between the Rt and Lt side at the atrial part is very low.
-pressure in Lt atrium more than pressure in Rt atrium but the pressure between them is very low.
-pressure allow left to right shunt occur but it Isn't allow murmur to occur.

- Small ASD>no murmur.
- Moderate ASD>no murmur .
- Large ASD >you may find murmur which is ejection systolic murmur at pulmonary area, it is grade 2or 3,that mean if you found thrill this Isn't thrill of ASD.

-third means grade 4 murmur.

Investigation of ASD :-

- ECG :-

-opposite to ECG of VSD .

-The blood go to Rt side from Lt side and during diastole (resting period)go to Rt ventricle ,so there is dilatation in the Rt side (Rt ventricle)so ECG reflect this testing stage by Rt axis deviation and right ventricle hypertrophy.

- Chest X-ray: -

-there is cardiomegaly due to Rt A&Rt V dilatation .
-prominent pulmonary artery.
-pulmonary congestion Isn't feature on ASD .

- Echo :-

-diagnose the case.

Natural History of ASD :-

- Spontaneous closure is very common In the 1st year of life

-small and moderate ASD may closed before discover it.
-spontaneous closure is common in secondum ASD .
-consul the parent to wait in 1st 3 years if the ASD is large or moderate.
-In 1st 3years just conservative treatment, don't interfere in 1st year .
-most of defect are closed in 1st year, if it's n't close in 1st year, the defect has low possibility to close after 3 years of life, after 3years you can do intervention.

- if the pt is unlucky and discovered late, he will develop Heart failure and pulmonary HTN.

-most of the pts are Asymptomatic so you find adult pt with ASD .it discovered accidentally.
-if the defect is large it cause HF,and P HTN

- Atrial arrhythmia in old children and young adult (10years and more).

-atrial arrhythmia either atrial fibrillation or atrial flutter .

- Infective endocarditis Isn't Complication of this disease at all because there's no pressure.

Treatment of ASD :-
-small defect >wait for 3years .

-moderate defect >wait for 3years .
After 3years you determine What you have going to do..
-regarding to secondum ASD >treatment is closure by device which inserted by catheterization .
#device closure by catheterization:-
The internus by the Rt side then go to Rt atrium through the defect go to left atrium, this under vision after it reach the left atrium you inflate the device and go back…..
-in the start you measure size of defect and select the suitable size of device to successful.
-not every pt is suitable to device closure so you need to go for surgery in some pt because the length of interatrial septum Isn't suitable to fix the device..
-surgery in ASD is open Heart surgery but more easier than VSD surgery .
-ICU facilities needed for this pt is not critical like pt with VSD .
-the time to stay in ICU in pt with ASD is less and motility rate following surgery more than mortality following surgery of VSD .

PDA

- it is common problem..
- more in female children, rather than male..
- it is common in premature babies.

#what is patent ductus arteriosus ?
- it is persistent patency of connect between left pulmonary artery and the descending aorta.
in the fetal circulation this ductus arteriosus is source of Fetal circulation continuation.
- it's carries blood from right dominant heart in intrauterine circulation to lower part of the body Fetus, Through this connection with descending aorta

- Left side of the heart hasn't function in Fetal circulation.
-after delivery this ductus arteriosus close immediately and physiologically some time it stay or become small gradually till closing after few days a but in normal babies the ductus arteriosus close physiologically after birth immediately .
-some babies record with patent ductus
For couple of days, then gradually it close. but in the diseased baby this ductus will not close and could be persistent open or patent.
- the pressure in the left side will increase after delivery and pressure in the right side will decrease so this ductus become Source of shunt from Lt Side to the right side. it act as VSD Typically .
All the data found in VSD ,found in PDA.
-Hemodynamic is same in PDA and VSD:- pressure of the Lt side as the same and pressure in Right side as the same the ventricular pressure and arterial pressure are the same ,the PDA has pathophysiology as the same of VSD.

clinical presentation :-
the same as VSD.
- small duct like small VSD the pt is Asymptomatic.
- large PDA cause respiratory infection and congestive heart failure.
-the history is the same as VSD.

on examination:-
- There are some different:-.
1) Bounding(large volumes) peripheral pulse because is arterial problem.
- PDA like fistula of dialysis >in 2 there is hyper dynamic circulation .
- one of the characteristic feature of PDA is Hyper dynamic circulation(large volume peripheral Pulse).

- collapsing pulse – large volume
2)wide pulse pressure is feature of hyper dynamic Circulation
- wide pulse pressure= wide different between systole and diastole.
(30 - 50) Normal pulse pressure.
when pulse pressure more than 50,This is wide pulse pressure.
3) hyper active pericardium.
4) systolic thrill at the upper left sternal Border which is usual area of murmur.

Investigation:-
- similar to VSD.
ECG , chest x-ray ,Echo …

* Treatment of PDA :-
- usually Treatment of PPA by using device Through catheterization .
- this is best way to manage PDA.
- This is Favorable way to child.
- This device introduce by catheter through venous circulation > go to the right side of the heart then pulmonary artery then ductus then go to aortic side inflate and then closed.
- usual is successful .
in some cases if the ductus is very large need to interfere with surgery .
- surgery is easy and doesn't need to bypass machine ،it is close surgery.
- the opening (scar) left lateral .
DD of left lateral.. scar :-
> PDA closure
> valvotomy
- not every single baby or child is suitable
for closure by device, the catheterization need the pt consideration ,weight of child, it is not important regarding to the age , but it is very

important regarding the child weight , usually. It done after the weight of 5kg.

* complication : -
- Like VSD .

*Natural history of PDA:-
Little different from VSD .
-usually it's not close.
-it closed physiologically in first few weeks .
-after the age of 6weeks ,decide what the intervention.
Coarctation of aorta
- it is stenosis or narrowing at the area of isthmus in the descending aorta.
- presentation of it is vague presentation,it has no Typical cardiac sign.
-don't forget to examine femoral pulsation of every child to suspect, detect or to diagnose coarctation of the aorta.
-diagnose may be by accident.
- examine femoral pulsation for small F pulsation and radio femoral delay.
- Tx : catheter or surgery.

Cyanotic congenital Heart diseases

Fallot Tetralogy :-
it is most common cyanotic CHD
* Pathology :-
The main pathology is anterior deviation of
conal septum.
- part of inter ventricular septum is called conal. septum et attaching to it aorta and pulmonary

Artery.
- Anterior deviations means it go to Right side
because the heart divided as anterior (Right side) and posterior(left side).
- when conal septum deviate to right side taking with it the pulmonary artery and aorta .

Interventricular septum formed from two part:-
- Conal part.
- Apical part.

Conal come from above,apical from below ،they converge in middle.
- when the conal septum deviate to right side, it doesn't meet the apical part so there is gap this gap is VSD.
Previously TOF >the conal go to Rt side and make VSD and take with it aorta result from that stenosis in Rt side .

*Anterior deviation of conal septum :-
1- large VSD .
2- Rt ventricular outflow obstruction :-This is obstruction is subvalvular mainly , the obstruction at the area of infundibulum (subvalvular area called infundibulum),so this is infundibulum stenosis .
Obstruction with different degree of severity .
3- RVH.
4-overriding of aorta ..part of aorta in the Type left side ,part go to the right side 50%
-usually not more than 50% (more than 50% this is due to another disease)
:this is 4 problems.

*consequences of TOF:-
1) Right to left shunt :-

- the cause of this shunt is pressure on right side exceed pressure on Left side because the obstruction increase pressure
- sever obstruction cause this shunt. -
-if there is VSD without obstruction the shunt from left to right.
- silent >this shunt has no murmur.

because the defect is large and pressure gradient Isn't sever, side there is no murmur at the area of VSD .
- if murmur present it is due to obstruct not murmur of VSD
2) reduce pulmonary blood flow.
- sever obstruction cause less blood to the lung.
3) small left side .
*clinical presentation :-
c/p of Tof need time , doesn't immediately after birth.
- this time is variable according to severity, of obstruction.
- in some children you discover murmur in age of
3or 4 years ,some children, symptoms of cyanosis appeared early .
-There is large spectrum of presentation of this disease according to degree of obstruction.
- Timing and severity of presentation depend on the degree of RVOT obstruction.
- the main symptoms and main sign is cyanosis which is major and main problem of this child.
- presentation is cyanosis and complication is

<u>complication of cyanosis</u> .
1)dyspnea on excretion & due to cyanosis.The dyspnea is due to hypoxia and it complication,
2)squatting :- Also due to cyanosis.
- it is compensatory mechanism done by the child naturally so as to compensatory hypoxia.

- squatting mean complete flexion of upper and lowerlimb.
-the child after walking they feel sensation of hypoxia.
- The child flex his knees and upper limbs.
- squatting cause increase in peripheral vasoconstriction, so reduce venous return
to the heart , so reduce deoxygenated blood to the heart ,that lead to increase oxygenated blood in arterial circulation .
3)hypoxic spells:-

<u>when you examine the child you may ! Find</u> :-
1)Finger clubbing which is important sign, it isn't found in every pt.
- Finger clubbing caused due to long Standing cyanosis .
- Also severity of the obstruction indicate The timing of presentation of clubbing.
كلما كان الاوبستركشن كتير كل ما كان الكلبينغ بدري.
Clubbing may appear in 8 months, 8years ,or when he become adults.

2) systolic thrill and murmur at upper left sternal border.
- it is ejection systolic murmur at upper LSB .
- it is not related to VSD.

<u>investigation</u> :-
1)Ecg :-
– Typical feature of RVH and right
axis deviation -
2)chest x ray :-
size of the heart usually is normal.
the size may be small.
Oligaemic lungs: -
plethoric lung means congested lung - oligaemic lung means that lung Film is lack due to pulmonary blood flow.

The Typical shape of the heart on chest is Boot shaped heart ,because hypoplasia of a
main pulmonary artery.
3) Echo:-
=
- till you about details of diagnosis and
anatomical discerption of the defect and severity of the disease and other associate problem .
*pulmonary artery in Tof is smaller than usual , this hypoplasia causing absent of shadow so heart become like boot shape

TOF complication:-

- hypoxic spells :-

the most important and serious complication of TOF.
- it causing mortality in this disease

- growth retardation : -

- it is not sever as growth retardation in Large VSD.

- Brain abscess and CVA :-

it is bad complication following sever cyanosis to the brain.
-pt may come with hemi or quadriplegia due to CVA

- infective endocarditis : -

it is one of complication at the
area of stenosis(pulmonary area).
- Not at area of VSD.

- poly :

a result from chronic long standing sever cyanosis
relative iron deficiency: -
- if you take blood for Hb it is very high
17or 20 but there is relative iron D in every child with cyanotic problem.

- Coagulopathy :

- bleeding disorders may occur usually sub clinical , bleeding isn't feature of TOF.
- usually coagulopathy is subclinical but during or ICU pt may develop bleeding .

Management of TOF : -

- curative treatment is surgery , but before Surgery you need to have some --
- medical treatment is important to relief the symptoms and to improve the condition of the pill
- Treat spells.
- oral propranolol as vasodilator and it

given for every child with symptoms. - -propranolol is improve the condition of the child and resolve Spasm at area of valve

- good dental hygiene to guard against iE.
- Treat iron deficiency by iron supplementation
- surgical : -

it is solution of this children.
it is open heart surgery major heart surgery
- Surgery to close every thing >VSD should be close + resection of tissues at area of Stenosis +aliening of the septum , and if there is involvement of pulmonary artery, value should be Also Treated by dilatation .
- in some pt the surgery is more complicated and you may need to do some graft .
- open heart surgery is Treatment of choice, and it is successful surgery for majority of the pts if it done in good situation with a good staph.

Transposition of the great arteries: -
-great arteries means pulmonary
and aorta .
- Transposition means exchange in position.
- more common in male rather than Female 3:1

* pathology :-
– simple but serious , the aorta arising y From right ventricle and pulmonary artery arising from left ventricle.
- the blood come from lung is oxygenated and back to it oxygenated.
- the blood come from body is deoxygenated and back to it deoxygenated.
-the child can't survive unless there is shunt which cause mixing of deoxygenated with oxygenated blood .
-the pathology should involve shunt at the level of interventricular septum (VSD,ASD,PDA) .
-shunt is mandatory for survival.
Clinical presentation: -
-sever congestion and sever cyanosis .
-VSD caused congestion while TOF caused cyanosis ,TGA cause both of them.
-no cyanotic spells, but there is sever cyanosis >child is irritable, hunger of air . .
-so c/p of TGA are cyanosis and congestion since Birth.
- this is very bad disease and very serious!

investigation: -
- chest x ray :-

cardiomegaly ،egg-shaped heart and plethora (congested).
- the heart is dilated, the lungs are plethoric, shadow of the heart is egg shape ،because changing of positioning of great arteries .

prognosis :-
- 90% of children die before age of 6 montht due to hypoxia, acidosis and heart failure so intervention should be early in this babies

management :-
medical :

-Treat any HF , Treat Any chest infection of .
- oxygen
- the diagnosis should be very early in neonatal period So if it discover early the chance for the child is better .
-PGE1 is very important drug to survival of this baby ,it causes the ductus to be Patent.
*ductus may be open in first days, so when you give PGE1 ,it save it open .
- if the shunt is small, it isn't satisfactory For survival so PGE1 is very important .
- PGE1, is very expensive drug and doesn't available in Sudan.
Rash kind procedure :
do some sort of atrial sept ostomy which means to create artificial ASD by using catheter for increase size of an original small ASD.
surgical Treatment:-
- is solution .
-should be done very early in first 3
month of the disease.
- after surgery if the baby pass All period of critical time on Icu, and discharge with good proper surgery,he will live normal life.
- <u>why we have poor prognosis?</u>
1)presentation of child isn't early.
2)the child come early but there is no
Facilities to do anything for this child.
- surgery of TGA isn't easy surgery, need very complex set up ,theater, Icu fir small neonate and also surgery need well trained pediatric cardiac surgeon and anesthesia people who deal with this small child for 8 hours in operation ford.
- It is very promising surgery if successful
- this surgery called Arterial switch.

Total Anomalous pulmonary venous return

- we have 4 pulmonary veins which drain, on the left atrium ,taking oxygenated blood from the lung.
- in this disease the 4 pulmonary veins are drain abnormally in the right side rather than the left side.

- The Anatomical Types :-

1) commonly the 4 Pulmonary veins will drain in the channel which drain in superior vena cava then in the right atrium. 50% or more form this Type which called supra cardiac Type .

2)cardiac Type

have different Type:-

- -drainage in pulmonary sinus which take venous of pulmonary circulation to Right atrium.
- direct drainage on the right atrium go directly through channel which open in right atrium.

3) infra cardiac Type :-

it is rare and serious Type .

- the 4 pulmonary veins drain into inferior vena cava .

The Anatomical variation of Total anomalous :-

- supra cardiac Type.
- cardiac Type ..
- infra cardiac Type.

-the most common is supra cardiac .

-The less common is infra cardiac which is sever.

Hemodynamic and c/pa:-

it is almost similar to TGA with less degree of severity because it venous r problem not arterial.

the time of presentation is almost start early like TGA .
common presentation is feature of pulmonary HTN .
-Right side usually dilated in this pt so they have Right axis deviation on Ecg.
The shunt is very crucially for Life , usually The shunt is ASD.
- The Treatment is surgery.
in some pt this Anomaly isn't total, it is partial which means some of the veins drain Normally and other drain on right side .

* cyanotic spell :-
it is one of the serious presentation of cyanotic congenital heart disease.
- it is called hypoxic spell.
- it is an emergency
1) paroxysm of hyperpnoea (rapid deep respiration).
* The common Type of dyspnea , regarding the cardiac problem and respiratory problem is shallow rapid breathing .
- breathing in hypoxic spells is deep rather the shallow .
rapid deep breathing due to acidosis + called Kussmaul breathing on acidotic breathing).
Kussmaul breathing present in :-
- DKA
- complication of sever dehydration
- liver failure
- renal failure
- brain insult
- there is rapid shallow breathing in heart failure.
2) the child is irritable and +cyanosis. 3)decreasing murmur intensity .
- pt with TOF come with cyanotic spells and minimal or absent murmur.
4) pt may come comatose.

5) predisposed by crying, feeding and defecation, predisposed by effort.
5) may cause convulsion, CVA or death.

Treatment :-on Icu :-improvement

- squatting :-it is one way of condition

- squatting on bed there is 2 option:-

if he is young enough and the mother can hold her child on her leg that is best, this mother helped by assistance.
other option for All children in Icu bed
create squatting position from bed it self

- morphine sulfate ,2 mg/ Kg

- morphine is the best, you can use .....if no morphine .

- oxygen.
- sodium bicarb >ABG Analysis.

1 mE per kg.

- A lot of pt may improve by this step of management :-

- some time if the case of cyanotic spells is difficult you may need do further management
what is further management ?

- intravenous propranolol ,05 mg / kg.

– propranolol is potent drug relieving Spasm at Fundibular area.

- vasoconstrictors are also one of the option
- Ketamine 2mg/kg > generally use in general anesthesia.

Ebstein 's anomaly: -
Tricuspid valve very descend and take large part of Right ventricle > this called atrialization of a large portion of right ventricle .
- the functioning right ventricle become very small .
-this abnormal tricuspid vale may cause regurgitation , pt may have ASD and cyanosis .
- it is very bad disease & but not fatal

earlier.
- No Treatment at level of the world .

Feature :-
-every huge heart on x ray.
- the most large heart is heart of
Ebstein's anomaly.

Truncus arteriosus :-
- instead of pulmonary and aortic artery the child has single Trunk.
- this is trunk usually over ridding and there is VSD .
- From this trunk originate the pulmonary artery (Lt& Rt branch).
- Then the main trunk arched as aorta - there is mixed blood.
- very early congestion and very congested heart failure and some sort of hypoxia.
– cyanosis is not sever in this pt ,the main problem is congestion.

Tricuspid atresia :-
- it is obliteration of continuation of circulation the right side.
- atresia mean close.
- Tricuspid value is closed.
- the right atrium is congested with blood so it is enlarge > there is bypass to Left atrium through ASD> blood go to left ventricle>so there is dilatation - right atrium and left ventricle , so it is characteristic on Ecg.
- it is one from bad cyanotic CHD
pulmonary atresia : -
This is bad cyanotic CHD , unless there is collateral.

Acute rheumatic fever

- ARF means Acute insult by which some organ are affected acurly Following certain conditional and after that his appear problem will resolve either with or without complication may be ARF as History or ARF with complication.

- Rheumatic heart disease RHD means there is perminant damage to the values which Following an attack or attacks of ARF.

- The heart may be Normal after attack of ARF, or may be involve , even if the valve is involve intially in Acute phase this isn't mean will continue as RHD - Some time this valuitis will resolve if the pt is well.

- causes
- ARF Follow immunological response to group A. Streptococcal of the pharynx - Tonsilitis & sore Throat .
- * predisposed by :-
- - FH of RF
- - Low Socio economic status ARF it is disease of poverty and bad hygine.
- * Typical Age of the disease :-
- 5-15 years but..it can occur before This age and after that.
- very rare after Age of 30 year.
- - peak at 8 years.
- * pathology :

There is 4 organs involved :

- heart very important because those is area of complication ..
- other organs have Transient manifeste.

- Heart , Joint, brain and skin a insult of other organ rather than heart is Acute and resolve completly immediatly or afterwise .
- The classical pathology in the heart is pancarditis" which mean involving 3 layers of the heart (pericardium, myocardium, endocardium).
- we concern mainly on endocardium because the value is value of endocardium.
- a pt with RF unlikely come with feature of L myocarditis because involving of the muscle in This disease is unlikely & RF mainly involve the endocardium .
- may involve the pericardium and pt come with pericardial effusion usually effusion of RF is mild or moderate , unlikly pt to have sever or large PE which need to intervention or cause problem to pt .
endocardium أهم حاجة نركز على
- in the endrocardium the main affected valve is mitral valve apparatus..
- apparatus = From Tip of Leaflet till___.
- the most affected area in value is leaflet mainly The Tip of the leaflet.
- * Aschoff bedies can be found as pathognomic
- * Histological finding :-
- .....

- clinical manifestations : -
- The Typical History is problem following History of strep-pharyngitis usually around! 3 wks , may be 1-2 wks or even 4 wks .

- *jones criteria :-
- - it is modified from time to time.
- - major manifestations -

1.carditis :-

- - it is main criteria.

- occur in 40% of pt.
- clinical presentation of carditis:-

1. pt may present with feature of heart failur.
2 - Tachycardia, particularly during sleeping.
- لمن الطفل يكون نائم مفروض نحسب ال pulse rate حقو.

3– murmurs are usually clue for diagnosis & of carditis .
- - the common murmur is pansystolic murmur at mitral area & murmur of MRU.
- - you may find diastolic marmur at aortic area due to aortic involvement.
- _ in some pt may you find mid diastolic murmur at mitral area due to valvitis (cary Coomb's mussen
- _No mitral stenosis in ARF ,stenosis it is need years to appear. if the onset start at 5 year may you face child with 10 year with ms.
- this cary coombos murmur may be transient murmur , after remmision of Acute phase this murmur disappear : 1. pericardial rub due to involvement of pericardium it is not common

- *investigation :-

1. prolong PR interval in the Ecg.
2. cardio megaly on x ray .
3.Echo -__ to see the heart.

- *sub clinical carditis :-
- _ No feature of carditis but manifestation on Echo .
- _this is real story in pregnant lady .
- you may find suddenly ms without previous History of involvement of pt.

2. poly arthritis: -
- _ involvement of multiple joint .
- * sign :–
- _in flaimmation and swelling , painful

- swelling of joints.

- this joints Typically large or big joints and lower limbs & elbow, wrist, Knee, ankles .

- - The knee is most common joint involvement in Rheumatic fever.

_Knee, elbow, Ankle and wrist __ الترتيب

- _ the central and prephral ,join are rarely to affect.

- Not Typicaly involve the Tempero mandibular joint and vertebral joint and small joint rare to be affected.

- _it is Acute arthritis long arthritis tis not ARF.

- - it is migrating flitting and it is involving one joint then affecting other joint and resolution of 1st joint.

- _migrating arthritis means arthritis start in the join then improving then involement of other one joint, may be 2 joint at the same time one is flaring and other is improving, involvement of the 4th [joint and the one point is completly resolve the other joint .

- *Acute flitting arthritis involving the big joints and it is self limiting arthritis and no perminant damage to the joint which is completly recover after resolve of Acute conditions, with dramatic response to NSAID or steriods.

– if pt receive aspirin within 24 hour the condition is improve dramaticaly .

3. syden ham chorea :-

– chorea it is abnormal movement originate from basal ganglia.

- * Basal ganglia have :-
- Type of abnormal movement
- 1. chorea.
- 2. athetosis which is manifestation of Kernicterus following baby with very high unconjugated hyper.
- Bilirubinemia First day or to 1 week after delivery.

- athetosis is abnormal movement due to difficult to control central part of the body trunks.

_ chorea is affect peripheral part of the body - (Face, feet, hands)
_chorea it is involuntary abnormal movement non repesitive of Bizard manifestation (have Two regular pattern).
_the chorea may be bilateral which is common manifestation or rarly may be unilateral.
_unilateral involvement called hemi chorea.
- pt with chorea start with early manifestation which isn't noted by his parent as problem.

- There is emotional disturbance and emotional liability ___ the pt laughing or crying in inappropriate way.
- يضحك بلا سبب بطريقة غريبة ويبكي بلا سبب بطريقة غريبة .
- Chorea is very common in female under puberty (12_ 15 years).
- Chorea is common in female more than male.
- The onset is gradual not like the other type of major criteria and continue to couple weeks or month up to 18 month.

4. Erythema marginatum:-
Erythema means redness over the skin
Marginatum means margin.

- It is rapidly macula around 1_2 cm over the trunk anteriorly or posteriorly.
- Start as homogeneous redness in area of geographical distribution but quickly it forming ring___ the red ness and skin.
- In central area is pale or normal colored.
- Start as complete erythema redness and after couple of hours it becomes as red ring and pale or normal color in center.
- The erythema marginatum change in shape , site and size rapidly
- تختفي وتظهر في حتات تانية .
- It is not common major manifestation it is rare and usually ass with other manifestation like carditis.

5/ subcutaneous nodule :-

- It is Small nodule raised usually lees than 1 cm in diameter typically found as bony permanent in the limb (wrist , leg).

It is pain less and not attach to the skin .

- skin over it is clear .
- -it ass with other manifestation, it isn't single .
- - it isn't common major criteria.
- - may contenous for wks.

*minor manifestation of jones criteria:-

1.Arthralgia :

- usually poly arthralgia .

_a pain of the joint without swelling .

2.Fever :

- more than 38 c.

elevated Acute phase reactant (ESR, CRP) .

_ prolong PR. interval.

- زمان كان ال total wbc معاهم.

3.evidance of GA strep:-

_ evidance of one of the following :-

- positive throat culture & elevated Aso ritert (stronge evidance)

4. Recent history of scarlet fever or Sore Throat (week evidance)

- *classicaly to say this pt has ARF the pt should have:-
- 1. 2major criteria + evidance.
- 2. 1 major & 2 minor + evidance

- *modified jones criteria :

1.subclinical carditis is evidance of carditis just by echo.

2. in highly risk areas monearthritis and poly arthralgia are consider as major criteria .

_ high risk area means: -

- - prevelance of 1 . 1000 of population.

- - 2 = 100,000 incidance of RF. 3.monoarthralgia consider As minor criteria. in high risk area.
- - one single joint feeling of pain consider as minor criteria.
- _ charea alone consider as a cute rheumatic Fever.

5/recurrance :-

- Any recurrence of ARF with more in 3 minor enough for you to diagnose this
- pt having ARF.
- x managment :-
- _Benzathine penicillin___ it is very important drug to erudicate Streptococcal infection .

- Bed rest :
- if the value is involve excercise will deteriorate the valve.

- Anti inflammatory medication : – _NSAID is drug of choice for ARF.
- _use Aspirin in Anti inflammatory dose.
- _Aspirin has 2 action: -

* Antipyretic dose in low dose -
*Anti inflammatory dose in large dose 70 - 80 mg/Kg day).

- حبة الاسبرين تحتوي على 300 ملغم.
- _ لو طفل وزنه ٢٠ كيلو جرام يحسب الجرعة كالأتي:
- 70 × 20 = 1400
- 1400/ 300= (number of tablet per day)
- وعدد الحبوب الجاتني دي بقسمها على اليوم.
- مثلا لو كان الحساب كالاتي tablet 6 = 1800/300
- ال٦ حبات دي بقول لي اهلو حبتين كل ٨ ساعات بعد الوجبة.

- <u>_you must be counsel the parent about details of disease and managment.</u>

_1 tablet of Aspirin consist 300 mg.

- give tablet after meal.
- Aspirin 70 – 80 mg/kg, 24 hour for .. 4 - 6 wks , this duration determine by response of the pt.

_ a give Antiulcer medication

* Few.pt may need given steriod .
- the only strong indication for steroid is poor or irresponse to Aspirin.
- This irresponse appear during follow up. For example
ESR is 100 for 2-3 wks__ in this case
you need to use steriode which judgement by it is time.

- steriod 1_2 mg/kg for 2 wks, There at the time of tapering you give aspirin tablet.

لمن أبدأ أنقص ال steroid , بدخل الاسبرين لمدة ٤ اسابيع.

_ after you give steroid predinsolone for 2 wks, then continue NSAID aspirin for further 4 wks.

* Treatment of congestive HF..
*managment of chorea :
- give sedation , The best drug is
halo peridol which improve the Life, continuous on this drug as the abnormal movement is there.

وممكن يستخدم لشهور و شهور.

* prevention:-
_ it is very important .
_RHD is sever valvular involvement and cadastroy for alot of pt .
_some pt have more than 1 value involve.
_surgery is Tx of RHD ،but follow in
surgery the pt still diseased .

أفضل congenital HD في بعض الأحيان

من ال RHD بعد ال SURGERY.

RHD surgery end by metalic valve which need alot of precation after surger (INR, warfarin , follow up complication ,bleeding, difficult compliance).

•
• primary prevention :-
• – proper diagnosis and treatment infection.
• _don't forget to examine throat of every child.
• _if child come with fever, don't depende on blood film, you must be examine the throat.
• *How you diagnose Throat infection ?
• examine the throat by using Torch For younger child.
• *in child less than 5 years :
• the mother should sit on chain and the child sit on his mother and hold the baby and you use Torch and Tongue
• depressed to look for Tonsil and throat .
• تخلي الام تقعد في الكرسي وتتجه عليك وتكون شايلة طفلها وهو كمان متجه عليك ، وتشوف ال tonsil وال throat .
• _ Treatment of throat infection:-
• *single dose of Benzathine penicillin.
• There is other option use__ oral antibiotic ___ use amoxicillin and 1st generation cephalosporine for 10 days.
• (use it in proper hose and proper time) -- hygine also from primary prevention - Secondary prevention:-
• 1 - to prevent further attacks of rheumatic Fever in pt who diagnose as Rheumatic fever.
• - So any pt with RF take prophylaxy of Benzathine pencillin .
• *Benzathine penicillin should be used every 3 wks single injection im & for how long ?
• - he continue for 5 years if pt have No cardiac involvement or minimal heart affected .
• _ 10 years if pt have significant
• cardiac involvement or for life .
• - senzatine penicillin is long acting pencillin so it given every 3 wks.
• _ what is different bw benzathine and Benzile ?

- _Benzile penicillin is short acting pencilin , Also called crystaline penicillin . بنسلين مائي
- ببقى crystal clear زي الماء.
-
- - Benzathine penicillin it is white color after dissolve the injection. لونو معكر
- Benzathine penicillin needs intially to be tasted for sensetivity ,intradermal
- Small dose of this drug should be given to sure this pt isn't hyper sensetive to insulin.
- - if give penicillin to pt who hyper sensetive you lead to anu phylactic shock and death.
- (يعطى الانسولين بعد اختبار الحساسية).

- _ Dose of B penicillin :-
- _in a in majority of children bw 7 - 14 year the dose is 1200, 000 Iu.
- - if the child is younger or weight is below the dose is 600 000 iu.
- - Young adult and more than 45 Kg. The dose is 2400,000 iu.

- *erythromycin is alternative drug if there is hyper sensetivity to Benzathine penicillin.

Child Hood Congestive Heart Failure

-causes of CHF divided acc to pathophysiology and hemodynamic :-
1.certain problem causing excessive workload either pressure or volume .
2.normal workload faced by damage myocardium
-this major problem cause CHF ,pressure overload ,volume overload ,myocardium damage .
-specific etiology of HF in children :
1.neonate:-
-the commonest cause of CHF in neonate is congenital heart disease.
-other cause are :
.dysfunction (damage).
.volume overload.
.pressure overload.
*cause of dysfunction in neonate is original fibrosis of muscle it self .
-there is some disease cause fibrous change in intrauterine life, this called myocardial fibrosis or endocardium fibrosis because this fibrosis may involvement the valve.
-this endocardium fibrosis specially called endomyocardial fibroelastosis .
-this fibrous changes in the muscle and endocardium may cause HF in neonate.

*volume overload in neonate:-
1.patent ductus arteriosus cause HF earlier rather than VSD which need 6wks.
-large PDA cause HF earlier.
2.truncus arteriosus presented earlier with HF.

-the most important example of pressure overload in neonate is coarctation of the aorta which present in first few days after delivery, it cause shock and heart failure if it very sever.

*infancy:-
-during infancy the picture is clear.
-this is the age of HF due to congenital heart disease (3 ,4 , 5 months), this is common age at which the child coming symptomatic with congestive heart failure (VSD, ASD) due to congenital heart disease .

-dysfunction:-
* also at this age child presented with problems in the muscle.
*typically at age of 5-6 month, a lot of infant develop viral infection which cause myocarditis and dysfunction of the muscle and may end with dilated cardiomyopathy
-cardiomyopathy occur in this age.
3.children:-
1.children beyond the age of infancy (…. And school age),also some of children with congenital heart disease(palliated CHD)

*palliated congenital heart disease:-
-the pt is symptomatic and receiving palliative treatment which is antifailure in HF eg: diuretics, and pt still surviving and struggling.
2valvular regurgitation:-
-this valvular problem may be congenital or acquired, the valve may be affected since birth by congenital valvular problem (c valvd-stenosis ,mitral valve cleft ,mitral valve prolapse).
-this congenital valve problem may be sever enough to cause damage to the valve and causing regurgitation this regurgitation if sever it causing HF at this age.
3.acute rheumatic fever:-
-HF may be as complication of acute RF.

4.myocarditis:
-the main cause of myocarditis are viruses.
-the common age of viral myocarditis is the age of dudlers and school age. "4-6 years"
-viral myocarditis may end with HF.
5.endocarditis:-
-specially infective endocarditis may complicated with HF.
6.HTN:-
-HTN isnot cardiovascular origin in children
-the most cause of HTN in children is renal or endocrine Renal hypertensive disease.
-essential HTN now is commoner than previously particularly in old children ,and obese children.
-the commonest cause of HTN in children is renal
More than 90% of HTN present is due to renal disorders.
-HTN may complicated with HF.
-if you find HF and high blood pressure, you should reach about renal disease .
7.thyrotoxicosis:-
-may complicate with heart failure.
8.arrhythmia: where is tachyarrhythmia or ……
-SVT is common in children at any age.
-heart block is bradyarrhythmia , it may be congenital , the neonate come only with heart block.
-even in intrautrine life you can calculate fetal heart rate, you find it below normal.

*clinical manifestation:-
-congestive HF means left and right HF
-majority causes of HF in children cause congestive heart failure.
-but there are cases still just present with left side HF, and other cases just present with right side HF.

*child: طفل كبير شوية
-SOB
-orthopnea
-wheezing which common in children rather than adult as symptom of heart failure.
Cardiac asthma= wheez in HF.
-pulmonary oedema:
May you find with HF have dyspnea, crepitation, wheezing but this due to pulmonary oedema.
*pulmonary oedema it have :-
*server respiratory distress.
*Finding on chest x-ray patchy ,congestion in x-ray.
*pulmonary oedema is emergency to give the child extensive antifailure medication.
*tachycardia :-
-you should consider the age of child to say there is tachycardia or not.
-if you find child with 12 year ,Hr 100 tachycardia .

*tachycardia:-
Neonate ….. 160
Infant and young dudlers up to 2 years ….140
school age children ….120
more than 12 years …..100
*gallop rhythm:-
-it is tachycardia +S3.
-it is one sign of HF.
*…….. pulse:
-it is small volume rapid pulse.
-not occur in every age because sometime you find pt with HF with large volume due to hyperdynamic circulation.
*tachypnea:
-circulate RR tachypnea.

-see for pt dyspnea.
-infancy less than 2 month ….60
-less than1 year ….50
-1-5 years ….40
-old children like adult ….30
*peripheral oedema also found in old children.
*LL edema.
*periorbital edema
* ….. edema.

*cardiomegaly:
-either clinically or by chest x-ray.
-to defined cardiomegaly on chest x-ray you need more than 60% of heart shaddow.
-the heart shaddow form… 60% of the thoracic shadow in chest x-ray film cardiomegaly.
*distended neck veins:
-raised jugular ……. Pressure.
*tender hepatomegaly:-
Tender and soft.
*all this classical signs of congestive heart failure, this are almost occur in majority of children and adult .
*clinical manifestation in infant:-
#feeding difficulties:
-inability of infant to fed .

#failure to thrive :-
-failure to thrive ass with HF if cause of HF is perminant cause.
#diaphoresis:
-excessive sweeting.
#tachycardia.
#tachypnea.

#hepatomegaly.

*the most important 3 signs are:- "3Ts"
1.tachycardia.
2.tachypnea.
3.hepatomegaly"tender hepatomegaly"
-majority of infancy presented with HF are presented with this 3Ts.
-JVP not be mention in infant pt. because infants have short neck.
-the suitable age to measure JVP is 6 years and above.
-lower limb edema isnot usuall sign of HF during infancy, rarely to find it, because oedema is gravity dependent and the infancy is usually bed ridden , you can find sacral oedema.
-basal crepitation not mention it in infancy.
*there is no JVP , LL oedema and basal crepitation in infancy.
*in Ecg you should know age of pt, to know there is tachycardia or not .
*SVT diagnose by sever tacky cardia.
*chest x-ray:-
There is cardiomegaly ……plethoric lung or …….. lung and large VSD.

-possibility of this x-ray:-
1.congenital:-
-HF
-cardiomegaly due PDA, VSD.
2.aquired:-
*treatment of congestive heart failure:-
-in acute phase you need to admit the pt.
-acute phase mean this pt is ill ,dyspnic.
#oxygen
#positioning:
-45 degree if the child is old.

او الطفل ممكن تشيلو امو 45 درجة وتكون هي مرتاحة وهو
مرتاح.
#monitoring :
-you need to monitor the vital signs, (RR, BP, AR, temperature).
-you monitor 2 thing also in HF:-
*urine out put: one of complication of HF is cardiogenic shock in which there is oliguria.
*weight of the pt daily.
-any pt with oedema you should monitoring the weight daily.
#anti failure medication:
#treatment of the cause.
#nutritional and home treatment.

*medication:-
-diuretics:
1. furosemide is drug of choice as loop ……
-the dos 55-1 mg/kg/dose acc to need of the pt.
-sometime you need to give furosemide 3times per day, or 6 hourly, or 4 hourly acc o need of the pt.
- frusomide may given as infusion.
نضعو في درب
2.chlorothiazide:-
-not given commonly in children.
3.spironolactone"alductone":-
-it is K+ sparing
*if you use Lasix for long time you need to use furosemide as adjuvant drug as combination.
-if the cause of HF is perminant you need to use spirolactone 1-2mg/kg/day "twise dialy".
صباح ومساء

*after load reduction :-

-captopril which is short acting use for child because it complication may resolve:
#captopril may cause hypotention.
ال hypotention استخدمنا لو يتلحق ممكن short acting استخدمنا لو بتلحق م لكن long acting.
-so prefer to use short acting after loud reduction for child.
-captopril is ACE inhibitor.
-start by small dose of captopril, 3mg/kg/dose as average dose 3 time/day
-الحبة فيها 25 ملغم .
-this intial dose of captopril can be tolerated by majority of children and you need to increase this dose which doesn't enough for maintenance. so dose gradually till reach 1mg/kg, usually reach up to , 8mg/kg.

*beta blocker:-
Known drug which use in HF in selected pt.-
-carvedilol is use in children.
-carvedilol is good ……. B blocker agen use in HF particularly in pt with ……..

*inotropic agent:-
-not commonly use.
-digoxin is one of inotropic agent not commonly use in children ,but sometime you need for this drug.
-use as anti failure medication in pt wit sever HF or arrhythmia, use it as maintanence dose rather than digitalization dose. ,01=maintenance dose

,01 once daily

-dopamine and dobutamine are use in pt with refractory HF .
-dopamine is use if there is hypotention
(HF with cardiogenic shock or hypotention).

-dobutamine use if Bp is normal.
-dopamine and dobutamine are use as infusion pump.

*nutriotion:-
-this problem need to special nutrition .
- calories.
-child many have problem in intake :-
#anorxia.
#فتور
#difficulty in feeding.
-give calories wHh …….. to avoid overload.
-give high calories in small diet frequently .
*Home Care:-
-advise the mother about:-
#what is problem of her child.
#what are expected complication.
#how deal with her child.
-the child should be normal life as possible as they can do , so facilitate optimal environment to the child.
-continous this drug at home:-
#when the child become well you discharge him with oral anti failure medication.

Clinical approach to cardivascular dz
_age of the pt
_age of the onset
Age
Young children most properly congenital
Young mean less than 5 y

Cardivascular dz in Young children most properly congenital but may be myocarditits or unusual presentation of Rheumatic fever
Old children most properlyacuired but may be congenital
Heart dz are discover later
Age of onset_
ممكن الطفل اجيك عمرو ٧ سنه كfirst presentation
بس الحاجه دي عندو من زمان
If the pt come at the age of 7 y since the age of 5 month this pt have repeat chest infection this is congenital rather than acuired
common presentation:
If the child is case of congenital Heart dz the common presentation depend on weather is cyanotic or acyanotic
In pDA and VSD with conciderable shunteither moderate or large this children present with repeat chest infection or CHF
Presentation of acquired cardiac problem depend on condition severity of condition usually related to Clinical event
_PMH
Ask about past history of admision
RHD is complication of Acute Rheumatic fever it may be Mild or sever this severity detect y admision
This pt with RHD may admited withHF or or infective endocarditist or ARF
How you know this pt admited pastly with HF?
Symtom dysnea congestion
Drug Use at that admision dieuretic
How you now this pt have PMH of ARF?
By Symtom attack of arthritist
Come with HF
Drug Use at that time asprine

How you know this pt has PH of infective endocarditist?
1_prolong fever

2_prolong Use of injectable antibiotic for couple of week
Infective endocarditist ttt by injectable antibiotic for 6 weeks
*commonest cause of admision
HF
ARF
Infective endocarditist
The admision indicate serious and sever condition of the pt
You Ask about recent attack of tonsilitist duration of tonsilitist to cause RHD need 2_4wees
لكن مابسال من الcurrent tonsilitist لأنها مامهمه
In pt with RHD in PMH Ask about
Admision
Recurrent attack of tonsilitist
*examination
General examination
Pulse
_volum small volum expected in any pt with HF
Small volum expected in any pt with mitral stenosis which is not common problem in children
Small volum related to aortic stenosis
Small volum in lower part of the body may indicate coarctation of aorta
Large volume Pulse is very Important diagnostic criteria of certain cardiac problem
_hyperdynamic circulation which May be Physiological or pathological
_cardiac problem with large volum
AR 1 _طفل كبير
2_PDAطفل صغير
examination of colapsing Pulse*
Itis very usefull colapsing Pulse is advance degree of large volum Pulse

Not all pt with large volum Pulse have colapsing Pulse
The upper limb should be extended when you do colapsing Pulse this manover should not repeat more than twice because it may disapear
Colapsing Pulse اما النبض كان ماواضح وظهر أو كان واضح وزاد
*cyanosis
Cyanosis may be prepheral or central
No central without prepheral
Pathological Cyanosis incvolve both central and prepheral
Prepheral Cyanosis without central may be duto weather and may occur in neonate
Any central Cyanosis is Pathological
Cyanosis dosnot usually cardiac it may be respiratory or dueto shock but majority of cyanotic children is cardiac dz
Any child with central Cyanosis should excluded cardiac dz(pulmonary or Tricuspid atresia)
Finger Clubing *
It related to severity of cyanotic congenital Heart dz
The most common cause of Finger Clubing of child with cardiac problem is cyanotic congenital Heart Diseases
Pt may have Finger Clubing without cyanotic congenital Heart dz as In infective endocarditist
*pericordial examination
Inspection
Contour of thorathic 1
Bulging pericordium retrosternal Bulging is common pulging
Some time pulging central to the left
Bulging is indicate chronicity and severity
5y child has pulging chronic and sever and pansystolic murmur most properly this is VSD
2visible pulsation

Accordig to area you found visible pulsation need Good light Good observation for area epex Tricuspid epigastric base of the Heart or it may be all over pericordium
Surgical scar3
Midsternotomy scar indicate open Heart surgery
Left later scar indicate Surgical ttt of PDA or valvotomy whice is very rare in children usually valvotomy is adult surgery
4 cautery mark
Done at area of complain eg visible pulsation
Chestpain is rarein children as sign of cardiac dz
#palpation
1epex
2palple s2
3thril
4 left parasternal heave
*Apex
Site and character
Site of epex when you find are of the epex but your finger at the area of the epex and calculate to know the Site normal epex at the 4th intercostal space mid clavicular line
Displace epex duto left ventricular dilitation
Character of epex
_heaving is most common specialy in children RHD
Vsd dilated chamber
Taping epex is not common specialy in children
Any things cause dilitation of the left ventricle can cause heaving epex
Left parasternal heave
Just but your hand at the left parasternal area donot press
Left parasternal heave indicate biventricularhypertrophy
Palbaple 2nd Heart sound at pulmonary area in children indicate pulmonary hypretention

In adult may palpate 2nd Heart sound in pulmonary area and aortic area
Palpale s2 at pulmonary area indicate pulmonary hypertention
Palpable s2 in aortic area indicate chronic systemic HTN
Chronic systemic HTN not common in children
*auscultation
S1
at the epex either normal muffeled or loud is heard
S1 loud common in children
S1 loud indicate hyperdynamic circulation
Loud s2 at pulmonary area indicate pulmonary Hypretention
Loud s2 indicate very complicated cardiac problem
Loud s2 and Palpale s2 indicate pulmonary hypretention
loud s2 في الأطفال الصغار شديد بتلقى بس

Murmur and thril#
Thril is Palpale murmur
Thril indicate grade 4 or more murmur
Thril may be systolic or diastolic murmur
any diastolic thril or murmur is patholoical
No Physiological diastolic murmur
Systolic murmur may be Physiological or pathological
Systolic thril +murmur =pathological
Thril indicate pathology in the heart
In children common to find inocent murmur
This inocent murmur not grade 4 or more
Thril is very usfull sign to localize the lesion
Benifit of thrill
1_indicate pathology of the Heart
2_localize the lesion
3_help to reach the diagnosis
How to differntiate between ps VSD?

Ps are ejection systolic murmur at upper left sternal border at pulmonary area
At 2nd intercostal space
Vsd is pansystolic murmur at 3rd and 4th intercostal space
2الكن الnd 3والrd بين ال بفرق شديد عشان كدا قراب الأطفال فيps والvsd بي الthrill
Upper thril ps
Lower thril vsd
Examine the thril by palm of the hand
The area in which you palpate for thril are area at which you auscultate for murmur at epex Tricuspid EPIGASTRIC LLSB 2nd intercostal space

<u>Murmur#</u>
Timing
Intinisity loud or low pitch
Area of maximum intensity
Area of maximum Intinisity are area of maximum Palpale thrill
Majority of systolic murmur are heard all over pericordium
Pansystolic murmur from S1 to s2in whice there are muffeled S1 لانو الmurmue مابدي فرصه لل s1 يظهر
To MR presentation typical
Pansystolic murmur muffeled s1
Ejection systolic murmur in pS
The common cause of pansystolic murmur in children is vsd
continuous murmur the strigth of the murmur heard at all systol all Ejection systolic
Donot forget sign of complication sign of HF 0and infective endocarditist

<u>Diagnosis</u> for eg Acynotic congenital Heart dz most properly Vsd then complicated or not chest infection
RHD MR on sinus rythm no arrythmia no HF no complication

BLEEDING DISORDER

DR.HISHAM ALAMDHA

- What is body response towards bleeding?

1. Vasoconstriction: To decrease blood flow ,that decrease bleeding.
2. Clot Formation: platelets aggregation + Von willebrand factor lead to plug formation.
3. Activation of coagulation cascade :through intrinsic pathway or extrinsic pathway.
4. Fibrinolysis.

*This is the normal hemostasis to stop bleeding.

- What disorders that occur due to blood problems?

• Either:

1. Coagulation defect
2. Platelets problem
3. Vessel wall problem

- Coagulation cascade :

-we have 13 coagulation factors.
-Intrinsic pathway with extrinsic pathway Common pathway.
-Then bind with calcium and last to fibrin formation to stabilize plug.

- History of patient with bleeding disorder:

Any child presented with bleeding disorder , you have to take a good history examination including:

1. Age of the patient : bleeding disorders in neonatal period may differ from other periods.
2. Sex of the patient: very important because there are inherited disease.
3. Frequency of the bleeding: once a day, short frequent or prolong frequency.

4. Location of the bleeding. Either in: mucus membranes , cutaneous , muco-cutaneous membranes, or internal organ bleeding.
5. Duration of the bleeding. Very important
6. HX of the medication(DH). There are many drugs that affected coagulation cascade : e.g : Aspirin, warfarin , heparin, NSAID and other.
7. HX of bleeding disorder in family members , Also sickling in the family.
8. HX of infection: such as vasculitis(HSP).Also Hx of Meningitis.

EXAMINATION of patient with bleeding disorder

1. Area of bleeding
2. Nature of bleeding. Is it petiche , echomoysis , purpural rashes , hemoarthrosis ,or deep hematomas "in mucscles".
3. Examination of reticuendothelial system. E.g: Hepatosplenomegaly, lymphadenopathy and other.

Specific Investigation: "To reach the diagnosis"

*Platelets disorders:

Decrease in platelets number"Thrombocytopenia"(commenest)

*Normal:15,000-450,000.

#If the platelet number is normal, think of defect in platelet function which called Thrombosthenia.

*platelets precarsors in BM are megakarycytes.

*The life span of platelets is 9-10 days.

#what are the function of the platelets?

If there is vascular injury:

1. Adhesion to the vessal wall.
2. Aggregation;primary hemostasis.
3. Secretion of vasoconstriction substance.

Causes of thrombocytopenia:

1) Bone marrow failure: caused by:
a) Congenital bone marrow hypoplasia :such as fanconi syndrome.
b) Infiltration with malignancy. "malignant cell"
c) Infiltration with viral infections such as :HBV , HCV AND HIV.
d) Cytotoxic drugs: chemotherapy and radiotherapy.

2) Platelets consumption: e.g :
a) DIC.
b) ITP.

ITP

ITP=Immune thrombocytopenia purpura.

ITP is bleeding disorder caused by autoimmune substance destroying the platelet by phagocytosis mainly In the spleen and some extent in the liver.

ITP usually proceed by URTI.

In HX : flue-like symptoms: mild grade fever, sneezing, cough.

This is very important in HX.

*Epidemology:

_Children are affected by acute type of ITP which is usually following URTI specially viral infections .(in adult chronic type).

-the incidence in male and female are equal.

*Clinical feature of ITP:

1) Cutaneous bleeding:
a) Purpura
b) Echymosis
c) Petechiae.
2) Mucus membranes of mucocutaneous bleeding:
a) Epistaxis (nasal bleeding)

b) Gum bleeding
c) GIT bleeding
d) Haematuria
e) Menorrhagia in female
3)Intracranial bleeding which is rare but is fatal (the cause of death in ITP).

*DIAGNOSIS
The most important thing to diagnose ITP is the absence of other constitutional symptom(presented by bleeding only)
NO splenomegaly, no hepatomegaly and no lymphodenopathy.
If they are present think about other causes.
*Diagnosis by HX , EX and Investigation.

INVESTIGATION
1) CBC:
*Platelets count: according to platelet count ,the amount of bleeding occurs.(the number of platelets directly proportional to the amount of bleeding).
#platelet count:
-less than 20,000 lead to spontaneous bleeding without any trauma.
-20,000-50,000 lead to bleeding after minor trauma.
-50,000-100,000 lead to bleeding after sever trauma.
-more than 100,000 bleeding due to major surgical operation.

*The other CBC "Hb ,WBCs, RBCs" are normal in isolated ITP, unless there is poncytopenia.
2)BM: increased megakaryocytes.
3)specific test(confirmatory test):donot need it in all patient, but we need to do it sometimes.
Either: A) Serum Anti platelets antibody.
B)Anti platelet glycoprotein Antibody.

Management of ITP

*70% of cases with ITP recover within4-6weeks without treatment or intervention because it self limiting disease.

*30%of cases need treatment and intervention.
- these 30% treated as :
1) platelet trans fusion: if there is active bleeding.
2)Steroids: to reduce the destruction of platelets.
*We give predinsolone 1-2mg/kg single daily dose , then tapper the steroid if the patient is responding within 2-4 weeks.

#HOW do you know the patient is responding?
1)clinically by the stopping of the bleeding.
2) laboratory by increase of platelets count.
*We find there is 2/3 of the patient respond to steroids.
3)IV immunoglobulin : IG decrease platelet consumption and accelerate antibody catabolism and reduce antibodies production.
*Daily dose for a full 5 days .
4)splenoctomy: if there is hyper splensim.
Von-wellebrand Disease(VWD)
It is common inherited bleeding disorder and inherited as autosomal dominant.
-Occurring in about 2% of population

-Von-wellebrand factor has an important role in hemostasis because:
1. IT act as bridge between the vessel injury and platelets.
2. Binding with factor 8.

*Symptoms of VWD:
1. Easy bruising
2. Epitasis

3. Gum bleeding
4. Menorrhagia in female
5. After surgery(major or minor)

*VWD has 3 type(م ح نركز عليها)

Investigations of VMD

1)Prolonged bleeding time & to some extent : prolonged PTT
(PTT has a relationship with factor8 which have a relation with VW factor)
2)Normal PT.
Treatment

Give von-wellebrand factor& sometime we need to give factor8
(في preparationفيها الاتنين سوا)

HEMOPHILIA
It is coagulation factor defect, either:
Factor 8: hemophilia A
Factor 9: hemophilia B
Factor 11: hemophilia C

- Hemophilia A&B are x-linked recessive.
- Hemophilia C is autosomal reccesive.(rare)

*the commonest one is hemophilia A.
*the Severity of hemophilia level (A) depending on the reduction of factor 8.
(بسيط قدر م كان النزيف كتيرFACTOR8 قدر م كان)

- Activity of factor 8:

1. Less than 1%: spontaneous bleeding even in early infantile period &this is severe hemophilia (A).

2. 2-5% Activity: bleeding trauma , surgery & in joint (hemoarthrosis).this is moderate hemophilia.
3. More than5% mild type.

- Clinical manifestations:

A) Hemarthrosis:
Is most common presentation ,due to bleeding from synovial vessels in the synovial space , this lead to joint destruction & the patient presented with painful joint during activity even walking.
*This may lead to disable joint . (deformity)
*hemarthrosis affected big joints especial : knee, Elbow, wrist and Ankle.
When we examine the affected joint we see : swollen – Edematous-Erythematous and worm.
*Joint disability is caused by bone destruction and mainly be cyst formation.
B)Hematomas: (collection of blood)

Usually deep hematoma , i.e : muscular hematoma compress the near by structure.
{Hematoma occur in these muscles:" calf , thigh muscle, Buttocks, psoas muscle" cause Retroperitoneal hematoma.(very serious)}
*Hematoma can occur in tangue.
*Intra cranial bleeding may occur and it is the leading cause of death in hemophilia.
{intra cranial hematoma=subdural hematoma, Epidural hematoma, or intracerebral}
*you have to suspect IC bleeding in patient with hemophilia and ITP if the patient develop persistent headache. So you have to scan the brain to exclude IC bleeding.
C) GIT bleeding "mucus membrane bleeding"

INVESTIGATION

1. Prolonged PTT
2. PT is normal
3. Bleeding time is normal
4. Factor Assay: is confirmatory test in hemophilia.

Decreased factor 8 "9?"

Treatment:

give the patient factor 8

جمعية اصدقاء الهيموفيليا:

1. Family screening
2. Investigation
3. Nursing
4. Social worker
5. Physical (orthopedic) therapy
6. Labeling the patient with detailed information and provide factor 8.

Hemorrhagic Disease of the Newborn:

*Occur at early neonatal period.

*Occur due to vitamin K dependent factor : 2,7 ,9, 10.

- The classical presentation occurs within 48-72 hours after delivery
- Vit K deficiency occur due to deficient vit K in breast milk & also because the intestinal flora is inactive .which help in formation of vit K.
- الأطفال البيأخذو Artificial formula milk م بكون عندهم لأنو مدعوم بي VIT K , وبتلقاو اكتر في الاطفال المابيأخذو ARTIFICIAL MILK وبعتمدو على breast feeding بس.

- Site of bleeding hemorrhagic disease of the newborn :

1. Umbilical cord
2. GI bleeding

3. Nasal bleeding
4. Hematuria
5. Intra cranial bleeding

*The classical presentation occur at 48-72 hrs but in rarely case it can occur in the 1th 24 hrs (early onset) and you should ask about :
a) Drug on the maternal side :anti-epileptic drugs whish interfere with vit K ."Phenobarbital-phenyl"
b) Any drug used for coagulation defect :warfarin
c) Center anti tuber callous : INH & Rifampicin.
*عشان تقدر تتعامل مع الطفل لو جاك في ال24 ساعة الأولى ANC دي من الحاجات المهمة في.
-the patient may come late up to 6 month ,in this case you must think about Vit K deficiency.

- There are some thing the enhance vit K consumption and make the disease worst :

1)Mal absorption 2) Bellary Artesia 3) Cystic fibrosis
* The patient come late up to 6month may presented with bleeding or hematoma at the site of injection after vaccination.

INVESTIGATION
1) Prolonged PT &PTT
2) Coagulation factor Assay (2,7,9,10)

*PREVENTION
WHO recommendation : any baby should receive a prophylactic dose of vit K 1MG IM single dose immediately after birth.

TREATMENT OF HEMORRAGIC DISEASE OF NEWBORN (VIT K DEFICIENCY):
1- Give vit K theraptic dose
2- Fresh frozen plasma(FFP) because it contain coagulation factors

3- Blood transfusion : if severe bleeding.

FACTOR 13 (X III) DEFICIENCY

- Factor 13 is stabilizer factor of the clot.

(هو اخر واحد في coagulation cascade)

- It is an Aut somal Recessive .
- The patient presented with delayed bleeding.
- It is disease of exclusion

يعني تفتش أي سبب للنزيف وما تلقى او من البدايه تلقى في نقص في factor 13

*<u>DIAGNOSIS</u> :FACTOR 13 IS DECREASE

*<u>RX</u>: FACTOR 13 REPLACEMENT

Heuoch-schonlein purpura (HSP)

It is type of vasculitis &this purpura is Anaphylactoid purpura.

- Ussually is occur in small vessels , causes small vessel vasculitis

 <u>Clinical presentation of HSP</u> :

1. Acute onset of purpuric rash that could be proceeded by URT infection.

*the typical rash of HSP occurs in the skin of : thigh , buttock , & legs.

2) Vasculitis can affect GI: patient develop GI bleeding then presented with ACUTE ABDOMEN.

3) Vasculitis can affect the joint :patient develop arthritis then presented with joint pain .

4) Vasculitis can affect the kidney :patient may develop renal failure and sometime secondary nephritic syndrome , then patient presented with hematuria.

 <u>TREATMENT OF HSP</u>:

1)Good hydration of patient.

2)Analgesia for pain.

HEMOLYTIC UREMIC SYNDROME

It is acute disease of infants & childhood& usually proceed by acute gastroenteritis (bloody diarrhea) which is lead to :

1. MAHA: Micro Angiopathic Hemolytic Anemia.
2. Thrombocytopenia .
3. Renal insufficiency (due to renal necrosis)

INVESTIGATION :

CBC: Anemia, increased Rustics, Decrease platelet.

TREATMENT

1. Blood transfusion .
2. Platelet transfusion.
3. Anti-coagulation (heparin)

{hyper coagulable state عندهم لانو}

4. Dialysis if there is renal insufficiency.

ACUTE post streptococcal nephritis

Functions of kidney:
1_ excretion of waste products and urine formation
2_ regulation of acid_base balance
3_ endocrine function (vitamin D & erythropoietin)
4_ regulating blood pressure through Renin _angiotensin _ aldosterone system
5 _ maintains water and electrolytes balance
Any problem occurring in the kidney can affect the functions of the kidneys

The pathology in the kidney either :
1_ glomerular disease
2_ tubular disease
The first one in glomerular disease is acute post _ streptococcal glomerular nephritis
Acute post _ streptococcal glomerular nephritis
Acute PSGN is glomerular disease cause by bacterial infection (group A & B _ hemolytic streptococcus)
These group A & B hemolytic streptococcus has different serotypes which are :
1_ serotype no "12" :
Present about 10 % >>> incubation or latent period " 1_2 weeks " _ related to sore throat and pharyngeal infection "in winter_
2_ serotype no "49":
_ present about 25 % >>> incubation period "3_6 weeks"
_ related to skin infection " in autumn " 25%
So when said acute post _ streptococcus nephritis that means it preceded by bacterial infection
What is latent period :

It is the time between the pt having the infection up to appearance of symptoms and signs and this is important point in past medical history
>>> in throat infection it is short latent period
>>> in skin infection it is relatively long latent period

Pathology :
The background of immunological theory of acute GN is >>> certain antigen antibody reaction and complexes >>> bacteria go to kidney and affect the glomeruli and lead to inflammatory process which result in :
1 _ the glomeruli get enlarge in size , so kidney becomes enlarge and bulky 1
2_ this enlargement causes transient compression of vessels " transient ischemia " >> reduced renal perfusion >> decrease urine output
3_ there is loss of RBCs "due to hemolysis or decentgration of RBCs which pass through glomeruli and enter tubules >> pass in urine and cause hematuria

Clinical features :
1_ hematuria :
>>It is main complain >> it is a universal clinical presentation in acute nephritis so , there is no nephritis without hematuria
>> starts from asymptomatic microscopic hematuria " common presentation and discovered accidently " up to frank hematuria " change in urine color >> occur in few cases "
2_ peri _ orbital edema :
>> due to fluid retention
>>increase mainly in early morning and subsided by the day room >> this is typical presentation
3_ high BP " HTN" :

>> Due to disruption or disorder of renin _ angiotensin _aldosterone system
4_ decrease urine output :
>> because kidneys undergo hypo perfusion state
5_ + or – impairment of renal function

Investigations :
1_ urine analysis :
Should be examine :--
>>>> macroscopically for :
>>> change in urine color " smoky urine , cola like urine , tea like urine "
>>> amount of urine " oliguria "
>>>> microscopically for :
>>> RBCs : " either countable or un countable for hematuria
>>> RBCs cast " mainly granular cast which indicate decentgrated RBCs >> not hemolytic
>>> proteinuria :" mainly albumin but does not exceed 3 crosses "+ + + " >>> mild proteinuria
>>> pus cells : " evidence of infection :
2_ RFT & electrolytes :
>>> RFT " urea and creatinine to assess the functions of the kidneys
>>> electrolytes " mainly hyperkalemia "
3_ abdominal & renal u/s :
>>> bulky kidney
>>> if not impaired RFT " preserved cortico –medullary differentiation
>>> if impaired RFT " not preserved "
>>> u/s also for differential diagnosis , because hematuria may be caused by renal stones
4 _ Evidence of streptococcal infection :

>>> in throat " post pharyngeal infection :

" >>>> increase ASO titer " immunological land marker

>>>> high ESR

>>>> + ve CRP

>>>> throat swab

>>> you may find high TWBCs or decrease Hb if excessive hematuria

>>> in skin infection :

>>>> + ve anti DNAase B

>>>> Skin swab

5_ other investigations :

>>> mearure C3 & C4 " low " why ?? >>> because activation of antigen – anti body reaction >> immune complexes which reflected by decrease in complement system " C3 & C4"

>>> for differential diagnosis :

>>>>> e.g " SLE you do ANA ":

If ANA :

>>> - ve >>> ok

>>> + ve >>> do anti dsDNA

<u>Complications</u> :

1 - hypertension :

>>>may be a just HTN without complications

>>> or HTN with organ involvement :

A – heart >> congestive heart failure

B – brain >> hypertensive encephalopathy

C – kidneys >> increase the damage lead to renal failure

D –lungs >> congestion of the lung and cause pulmonary edema

2- complications of electrolytes :

>>> mainly hyperkalemia " serum k+ > 5.5 mEqu/l "

Treatment :

Actually the treatment of nephritis is the treatment of complications :

1 –admit the pt

>> whenever there is :

-high BP

-Renal impairment

-oliguria

2 – do renal chart

3 –give antibiotics

>> penicillin for 10 days either injectable or oral

4 – give antihypertensive :

>> it is better to start with calcium channel blockers orally " nifedipine " or beta blockers

>> if BP not controlled " you can use double therapy "

>> in very serious cases " if pt developed hypertensive encephalopathy or not response after reaching maximum dose of oral drugs >> you give parental I.v antihypertensive drugs such as hydralazine , Na+ nitropurrside , or diazoxide " beta blocker"

5 – dietary control :

>> you have to restrict fluids , salt , and potassium

6 – treatment of hyperkalemia :

>> eliminate k + from the diet

>> you have to protect the heart from hyperkalemia by giving calcium gluconate

You can use :

>> sodium bicarbonate " NaHCO3-"

>> B2 agonist " salbutamol "

>> dextrose 10% with insulin

>> oral chelating drugs for potassium " calcium resionum , or sodium resionum orally "

You do dialysis if :

>> medical treatment fails
>> increase urea and creatinine
>> there is pulmonary edema

UTI

- Ideally there is no bacteria in the urine.
- What is UTI?

It is infection throughout the urinary tract.

- Bacteria (pathogenic or normal flora)

Come from peri-anal area->go into the urethra ->bladder ->kidney (ascending).

Organisms:

1- Gram -ve bacteria.
   Commonly: E-coli (>85%)
   Others: Klebsiella ...Proteus... Salmonella.
2- Others virulent organisms do not cause UTI unless there is low immunity.
3- Fungal UTI: occur in pts with frequent catheterization -using antibiotics for long time- or low immunity.

Bacteria in:

1- Urethra: urethritis
2- Bladder: cystitis
3- Renal pelvis: pyelitis
4- Renal tissues / tubules: pyelonephritis.

Clinical presentation:

Urethritis: painful urination (dysuria).

Cystitis:

1- Suprapubic pain.
2- Mal-odor urine.
3- Hesitancy, urgency, frequency

Pyelonephritis (اكثر حاجة مزعجة):

Pathology: formation of small abscesses in the kidney and if not treated ->all kidney is full with pus.

C/P: High grade fever with rigor – severe loin pain- vomiting and malaise.

U/S: enlarged or bulky kidney with small abscesses.

Urine analysis: hyaline casts.

Clinical presentations: depends on age

Neonatal period:

Non specific presentations

Pt(patient) may present with :(abnormal crying -high grade fever- refuse breast feeding).

More than 2 months:

May present with crying on micturition.

More adult (يعني اقرب انو يكونadult):

May present with:

1- Suprapubic pain.
2- Burning micturition(dysuria).
3- Voiding problems (urgency- hesitancy -dribbling).

Asymptomatic (sterile)bacteria: the presence of bacteria in the urine without clinical symptoms (benign conditions).

Risk factors for UTI:

1- Female gender.
2- Neurologic problems: atonic – bladder neural tubal defect.
3- Urogenital malformation:

Posterior urethral value (PUV).
Vesicoureteral reflux (VUR).

4- Constipation -> due to voiding problems.

Natural defense mechanisms against UTI:

1- Voiding
2- Acidity of the urine
3- Antireflux mechanism of the genitourinary system.

Complication of UTI:

1- May cause renal scarring
2- Pyelonephritis: kidney is full with pus ->drainage / evacuate this pus and may need nephrectomy.
3- Septicemia
4- Peri-nephric abscess.

Diagnosis of UTI:

1- Urine sample:
A- Midstream urine
B- Catheterization.
C- Suprapubic aspiration
D- Adhesive urine pack.

Urine sample for:

A- Microscopic examination for pus cells – hyaline cartilage – proteins.
B- Culture and sensitivity by colonies:
It is confirmatory test
To diagnosis :>100,000 colonies.

2- Supportive investigations:
a- T WBCs

b- RFTs
c- Abdominal U/S
d- Scanning for urogenital defect by MCUG: micturating cysto-urethrogram ->for vesicoureteral reflux and posterior urethral valve.

Treatment:
According to culture and sensitivity
Whom patient need hospital admission?
For any neonate.
Whom patient needs parenteral?
For any neonate.
Whom patient needs oral?
For who can tolerate oral treatment (الكبار شوية).
So, any neonate for admission and IV.
Oral treatment intolerance (vomit the drug) is also for admission and parenteral.
Pt with UTI and prepare for surgery ex: PUV, VUR, also for parenteral.

Indication for parenteral RX:
1- Neonates
2- Oral Rx intolerance
3- Who prepare for surgery

Combined treatment of UTI:
The most common causative organisms are gram negative bacteria.
Drugs sensitive to gram negative:
1- Injectable gentamycin, aminoglycoside ->for severe UTI clinically and laboratory.
2- Oral amoxicillin or co-trimoxazole (septrin) or nalidixic acid-> for mild UTI in Pt <12 years.

3- Quinolones: ciprofloxacin or norfloxacin->for mild UTI in Pt >12 years.

Cephalosporines are not used as combined treatment (الا باختبار الحساسية لانها ما كلها بتشتغل في ال UTI).
But you can use combined: Amoclan (amoxicillin and clavulanic acid)
You give prophylactic antibiotics for Pt that planned for surgery to correct congenital malformation (PUV/VUR).

سبحان الله والحمد لله ولا اله الا الله والله أكبر
أخوكم: مجتبى محي الدين

Malaria

It is the commonest parasitic infection

Live cycle:-

- sporozites occur in the salivary glands of mosquito
- after a mosquito takes a blood meal these sporozites injected in the blood and go to the liver
- then emerge in the bloodstream as merozoites these merozoites enter RBCs
- then RBCs ruptured and give gametocytes
- these gametocytes are taken by mosquito in a blood meal
- then go to the salivary glands of mosquito and change into sporozites.
- The whole pathology occurs within the RBCs.

Pathophysiology:-

-the whole pathophysiology takes place within the RBCs.

-When the malaria parasite enter the RBCs:-

1. It reacts with HB & the component of RBCs & releases cytokines which produces TNF which is responsible of fever… (Cytokines also increases the capillary permeability).

2- It lead to release of cytoadherent which change the RBCs into knobs. ((تبقى زي الصمغ وتلصق مع بعض

- Cytoadherent attack the normal and abnormal RBCs
- clump together.

3. Rouset formation:-

 It's due to clumping of infected RBCs (RBCs gather and make lines) these RBCs can obstruct blood vessels.

This rouset formation is responsible for cerebral malaria & pulmonary edema (also may be due to increased capillary permeability caused by cytokines).

Presentation:-

A.Muscles:-

Myalgia (muscle pain) malaria parasite causes damage of the muscles lead to release of CDK (creatinine phosphokinase) lead to myalgia.

B.Lungs:-

Pulmonary edema edema due to increased capillary permeability.

C.Kidney:-

Renal failure malaria lead to tubular necrosis & damage the glomeruli (so malaria doesn't cause HTN)

D.GIT:-

(Can cause everything in GIT)

1. Abdominal pain.
2. Vomiting & diarrhea.
3. Malabsorption.
4. Hypoglycemia

Prevent sugar absorption

Stimulate pancreas to release insulin

E.Brain:-

- Brain edema
- Cerebral malaria "unknown mechanism"

F. Orthostatic hypertension:-

Pt can't maintain his normal bp when standing up which lead to dizziness & syncope.

G.Severe metabolic acidosis:-
lactic acidosis.

*Laboratory indicators of severity of malaria (indicate poor\bad prognosis):

1. Low blood sugar
2. Increased cpk
3. Lactic acidosis
4. Low HB

 Diagnosis:-

1. Blood film:-

-by seeing Trophozoites inside the RBCs (ring stage)
blue
Chromatin
pink
-it's the definitive diagnostic tools

-Advantages of BF:-
Cheap.
Tell you about response to treatment.
Tell about the degree of parasitaemia.

-Disadvantages of BF:-
The minimal number of …. That should be seen to diagnose malaria is 200 …….
need honest technician .

1 cross (+) in BF = in every 100 film 10 parasites
2 crosses (++) you see from 10-100 parasites
BF also show you the type parasite:-
-P.falciparum 90% of cases

-P.vivax 10-12% of cases
-P.malariae none in Sudan
-P.ovale none

2. ICT
Rapid test .
Tell you if there is malaria or not but doesn't tell the degree of parasitemia.
Contain antibodies for Vivax & Falciparum
tells you which one is positive but doesn't tell the degree of parasitemia.

Malaria is divided clinically into:-
1. Simple malaria
2. Complicated malaria

Simple malaria:-
Its characterized by:-
1. Fever : the most important feature & sometimes simple malaria is called malaria fever.
2. Headache.
3. Convulsion (febrile convulsion).
4. Malaise.
5. Aches & pain.
6. Diarrhea & vomiting.
7. Coughs.

Complicated malaria:-
Severe complicated malaria is a medical emergency, and if the patient not treated properly ,the patient will die in few hours.

Criteria of complicated malaria :-

1.Cerebral malaria
Repeated convulsion & coma.
2. Hyperpyrexia : temp>41
3. Hypoglycemia
It's a serious complication
It's more common in children & pregnant women
4. Pulmonary edema .
5. Renal failure.
6. Hypotension.
Can't be treated unless you treat malaria
doesn't respond to fluids & if you give the patient large amount if fluids
increased risk of fluid overload
P.edema.
7. Acidosis.
8. Severe anemia
Sudden & very low HB.
9. DIC
10. Jaundice (more in adult).
11. Hyperparasitemia

If you find one or more
this is complicated malaria.

The commonest presentation of complicated malaria:
1. Cerebral malaria (coma).
2. Anemia HF.
3. Hypoglycemia
4. Hyperpyrexia associated with hyperparasitemia.

Investigation for complicated malaria:

1.CBC
- for anemia

2.Urea & electrolytes for RF
3.Chest X-ray for
- pulmonary edema
- to exclude severe pneumonia : because its DD for malaria

4. RBG for hypoglycemia
5. ABG for acidosis
6. Urine general
7.Lumbar puncture if you suspect meningitis as DD for convulsion and fever.

Treatment of malaria:-
1.First you have to diagnose malaria
2.Classify it simple or complicated(both are dangerous because simple malaria can change to complicated within few hours)
3.Give medication according to the national protocol:
- first line: coartem
- second line: DHAP
- third line: quinine\artesunate (iv\im)

*Coartem= artemether + lumefantrine
*DHAP= dihydroartemesinin+piperaquine

For **simple malaria** *start with the first line*
- coartem
- If no response give DHAP
- Also yet no response give quinine or artesunate

For **<u>complicated start</u>** *with the third line*: either quinine or artesunate iv\im

<u>Coartem</u>
- 6doses
- Given at…….,then after 8 hours then after 12 hours

Artesunate
يجي في شكل 3 فتايل :
Powder, NaHCO3, NS

- powder has 3 concentrations:
  - 30, 60,120mg
((اخضر,اصفر,ازرق
Dose
Weight ≤20kg 3mg\kg
Weight>20kg 2.4mg\kg

<u>How do you give it?</u>
1.You dissolve the powder in NaHCO3 (pour NaHCO3 in the powder till it become crystal clear & don't start with NS because powder doesn't dissolve in it)
2.Then add NS . This dose should be given within 30 minutes ,because after 30 minute it becomes unstable.
- give it IV (preferred) but you can give it IM (we don't prefer im)
- SE: no SE unless you give a large dose (eg.80\kg)
- this causes severe hemolysis.
- how many doses? 3doses
- When do you give it?

At….time then after 12 hours & then after 24 hours
- if you come after 24 hours & you find the pt has improved clinically (no symptoms) &laboratory (-ve BF)

you have to give him coartem before discharge (take the pills home)
if you come after 24 hours & you find that the pt is febrile & not improved , what shall you do?
Give him another dose of artesunate , yet not improved
 give another dose
 till you reach 7 doses
& don't give more than 7 doses. (Dx may be wrong). (max is 7 doses).

Quinine:-
You can give quinine but recommended is artesunate iv
dose : 10mg\kg\dose
Dissolve it in D5%
1mg in 1ml of d5% to avoid hypoglycemia (e.g.mg200 in200ml)
Give it slowly within 3hours
3doses\day: every 8hours.
you can give quinine im (deep im 60 mg in 1ml NS or d5%)

Duration of Rx:
3doses \day usually for 48 hours (pt usually improve after 48 hours ,no symptoms) & then the pt can tolerate quinine tablets
 so give him the same dose 8 hourly tabs for another 7 days
total duration quinine is 9days at least .
SE:
1.HYPOGLYCEMIA
2.Arrhythmia
3.Hypotension
123تسبب الموت لوالدرب مشى سرعة
4.Tinnitus\deafness
5.Blindness

Malaria vivax:
الحمى الحبشية
- account for 12% OF malaria parasite cases.
- cause malaria parasite resistant .
- also can cause :

1.Hyperparasitemia + fever
2.Thrombocytopenia
3.Coma
4.Relapsing fever characteristic for vivax because merozoites of vivax can stay in the liver for 70 years.
So once you treated vivax you have to eradicate merozoites from the liver by giving primaquine daily at least for 2-3 weeks otherwise you get relapsing fever

p.malariae can cause nephrotic syndrome.
غرب أفريقيا))
- if the pt is splectomized
- give fansider +artesunate as prophylaxis
- in simple malaria ,quinine is contraindicated, you have to follow the protocol.
- if you give the pt quinine & after 3 days three days there is no improving
- it is most likely other diagnosis or quinine resistant

ما عندنا في السودان

Other DD for malaria
1.Typhoid fever
2.Severe pneumonia
3.Brucellosis
4.meningitis

5.Hepatitis
6.Kalazar
7.Leukemia

Congenital malaria:

Neonates presented with criteria:-
1.Fever
2.Jaundice
3.Anemia
4.Splenomegaly
5.+ve BFFM

Treatment *:*
Quinine im 10mg \kg for 5-6days PD.
Give sugar through NG tube because these are very sensitive to low blood sugar.
prognosis :is very good

Malaria with –ve BFFM:-
-don't allow pt in Sudan die with fever unless you give antimalarial drugs because it's an endemic area
في غانا في ناس ماتوا وبعد الوفاة لقوا ادمغتهم ملانه طفيليات بالرغم من ان التحاليل كانت سلبية

Prognosis of cerebral malaria is good & we rarely lose pt due to cerebral malaria.

Quinine IM
Deep IM
Instead of 1mg in 1ml of d5% when you give quinine IV here 60 mg in 1ml of NS or d5% due to absorption is less.

Meningtis

- Meningitis can be‘‘causitive organisms’’:-
- Bacterial . the commonest
- Viral .
- Fungal .
- Parasitic .

Bacterial:
Neonatal meningitis can be caused by :
4-E.coli . from Birth
5-B . Hemolytic streptococcus . canal
6-Staph . during Labor
7-Listeria Mono cytogenes ‘’ very bad ‘’
[but any bacteria can cause meningitis]

*Infants :
Any bacteria can cause meningitis but the commonest are :_
1)Meningococcus .
2)Strept . pneumonia .
3)H. influenza .

*Adult :
Also any bacteria can cause meningitis but mainly meningococcus .

Viral:
Any virus can cause meningitis but the most common 2 viruses are :
 Mumps .
 Poliomyelitis virus .
Fungal:
•Candidia : candidia meningitis .

Parasitic*:*
Very rare

Symptoms and signs of meningitis :

 symptoms :_
1.Fever .
2.Headache .
3.Vomiting .
4.Convulsion .
5.Blurring of vision .
6.Coma .

Signs:_
1. Pt is very ill .
2. Ferbile .
3. In infants :tense Bulging .
Fontanelle before 18 month قبل تقفل)
4. Inflammation of the meninges lead to neck stiffness .
resistance في تلقى راقد والطفل neck لل Rlex نعمل نحاول
It is painful دي والحاجة muscles في ال spasm ال بسبب ودا

9. +ve kernig's sign .
راقد كدا Pt وال knee extension نعمل لما نحاول pain بحصل)
)Spinal cord في ال meninges لل stretch بحصل عشان

10. Triput sign : -
(الطفل بقعد على 3: يده اليمنى ,يده اليسرى,صلبه)
11. Papilledema : it is a late sign of meningitis .
(this in meningococcal meningitis)
???

Menigococcal menigitis has a feature that it can cause meningococcal septicaemia :
Sever Hypotension .
Adrenal insufficiency this adrenaline & noradrenaline lead to sever hypotension
Skin rash : subcutaneous bleeding .
Very sever & mortality rate 50% .
So ,
Meningococcal meningitis is associated with meningococcal septicaemia .
But ,
Pneumococcal meningitis is associated with brain Abscesses & can be recurrent .

Diagnosis of meningitis:
Meningitis is diagnosed by lumbar puncture (LP). (LP لو ما عملت meningitis ماف طريقة تشخص ال

Investigations are :
- LP .
- CBC : you find leukocytosis .
- CSF culture .
- Blood culture : because sometimes meningitis is associated with septicaemia .
- CT scan : for brain oedema .

 LP :
Should be done under a septic technique .
Between : L2 and L3 or L3 and L4 .
Firstly you measure the CSF pressure .
برش CSFال duraال تصل ما اول meningitis في
Secondly : color of CSF .

عشان ال CSF يبقى turbid لازم عدد الخلايا يبقى اكثرمن 100
(Turbid > 100 cells/mm2)
Thirdly : you prepare 3 containers of CSF and send for 3 departments :
A) Biochemistery department :
for sugar and protein .
- Sugar : low in bacterial meningitis .

*CSF sugar = 2/3 of blood sugar .
So , you should do blood sugar at the same time . (hypoglycaemia لان المريض يمكن يكون عنده)
- protein : High .

B) Hematology department :
*For cells .
- Normal cells in CSF is 0_5 lymphocytes .
- If Neutrophilia Bacterial meningitis .
- If lymphocytosis Viral meningitis .

C) Bacteriology department :
- CSF for gram stain and for culture .
- Also send blood for blood culture .

*meningococcus Gram –ve
*pneumococcus Gram +ve

	Viral	Bacterial
Sugar	Normal	Low
Cells	Lymphocytosis	Neutrophilia

- Sugar is normal in viral meningitis except in Mumps meningitis low sugar So , to differentiate Mumps meningitis from bacterial meningitis .

Do cells :
- Lymphocytosis in Mumps .
- Neutrophilia in Bacterial .

*You cannot diagnose meningitis without

lumber puncture .
Treatment of meningitis :
Antibiotics : according to the protocol of the hospital .
(cholormephenocol زمان كان بندي)
Now we give 3rd generation cephalosporins (cefitax , cefitazidime) .

Duration of RX :
In neonates : 21 days (3 weeks) .
Because neonatal menigitis affect the ventricles causing ventriculitis and so , it can cause Hydrocephalus .
In children : 7_ 10 days .
*You change your antibiotic acc to CSF or blood culture (usu afrer 48 hrs) .
*pneumococcal sensitive to penicillin لو قال ليك ال
penicillin غير طوالي لل

*ampicillin H.influenza sensitive لو قال ليك
ampicillin غير طوالي لل

Complications of meningitis :
Mortality :_
= 10% of Pt with meningitis death .
= 50% in meningococcal septicemia .

Epilepsy :_
= meningitis can cause scarring of the brain epilepsy .
Hydrocephalus in neonates .
Mental Retardation (cP) .
Deafness .
Blindness .

*RX of meningococcal septicemia : _

You give antibiotics plus Immunoglobulin
1 gram / kg infusion .

Prevention :
= by vaccination : Meningococcal , pneumococcal , H.influenza , Mumps & Measles in MMR .

Tuberculous Meningitis :
Gradual onset , not like acute meningococcal meningitis (acute onset)
- So , it always diagnosed late .

symptoms :
= Headache . = Depression .
= Vomiting . = Fatigability .
= Constipation .
- So it always late to diagnose .

Investigation :
LP : lymphocytosis , low sugar & culture .
CT Brain : you find brain abscesses .

RX of TB meningitis :
= 8 months in duration .
= 1st two months :
(1)Rifampicin . (2)INH .
(3)Ethambbeutol . (4)Pyrazinamide .
= rest 6 months :
Rifina (Rifampicin + INH) .
= children with TB meningitis usually recover with complications .
(نادرا ما يطلعوا منو ساي)

Diphetheria

- Caused by coryne bacteria (gram+ bacilli)
- Mode of infections by droplet
- Incubation period:1-6 days

Any non immunized person is exposed for diphethria
pathology
It causes necrosis of coagulation of tissue through exotoxin wich lead to formation of membrane this membrane contain:macrophage-lymphocytes-basophils-other inflammatory cells + necrotic tissues
This membrane gray in colour –smelly-attached to deep tissue and if you try to remove it it will bleed
This bleeding differentiate between diphethria and follicular tonsillitis (wich not leed to bleeding)

Types
Nasal diphethria (commonest)
Pharyngeal
Larengeal
Nasal
Mucousy bloody and smelly discharge
This type is highly infectious
Pharengeal
The patients is toxic very ill

Symptoms:-
difficulty in swallowing+pain in throat+low grade fever and the pulse is rapid
signs:the patient looked ill and bull neck appearance due to enlarged lymph nodes (cervical glands)the neck is swollen by one side (characteristic for diphetheria)

when you look to the pharynx you will find the membrane which bleeds when you remove it
larengeal
it affect the larynx /obstructs the airway so can lead to airway obstruction and stridor
*diphtheria also can affect :
1.eyes : causes conjunctiva diphtheria
2.skin
3.genital :esp. vagina

D/D of diphetheria
1.tonsilitis 2.adenoids}esp for pharengeal type
3.infectious mononucleosis
4.foregn body

Diagnosis of diphtheria
Diphtheria is a clinical diagnosis :when you suspect it by symptoms and signs you will start treatment But you can take a sample for culture or immunofluroscence(coloured by dye and seen under microscope)

Treatment of diphtheria
1.Give anti toxin therapy(pharyngeal:20000-40000 units/Nasal:40000-60000 units/Laryngeal:80000-120000 units{because it is the mst dangerous})
If there is complete airway obstruction:Do tracheostomy and take care about airway otherwise the patient wil die
2.Antibiotic because it is bacteria

Complications of diphtheria
1.myocarditis and arrhythmia (diphtheria myocarditis occur 3-4 weeks after disease

Arrythmia is the cause of death in diphtheria
2.Acute polyneuritis(mainly motor not sensory the patient sense normally)
Diphtheria causes: A.Bulbar palsy(because it affect nerves of soft palate) so cause nasal regurgitation (food exit by nose)
B.Loss of accommodation in the eye due to affection of nerves coming to eye
3.diphtheria nephritis

<u>prevention</u>
by penta vaccine (three doses)
penta vaccine {diphtheria /tetanus/hepatitis/whooping cough/H.influenza}
TD(tetanus+diphtheria) is for pregnant women and child bearingage women.(5 doses :2 in first pregnancy the time between them one month and then one dose every pregnancy until complete 5 doses)
T+D makes response of body better

Encephalitis

- Presentation is more or less similar to meningitis but with bizarre picture(Eg :coma common in encephalitis)
- You do csf:lymphocytosis or normal
- Culture for viruses
- Viruses causes encephalitis
- Herpes simplex (commonest)-mumps-measels(sspe)-polio
- You do ct scan: you find brain atrophy(after few monthes) because encephalitis . usually recover with sequlae
- Measles

Measles can cause:acute encephalitis bu measles itself

- Or late encephalitis(SSPE)

SSPE

It is subacute scelrossing pan encephalitis
Usually occur after 8 years from measles
There is brain degeneration so the child is dying infront of you
The early symptomps of SSPE(take about 3 to 4 years to appear)is change in mood(less interesting)(poor eating or sleepy)
In csf :lymphocytosis + culture
Serum IgM is high

Treatment of encephalitis :-

No treatment except for herpes simplex: we give acyclovir.

COMMON CONGENITAL ANOMALIES
CLEFT LIP and PALATE:-

- Familial
- It could be associated with other congenital anomalies (eg:CHD)
- There is feeding problem
- Has specific teat

)فيها membrane ال لانو فوق من soft muscles act as vaccum)

- Should be treated before 18month to acquire speech

<u>TRACHEO-ESOPHAGEAL FISTULA:-</u>

- There is fistula between the esophagus +trachea
- Milk go form the esophagus to the trachea + respiratory system +so PT distressed PT need NG tube .
- <u>Has different types</u>:-

1) (esophagu ال واصلماstomach)

2) H-shape دي اسمها
3)

- الطفل يرضع يشرق طوالي ودا بحتاج TG tube وبتدخل عادي
- Diagnosis by barium swallow
- Rx need surgical operation

Pyloric stenosis:-

- More common in boy than girls
- Inciedence 1in 500boy

- <u>Presentation</u>:-

1) Projectile vomiting :start with in the 3rd week
- This is vomiting does net contain bile

(الطفل بعد الاستفراغ يرضع طوالي)

2)Hungury child but well : vomiting with good appetite
3)constipation
4)peristalsis movement form left to right

- We do test feed :-

(تدي الطفل لبن قدامك حيستفرغو)

- Diagnosis :by U/S you find hypertrophy of the pylorus
- Rx:- 1) correct electrolyte imbalance by giving IV fluid [ed K+, ed Na+ ,sometime ed urea due to dehydration]

2)surgery :- gram_ state operation , done by most senior one

- بتقطع ال muscles الفيها hypertrophy بس لازم ما تصل ال mucous membrane عشان ما يحصل perforation
- After 6 Hr start feeding

Duodenal atresia:-

- More common in down syndrome
- Present with intestinal obstruction (vomitus contains bile)
- Diagnosis: by plain X-ray abdomen you find double babble sign (this is diagnostic for duodenal atresia)

Mal rotation:-

- Present with vomiting contains bile
- Diagnosis : by plain X-ray abdomen

)بتلقي كل الabdomen shadow علي جهه واحده)

- Rx surgery

Hirshsprung disease:-

- Agang loinic disease of the large bowel (no parasympathatic innervation this will lead to obstruction

- This start form the rectum + then ascend + can affect whole colon

- Presentation:-
- Any baby (new baby) should pass meconium in first 24Hrs
- D/D if the baby did not pass meconium in the 1st 24 Hrs:-

1. Hirshsprung disease
2. Meconium ileus
3. Imperforate anus

- You suspect the diagnosis of hirshspruny disease by not pass of meconium

- Investigation:-

1. Plain X-ray of abdomen Empty rectum (no air)
2. Rectal biobsy definite diagnositic test
3. Bariam enema rat tail appearance

- How to differentiate it from constipation??
- In constipation rectum is full by feces, so when you do PR you find feces in tip fingers

- But in hirshsprung disease when you do PR empty rectum

- Treatment :-

1. Colostomy for 1Yr till the PT reach 10Kg weight
2. Then do pull through operation

(يوصلوا ال sigmoidمع ال rectum)

Diaphragmatic hernia:-

ال intestine تطلع فوق و commanestبتطلع بال left side

- It could be diagnosed antenatally by U/S
- It should be diagnosed at the brith :-

1. The PT distressed

2. Scaphoid abdomen

- Do not resuscitate with diaphragmatic hernia by mask because he will die

O2 mask air will go to stomach
Form stomach to intesti compress the lung compress the heart +other lung death

But if you suspect intubation
You can hear bowel sounds at chest

- Dx :by X-ray

bowel فوق) تلقي ال(

- Rx : surgical

(تنتظر 24 ساعه غشا ن ال lung وال heart لأ ولا طبيعيات)

CONGENITAL HIP DISLOCATION:-

- Not common
- PT present with limbing
- You should look it in the 1st 24 Hrs

- Rx: pouble nappies

- Indication of immuneglobubins:-

1. GB syndrome
2. ITP
3. Meningococcal septicemia

[Ig 1gm/Kg]

- دي لو جاك طفل بالصباح ما قدر يقوم من السرير بعد ما صحي من النوم GB طوالي.

Iron Deficiency Anemia

Introduction :
Hemoglobin is formed of heme & globin
-Hb synthetized in bone marrow
-Hb synthesis need iron +folic acid +U B12+thyroxine "hormone" +amino acids +some organs(kidneys, liver, spleen, intestines :absorption of above, heart)
-there is anemia called nutritional anemia
-So if there is chronic disease eg. Renal failure or systemi disease eg. Rheumatoid arthritis >>can lead to anemia
-Hb is destruced in spleen (120 days)
-Normal Hb >>> 11-15 g/l
-In neonates up to 22g/l

Anemia : is a low Hb in comparison to the mai Hb of the community
°pre-mature destruction of Hb:
1-anything affects RBCs because Hb occurs inside RBCs
2-Disorders affects Hb itself
This is called hemoglobinopathies:-
>Thalassemia
>sicke cell anemia
(Hb pathy=abnormal Hb)
3-Deficiency of some enzymes like G6pD & pyruvate kinase
4-Abnormal membrane :-e.g: congenital spherocytosis
5-From outside :- e.g:malaria, antibodies (ante immune hemolytic anemia)

-CBC: or complete hemogram
>WBCs :- 4-10 thousands
- lymphocyte: if elevated= viral infection

- neutrophils : ¬ ¬ ¬=bacterial infection
- monocytes
-esinophis
- basophis

*there are many causes of leukocytosis

>RBCs :- 5 millions

-Hb: 11-15 g/l

-PCV: 45-50

-MCH, MCV, & MCHC

>Platelets : 150- 250

-Reticulocytes: 2% (maximum)

-blood film : micro/ microcytosis, hypochromia

Iron Deficiency Anemia :-

Source of iron:

-green vegetables, meat & cereals

-iron is absorbed by upper GIT

-iron is absorbed in form of ferrus

-we need about 1 - 2 mg/day

(less in males and higher in females due to menses & B. F)

- لبن الام فيهو حديد كتير وامتصاصو ساهل عشان كدا معظم الاطفال البدوهم لبن علب في السنين الاولى بتجيهم IDA

-نسبة امتصاص الحديد من لبن العلب حتى لو اضفت ليهو حديد هي ٢٪ بس

-اما نسبة امتصاص الحليب من ال breast milk هي ٥٠٪

-ممنوع لبن من برا في السنة الاولى بعد ستة شهور ياكلو بس

-Iron absorbed>>>apoferritin>>>transferritin>>>ferritin(storage form)

-iron stored in liver in the form of ferritin

-65% of iron occurs in Hb, 15% of iron occurs in myoglobin, 1% of iron occurs in cytochrome, the rest of iron is stored

-iron in human body=only 4 gram

Clinical features :-
>Symptoms : fatigability, headache, palpitation, Dizziness & loss of energy

-signs:-
1-pallor
2-kolionyclia
3-spleen enlargement
4-gongestive cardiac failure or dilated cardiomyopathy if there is sever IDA

-Investigation :-
1-CBC : low RBCs, low Hb, low MCV, low PCV, low MCH, low MCHC but high platelets (thrombocythenia)
2-S.iron :low
3-TIBC: high
4-transferrin
-Blood film: hypochromic microcytic anemia
5-stool for parasite, UG, malabsorption and this to know where the iron loss
*Celtac disease :intestinal (duodenal) biopsy

Neonatal jaundice

Serious in newborn due to it's complications (..... Liver damage)
-In newborn it detect when exceed 5-7 mg/dl
-about 60%of term babies and 80% of preterm babies
-will have hyperbiluribiemia in the first week of life
-UCB can cross the BBB and deposits in basal ganglia and causes kernicture
-Urobilin is colour less, when therer h.jaundice the urin colour is normal
-eutero hepatic circulation can be affected by any condition that can causes stasis of the intestine or any condition affects the normal flora of the intestine e.g intestinal atresia or decrease euteral intake or mechonium.....

<u>-why babies develop hyperbilrubinemia:-</u>
1-immature liver enzymes
2-high hematocrit average = 20 mg/l while the maximum levels in adult is 13g/l
2-decrease RBCs life span "90 days"

<u>Classification according to time and causes</u>
-according to time :-
Physiological jaundice usually occurs in the first 72th hours which is serious
Causes in day 1:-
1-Rh iso immunization and ABO incompetability
2-congenital cpherocytosis and eleptocytosis
3-G6PD
4-intrauterine malaria, bacterial, viral infection

Day 2:-
1-physiological
2-sepsis
3-any disease that can cause polythycemia
Day3:->>>> lead to prolong jaundice
1-neonatal hepatitis
2-extra hepatic biliary atresia
3-breast milk jaundice. Maximum at day (10-15)
4-metabolic disease and occurs after day 4

Any collection of blood "hematoma" end with hemolysis and cause jaundice
Most common causes of obs jaundice in neonates is extra or intra hepatic biliary cirrhosis
.....

Note:-
Inborn errors of metabolism =galactocemia
Approach=hx+examination

History:-
1-Birth Weight >>>>>low body weight develop complications similar to preterm babies and develop complications at lower level
2-GA
3-Maternal complications >>>>>GDM +TORCH
4-FH >>>of jaundice, liver disease
5-Previous hx of jaundice, malaria, kernicterus
6-colour of urine, stool, onset and duration
7-breastfeeding
8-birth asphyxia
Bleeding tendency by itself "hemorrhage" doesn't cause jaundice

Sepsis can cause jaundice thus we must perform sepsis screen which include:-
1-blood culture
2-CSF analysis
3-CXRY

Complications :-
1-acute encephalopathy
2-chronic encephalopathy
3-kernicterous

CP وحصل ليهم neonatal period ديل طلعو من ال
Called athetoid CP with sensory neural deafness
Which affects :-
1- Movements >>>cerebral palsy +athetosis+dystonia
2-ear>>> high frequency hearing loss
3-eye>>> paralysis of upward gae
4-teeth>>> dental Dysplasia
5-mild mental retardation +disability, they can understand the orders for example.

Kernicterus :-
Is a pathological diagnosis and refers to yellow straining of the brain by bilirubin +neural injury
This neuronal injury is due to necrosis, neuronal loss & gliosis
It leads to apasticity

-Features in the first year:-
1-poor feeding
2-highly pitched cry

3-hypotonia
4-good deep tendon reflex
5-persist tonic neck reflex and righting reflex
6-slow motor skill
-after the first year extrapyramidal features occures

-causes of death :-
1-end stage coma
2-respiratory failure
3-intractable seizure

-Management :-
The goal standard of mx is prevention of further deposition, and prevent complications especially if there's causes of hemolysis"Rh, ABO, hematoma " as much as possible
-you have to prevent complications by converting th UCB into a conjunction bilirubin by using a phototherapy which convert it into isomer that can be easily excreted in urine and stool

-Prevention of hyperbilrubinemia can be achieved by:-
1-early and frequent feeding. من ما يلدو هو
2-adequate dehydration

-I put the baby in the phototherapy according to the bilirubin chart which contains :-
-the bilirubin level
2-age by hours

SIE of phototherapy :-
1-loos green stool
2-skin rash and bronze skin syndrome
2-affects the retina and may cause blindness

3-affects genitalia >>>subfertility
4-hyperthermia>>give small hydration and breast milk through NG tube
-Prolong jaundice (indirect) :-
More than two weeks
-could be direct causes by:-
1-.....syndrome
2-breast milk jaundice
3-hypothyroidism
4-.....
Pyloric hepatitis, can causes conjugated
5-.....

<u>Defined by:-</u>
1-direct bilirubin more than 2
2-any conjugating bilirubin exceed 10-20 % of the total serum bilirubin.

RDS

- RDS occurs primarily in premature infants
- When GA is advance the occurance of RDS is less and when GA is early the occurance rate increase
- RDS rarly occurs in term
-

 RISK FACTORS

1. Preerm baby
2. Precipite labour
3. Cs
4. Asphyxsia
5. Multiple pregnancy
6. Previous history
7. Maternal DM

FACTORS THAT DECREASE INCIDENCE OF DM

1. Antenatal cortico steroid prophylaxis so once you suspect preterm labour>>>> is better to give steroid
2. PIH : it is stressful condition leading to release of endogenous steroid {this steroid enhance the lung maturation }
3. Prolong rupture of membrane
4. Maternal opiate condition

 Surfactant produce by type 2 pneumocyte of the GA

 It distribute at GA =30 weeks

 It is less distribute before 30 weeks

Main function of the lung surfactant >>>is decrease the surface tension ,increase the compliance of the lung and thus improve the functional capacity of the lung

 No surfactant >>>>high surface tension and collapse or atelectasis

- Competent of the surfactant >>>protein + fat
  Protein = lecithin ,surfactant protein =appoprotein A. B. C.D
  Fat = cholesterol
- Mutation in gene coding sp B+C >>>>FH of RDS
- The synthesis of surfactant depends on {mandotory }

1. Normal PH
2. Normal tempreture
3. Normal perfusion

- Asphyxsia ,hypoxemia ,ischemia >>>>>decrease surfactant
- High o2 concentration or 100% o2 >>>>injur lung >>decrease surfactant
- Ventilation >>>>injure lung >>decrease surfactant
- When the surfactant decrease >>>the lung appear like the liver + deep pureplish red

- C/P

1. Tachepnia ,,,,,,,RR >60
2. Cyanosis,,,,,,,(central or by o2 concentration less than 94
3. Prominent
4. Intercostal or subcostal retraction =chest indrawing
5. Breath sound normal or diminish +fine harsh may be heard posteriorly over the lung basis
6. Mixed of respiratory metabolic acidosis
7. The hypoxia affect the perfusion of the other organ eg kidney and GIT leading to oligorea and ileus
8. Oedema due to ….

- Death rarly in day 1 >>>>>destroid after that {day 1 is better than day 2 >>>worse and worse till day 7 }
- If they pass the first week >>>things are getting better

And is associated with >>>

1. Pulmonary Hge

2. IVH
3. Alveolar air leaks (emphysema +pneumothorax)
- The natural course of un treated RDS is characterized by progressive worsening of cyanosis and dyspnea

History >>>>

1. GA (prematurity) 2. Risk factors

Diagnosis >>>>>

- By hx , examination the investigation like

1. PH
2. Acid base values and blood gas
3. Chesr x ray to show the feature of atelectasis >>>> more prominent in the ll lobe

- reticular shadow
- air bronchogram

DD :

1. Sepsis causes respiratory distress
2. Cyanotic heart disease can mimic RDS
3. AIR leak eg pneumothorax
4. Biochemical disorder (eg hypoglycemia . anemia. poly cythemia .metabolic acidosis)

Management :

- General = supportive

Heat hypoxia ,hypotension,hypothermi,

- Specific =o2+ surfactant

02 >>>saturation 85-95 don't exeed that >>>worse the condition

- Surfactant >>>intranasal
- CPAP(continuous airway pressive)at 9 pressure of 5-10 cmH2o to improve the end expiratory pressure to make the alveoli open {cause when expiration ends atelectasis occurs}

- Ventilation >>if the baby have metabolic acidosis ,v ill , abnormal tempreture :
- Give the surfactant through the endotracheal tube , pull the tube outside and put the pt on CPAP machine
- But when use ventilator , give the surfactant and keep the tube inside the trachea and put the baby on the ventilator or machine

Complication:
acute
1. Apnnea
2. Pulmonary He
3. Respiratory failure
4. Death

Chronic>>> complication of disease
+management
5. Sub glotic stenosis due to intubation
6. If the baby receive o2 for a long time ,he will develop broncho pulmonary displasis (BPD) more than 6 weeks
7. PDA

Prevention:-
1. Whenever you suspect premature labour >>>give dexamethasone to decrease RDS
2. Avoid unnecessary elective c/s
3. Appropriate Hx of high risk group

How to counsell a lady present her baby to you with feature of ……on day 3 ?
- The next delivery should be at hospital contain neonatal unit and pediatric unit
- Good pediatric unit is important to identify the blood grouping and ABO incompatibility

Prenatal asphyxia

Def :impaired in blood gas exchange that leads to progressive hypoxemia and hyper capnia

- This asphyxia may affect the brain and cause encephalopathy(hypoxia-ischemic encephalopathy…HIE)
- Neonatal ancephalopathy =neurological disturbance which manifested by ;

1. Difficult respiration
2. Alter level of consciousness
3. Hypotonia
4. Abscent primitive reflexes eg mero and suckling
5. poor feeding due to abscent reflexes

criteria that collectively suggest HIE due to prenatal

asphyxia :-

1. occurs prenatal ,intra partom or post natal immediately
2. fatal brady cardia or absence of fetal heart rate
3. apgar score (0-3) any score less than 7 is asphyxia
   - less than 3>>> severe asphyxia
   - 3-4 >>moderate asphyxia
   - More than 4 >>>mild asphyxia
   - Apgar score done at birth (0 mins)and after 5 mins and every 5 mins
   - Maximum until 5 20 mins
   - Apgar score is prognostic factor
4. Early imaging showing evidence of acute non focal cerebral abnormality >>>>electrical imaging

5. Onset of multi system involvement within72 hrs of birth

- These 5 criteria important in anticipation

- Risk factors or causes :-

Antenatal

1. Maternal trauma
2. Maternal hypotension due to APH

Intra partom

3. Prolonge and obstructed labour
4. Cord prolapse
5. Any condition associated placental insufficiency eg DM,IUGR,MAL PRESEULATION ,pre eclamsia and multiple gestation
6. Post partom (related to the baby
7. Any cause of cardio compromised eg congenital heart disease ,respiratoy disease ,metabolic disorder
8. Small for gestational age >>>some of them develove asphyxia especialy if they have RDS

Pathophysiology of HIE:-

1. Hypoxia cause cerebral injury,leading to release of special metabolic substance eg glutamate which lead to primary engery failure (ischemia and edema)
2. High glutamate stimulate N-methyle-d-aspartate leading to cell death by necrosis >>> high ca2+
3. Reperfusion period with normalization witin 2-3 hrs
4. The oxidative metabolism returns to normal (takes around 15 hrs)
5. If not the primary failure stop >>>go into 2ndary failure and neonatal death in the first 3 days (from 15 hrs until 3 days)

- The adaptation is by shifting the blood flow to the heart ,brain and adrenal>>and still there is possibilities to develop ischemia in these organs due to hypoxia

- <u>c/p(multi organ damage):-</u>

1. renal system>>renal ischemia leading to tubular acidosis….s+s of acute kidney injury like electrolyte disturbance and oliguria (even haematouria)
2. heart >>>myocardial ischemia leading to poor cardiac outout ,hypotension and cardiomyopathy
3. autonomic disturbance :usually present with urine retusion rather than stool

- abnormal tempreture due to affection of the hypothalamus and autonomic NS

4. liver damage : damage to the hepatocyte (present with feature of liver failure eg bleeding tendency ,co-agulopathy

- feature of hepatitis eg high liver enzyme

5. GIT:- edema, necrotizing euterocalitis,feed intolerance (can not absorb)
6. Haematology=DIC

7. Metabolic :-

- Metabolic acidosis due to asphyxia >>>unmetabolic leading to production of lactic acid >>acidosis(deep+slow breathing)
- Birth asphyxia lead to both respiratory +metabolic acidosis
- Electrolyte disturbance
- The maximum period that the child needs to grow is 4 yrs
- So any problem affecting the CNS before 4yrs >>>serious and reversible permanent brain damage because the brain is not growing yet
- Hypothermial therapy (cooling)>>>they say it switch off the metabolic process and thus prevent further damage
- How u determine the severity of asphyxia ? by apgar score

<u>Sarnat scale to determine the grade of HIE</u>:-

It classify the baby according to the tone,convulsion,ANS affection by pubil assesss ,respiration to assess BS function,reflexes.

1. Grade 1 (mild) :-

- Normal muscle tone
- No convulsion
- Mydriasis
- Hyper alert (open eyes)
- Weak moro and suckling reflex

2. Grade 2 (moderate):-

- Mild hypotonia
- May or not ……
- Myosis
- May be affection to the level of consciousness>>>lethargic or drowsiness
- Weak reflexes

3. Grade 3(severe)
- Flaccid muscle tone (flobby)
- Intermitted decevbration
- Poor pubillary light response
- Comatouse
- Abscent reflexes

History:
m/c >>>baby not cried
1. Cry or not
2. Normal VD or prolonge labour
3. Instrumental delivery
4. GA (preterm baby or term)
5. Calculate apgar score
6. Complication during pregnancy eg DM,HTN,APH .
7. Cord prolapse during labour
8. Start breast feeding or not
9. PMH>>>>previous baby with asphyxia

RF for complicated delivery >>>>if she every time delivers a baby with asphyxia >>>CPD>>>poor antenatal care .

10. FH >>>CDM

Congenital malformation
Congenital heart or respiratory disease

11. DH >>>>Narcotics eg morphine or pithidin which given 6 hrs before delivery and cause respiratory depression
12. HPI >>>if home delivery ,you have to ask about symptom of systemic affection >>>>

- Breast feeding or not
- Pass urine / stool or not
- Abdomen distension or not
- Examination >>>> sign indicate encephalopathy
- Investigation >>>>

1. ABC for metabolic acidosis
2. Electrolyte for hypo ca2+,hypoglycemia .
3. RFT
4. LFT for enzymes and serum bilirubin level
5. ECG , X- ray and echo
6. Bleeding profile (pt ,ptt,IIVR)>>>>for co agulating
7. Renal Na+/urine and plasma osmolarity for SIADH
8. Cerebral monitor (modify ECG) for brain activity
9. Abdomen x-ray if you suggest necrotizing entero colitis (feature of ilius).

For investigation ,we don't do all investigation ,we check the system and investigate according to.
Management : >>>prevent the causes that lead to asphyxia

- The care of these babies (related baby with cp,haemoplegia, paraplegia) is very difficult >>>they need multidisciplinary team .
- The goal standard of Hx is optimal resuscitation .

1. Attend all high risk group (anticipate)>>>(DM,HTN,APH,emergency,c/s)to deliver as hospital with pediatric units and qualify by doctors that able to perform resuscitation of the baby.

- Then perform apgar score
- Perform high quality CPR
- Giving breathing -+chest compression

- Then follow up pediatric unit (special baby care unit)>>>>must contain the basic needs>>>(fluid,anti convulsion,antibiotics,o2)

2. Optimal resuscitation in the delivery room
3. Support and treat the complication
4. Hypothermia therapy or cooling >>>>minimize the metabolic damage (effective)

- Prognosis >>>>depend on the severity of HIE
- Severe damage >>>>>
  - Multi organ failure
  - Cp
  - Mental retardation
- Morbidity rate=30%
- Mortality rate =30%

• Colour	pink	acro-cyanosis	blue or central
• Tone	fully flexed	semiflexion	flaccid,flobby
• Res Effort	crying	……..	NO BREATHIG
• RR	>1oo	<100	no RR

Hepatitis

. Other HIV , CMV , EPV , HSV , adeno , …….. , pora virous
Rubella + varicella .
. Transimission :-
A+E by feco oral route .
B+C+D blood product +sexual route .
. prevention :-
A+B have vaccine .

HAV:_

- Highly contiguous disease .
- IP from (15_50) day avarage of 3 weeks .

Fecal excretion of the virous :

- Start late in the IP .
- Reach it's peak before onset of the symptoms .
- Continuous for 2 weeks after symptoms appear .

 3 stages of pathogenisis :-

1- Non cytopathic replication .
2- Cytopathic erosion +necrosis .
3- Hepato cellular damage infilration by the cells .

- HAV cause acute hepatitis .

 CIP :

1- Pre ectric prodromal symptoms .
2- Ectric may be ass hepato spleeno megaly .
3- Post arthralgia + evanescent rash

'' maculo – papular rash ''

معظم الناس بكون جاهم HAV وهم صغار

اي مكان

May have extra manifestations due to deposition of immuno complex within the tissue .

Investigation :

1. Ig M remain +ve after 6 months of acute infection .
2. PCR .

 Sometimes pt discover has HAV before it become infectious .

2 weeks before excretion in the stool one week after .

3. High liver enzyme

 have Direct and Indirect hyper bilirubinemia .

 Course of the …..

_ majority become well .

_ few develop complications :_

1) Fulminant hepatic faiture .
2) Prolonge cholestatic syndrome itching + steatorrhea + maL absorption .

- Mx supportive .

avoid fatty diet .

 anti pyretic .

hydration .

 glucose .

no need to give specific drug .

*early sign of liver failure is disturbed level of ……

 Indication of admission :_

1) Sleep disturbance .
2) Intractable vomiting .
3) Poor intake .

 Prevention :-

Vaccine + education .

*who need the vaccine ?

Pt with CDL .

HBV :_
_ has capsule inside it□cAg , eAg
Outside it□sAg .

*serious complications:_
Cirrhosis + HCC .
_ The virous found with high concentration in the blood , with lower percentage in all other secretion .
_ most important RF□ pre natal exposure
''trans placental '' to +ve HBV mother .
_ If the mother has +ve eAg □90% the percentage of the children to get the disease .
_ The immune prophylaxis (vaccine) □ 90% protection .
_ sAg is un predictable in the human milk □ so, advice her to continuo breast feeding .
_ sAg 7 6 month □ chronic .
…….
Pre rist to develope chronic infection is more . ……..
_ IP □ longer than HAV up to 4 months .
_ majority □ A symptomatic .
_ the …. Doesn't differentiate b/n the type of hepatitis '' it mean acute hepatitis ''

_ has extra hepatic manifestations : -
1. GN .
2. Poly arthritis +aplastic anemia .
3. Prolonge jaundice .

recovery rate :_
□ Less than 1 year □ 5%

95% □ develop peristant hepatitis , chronic hepatitis , cirrhosis , HCC .

□ 1_5 year □70%
□ …… □ 95%

Markers :-
_ eAg □ indicate infectivity and active replication .
*It correlate to DNA level .
*pt highly infectious .

_ eAb □pt is infectious but start to wash the virous out by developing Ab against e .
اعملو start to recover or notلوعاوز تعرف العيان
Chronic stage وبتلقاو في ال

* اول ما الزول يكون عندو eAb--->
نسبة ال eAg حتبدا تقل

_ sAg □
□ It serological marker of infection .
□ Found in all infected pt .
يعني لو عاوز تعرف روحك عندك HBV اعملو هو
□ Fall before symptoms appears .

_ sAb □ indicate recovery + protection .
_ cAg □ما مهم
_ cAb :
cIg M□ acute .
cIgG□chronic .
'' acute'' اهم واحد لانو بكون قاعد من بدري
يختفي sAgال حتى لمن course وبكون قاعد في كل ال

Mx :-

1. …….. as HAV .
2. Interferon alpha / lamivudine …….

Prevention :

- 2 inj ''vaccine +immune globulin ''

In different site ''IM'' in the first 12hrs after the birth .

*كلما كان بدري كان احسن

? Infected with HBV لو طعنتك حقنة

- If you are not vaccinated both as soon as possible 3 doses (0 , 1 ,6 month)

Immuno globulin temporary protection (3_6 months)

HCV :

- Worser than HBV .
- Same complication of HBV .
- No vaccine .
- Short IP (7__10 days) .

HEV :

- Recombinant HEV vaccine is highly effective .

AIDS

- Single strand RNA virous .
- Can transcribe ….gene into double strand DNA '' reverse transcriptase ''
- Outeside the capsule there is glyco protein .
- It composed of (lipid + protein) .
- Weak virous .
- It act by invading the other cells .
- Mainly vertical transmission in infant
- Maternal_fetal transmission at birth 25-30%
- Transmission via breast milk 10-15% .

*factors risk of transmission :-

1) advance maternal Dis .
2) high maternal viral load .
3) Low maternal CD4 count .
4) vag delivery and avoid it if mother is +ve HIV .
5) PROM .
6) chorio amnionitis .
7) STDs .

- Any PEM doesn't respond to feeding consider as HIV .
- Papules , nodules , ulcers dermatitis .
- Cause bacterial sepsis (maningitis , encephalitis , pneumonia) .
- Cardio myopathy , hepatitis .
- Herpes simplex and herpes zoster .
- Chicken pox .
- Interstitial pneumonitis .
- Hepato spleenomegaly .
- Parotitis .

- Clubbing .
- OM .
- *so , bacterial , viral , fungal infection .
- Malignancy .
- Wasting syndrome .

Diagnosis :-

o HIV Ab test usually +ve in the 1st 18 months after delivery if the mother is HIV +ve .
o Cannot benefit from screening in the 1st_18 month .
o So do it after 18 month diagnostic .

* False –ve :_
- in late disease .
- hypo gamma globulinemia .

o Screening before 18 month by PCR .

*2 واحد ما بنفع –ve

- PCR in the 1st 6 months :
  _ 6 month قبل PCR no screening
  _ 18 months قبل Ab no screening .

*If you suspected HIV Ab because the mother is HIV +ve :_
- wait 18 month for the Ab .
- اذا اختفت معناها ما عندو

*لو ما اختفت :_

- do PCR + CD4 viral load عشان staging ال.

o If you donot know the state of the mother better to do HIV Abs .
o Prevention is better than Mx .
o The risk of vertical transmission can be reduced to 1% by knowing the viral load and give anti retro viral theraby .

*Mx :-
combination of anti retro viral theraby .
=(Zidovudine) at 3rd trimester because it is the time to get infection .
= If you discover at labour give it
4 weeks course of anti retro theraby for the neonate .
avoid normal vaginal delivery , breast milk and …. STD .
…….
= P. carinii septrin for one year (till CD4 count decrease) .
= progresis improved with …..bination use of anti retro viral therapy .
= aim of nucleoside analogues prevent production and further replication (prevent viral elongation) .

= protease inhibitor prevent cleavage of gag – Pd poly Protein . use in :_
sever disease (advance ….)
HIV viral load .
low CD4 count .
e.g Nelfinavir .

_ combination 2 NRT is
NRT1 + NNRT
NRT1 + protease inhibitor
= Failure of response indicate :
- Poor compliance.
- Drug resistant .

= cost effective drug .

NLS

Condition that need basic life support :_

1. Asphyxia .
2. Sepsis . …….
3. Resp compromise .
4. Hypo volumic bleeding , burn , dehydration .
5. Convulsion .
6. Cardio pulmonary compromise .

= major cause of cardiac arrest in pedi hypoxia .

= maternal factor lead to placental insuffiency :

1. …….
2. …….. :

- O2 . - ……..
- Sucction . - canula .
- EIT . - umblical vein cathetr.
- Light sourse . - gloves .
- over head heater . - laryngeoscope .
  _ adrenaline .

aim make Pt warm and dry .

_ so you need to make good ventilation …

……

…….

……..

Head till chin left jawthrust

1. Keep Pt warm + dry most common treatment .
2. O2 .
3. Effective ventilation of resucitation.
4. Chest compression (cardiac massage) .

5. Adrenaline less common treatment .

correct position اسمو ترانس
= to make ……. air way .
= oro pharynx _ naso , pharynx _ trachea مستوى واحد
= head semi position .
Head head till . ……..

Good oro pharynx position by ….. The jaw jaw till thurst .
لو راسو كبير محتاج ترفع ال shoulder شوية
cover nose + mouth O2 mask bag mask ventilation .
= every 2 sec one breathing
One and two and breath .

 neonate .
Chest compression thumbinsertioning chest
 index + middle in the
Para sternal area .
*child one hand .
* adolescence two hand .
_ depth of finger1 1\2 of the …… diameter of the chest ……
1\3
…….
……. 3 compression fellow by one breath
لانو بعملوهم الاثنين سوا
(3:1)
_Old pedi (15:1)
Compression ventilation ratio
(30=2)

_ any step ثانية 30 عندي
_assess puls apical area/umblical cord الاحسن

Check resp effort (تنفس ولا لا)

……

في اول دقيقة بعد الولادة داير 5-10 دقائق

الدقيقة الاولى بكون 65% (ماcyanosis)

……

……..

_ … less than 60 need chest compression

_ more than 60 no need for chest compression .

two thumb technique

لازم يدك التانية تكون ورا عشان يكون ال

……. ……..

_ naloxane for floopy baby

……

Measels

- If the patient has skin rash you have to ask about --> the first eruption " were was it , when , colour , size , spread of the rash , ass itching or not ".
- Skin eruption is usually due to viral infection , and the viral infection is through the mucous membrane .
- You must ask about the vaccination in the history .
- If the mother say to you he give him antibiotic for fever , ask about :-

- Prescribed by whome .
- type of drug
- Who give him . منو البديهو الدواء

<u># case presentation</u> :-
1/ skin rash .
2/ cough , coryza and conjunctivitis .
3/ maculopapular rash on face and the trunk .
4/ white papule on erytheromatous base "koplit sign "

*it is the disease of overcrowdingat
koplic sign disappear at the day when the rash appear . it is pathogenomic for measles but its absence don't exclude measles .
The respiratory system is the most important complication .

<u>DD :-</u>
1- bacterial infection meningitis
Strepto scarlet fever
Staph toxic shock syndrome
2- viral most of the viral infection in the pediatric in early stage can attack the mucous membrane and cause slaf rash . eg : viral hgic fever

 rubella + roseola
 ifective mononucleusosis
 PV a + entero

- Roseola cause fever , rash , convulsion .

3- immunological diseases SLE + kawak

4- drugs whealing rash
 larg patches
 type 1 allergy

*In this infection the main immunological response early is by IgM , so , in the acute phase you may find IgM with specific level .

* Highly contagious disease .
* cause more mortality in the developing country .
* reservoir human .
* transmission respiratory airborne .
* Peak winter and spring .
* time 4 days before to 4 days after rash onset .

the reservoir is single strand RNA . so , we don't deal with it with PCR but we need an enzyme reverse transcriptase , to replicated and transcripted into DNA .
So the diagnosis is depend on the viral structural
Which have 6 major structural protein .

<u># the 2 most important in induction of immunity are :-</u>
1// H protein hemagglutinin .
2// " " fusin

TORCH infection

Intrauterine infection is a group of congenitally acquired infction that mean when the featus in utero or during his passage through the birth chanal . so transplacental or during the labour .

Others are :-
1- treponeame pallidum " syphilis " .
2- HIV .
3- Zica virus (latine amrica) result in microcephaly of the neonate .
4- varicella zoster paro virus (19) , measles , hepatitis , enterovirus and adeno virus .
LMCV lymphocyte chorio meningitis virus .

- They are putting together due to skin rash and ocular finding
- The baby born with 49 cm
- The indirect hyperbilirubinemia usually the cause is hemolysis or problem within the conjugation . While the direct bilirubinemia usually the cause is hepatic or cholestatic .

Toxoplasmosis

- Caused by protozoa (toxoplasma gondii)
- Definitive host domestic cat

- Transmission :

- Transmited to the baby Transplacentaly
Transmited to the mother by feco oral route ." oocyte will found in cat feaces " , contaminated soil , water ,under cooked meat + unpleasant milk .

- With increased GA : the risk of infection increased ^ The severity of infection decreased

- In the first trimester --- may cause abortion .
- " " second " --- has classical triad :-
  1- hydrocephalus
  2- intra cranial calcification
  3- chorio retinitis
- In the third trimester --- asymptomatic

- Case presentation :-

- Large head (the triad)
- Small baby
- Hepatosplenomegaly
- Jaundice
- Thrombocytopenia
- Rash

--- generally abnormal baby growth .

Mx :-

Pyrimethiamine for one year

-Other drugs sulfa diazine / leucovorin

Prevention :-

- Primary prevention : by increase the awareness of the mother about this infection which transmitted by cats .
- 2ry prevention : early diagnosis by proper history and examination .
- Tertiary prevention : prevent complication by using multi dysp team

  Neurosurgeon
  Pediatrician
  Nutritionist
  Ophthalmologist

Rubella

Family of taga virus., genus of rubi virus
-Transmission :
To the baby transplacentaly
To the mother through the respiratory secretions .

C/P :- baby gain weight , don't smile due to ocular problem .
generally it is developmental problem .

- In the eyes -- Cataract

Glaucoma

Salt and pepper retinopathy

- Ear -- sensory neural hearing loss .
- Microcephaly – neurological
- Cardio -- PDA

Pulmonary stenosis

- DM
- Others like toxoplasmosisi – jaundice , hepatosplenomegaly thrombocytopenia , IVGR .

Triad of congenital rubella syndrome :-

- Congenital cataract
- Deafness
- Congenital cardiac disease esp PDA

Mx :-

- Supportive nutritional
- Send him for ophthalmological department

في الحقيقة ما بركبو ليهو عدسة لكن بدو هو نظارة
to modify their vision and development .
active rubella is poor prognosis .

prevention by vaccination : not found because it is not common .
Mummbs , Measles , Rubella vaccination at the 9th month .

CMV

Family of herpes virus " 5 " هم 8 افراد
Transmission :-
- The baby --- transplacentaly

Contact with vagina during delivery or breast milk after delivery .
- Body fluids

C/P :-
- Usually asymptomatic at birth .
- Febrile convulsions --- abost the fits
- Delayed milestones
- Brain scan : periventricular calcification .
  Dialated ventricle
- Some to the previous one
- In the eyes : retinitis
- In the skin : PPE , jaundice. - Microcephaly

Diagnosis :- 1/ culture from urine , saliva , secretions 2/ PCR …
* generally it is developmental problem

Mx :-
1- good nutritional support
2- all the herpes group are respond to acyclovil and gancyclovir .
Prevention :- by immunization

Herpes simplex virus

Herpes virus 1+2

Transmission :-
الام بجيها برجم في الاسبوع الاخير
During passage of the baby through the canal the visicle rupture .
The disease appear in the 5th or 6th day after the incubation period end .

C/P :- usually a symptomatic but has 3 pattern :-
1- localize to skin eye and mucosal
- Skin – vesicular lesion on erythematousbase .
- Eye – kerato conjunctivitis
  Cataract
  Retinitis
- Mouth – ulcer

2- CNC disease – causing encephalitis or panencephalitis .
3- disseminated disease – characterize by end organ damage " renal , lung , liver , intestine , CNS , adrenal , skin , eye , mouth " every where .

Diagnosis :-
-From the mother – culture from the lesion
-From the baby – lesion , CSF , nasopharynx

Treatment :-
1/ respond to a cyclovir and gan cyclovir .
2/ supportive management .

Syphilis

- Organism – treponema pallidum .
- Transmission – transplacental + sexual activity .

C/P :-

1- majority are asymptomatic at birth

2- early congenital syphilis – age of 1-2 months

- Maculopapular rash
- Reticuloendothelial system involvement .
- Osteochondritis + retinitis + meningitis .

3-late congenital syphilis – after age of 2 yrs

- Hutchinson teeth
- Mulberry molars
- Perforated hard palate
- Phagades – fissure in the skin around the mouth
- Saber shins
- Interstitial keratitis
- Saddle nose
- Sensory neural hearing loss

Mx :-

Penicillin for short period

Overuse of penicillin decrease the incidence of problem .

the first DD for TORCH is neonatal sepsis

Whooping cough

Pertisus and parapertusis

-Clinical feature :-
Pt less than two months has
- Cough
  Repeated cough without taking a breath
After he push every air out he take breath (whoopy).
C/P :- depend on seveity
يا امو بكح كيف ؟؟؟؟
Usually a febrile
Have high mortality .

Complication :-
1- Brain hemorrhage
2- eye hemorrhage
3- cyanosis
4- respiratory failure
5- rectal prolapse due to ^ abdominal pressure with cogh .

Diagnosis :-
1- Suspension --- بالكحة
2- Hx and exam
3- throat swab
4- culture

Mx :-
Macrolides :-
1- azithromycin --- 3doses
(اي نسبة دهون بتمنع امتصاصو و تاثيرو عشان كدا بقول للام م ترضعيو الا بعد ساعتين) .
Solid 2-3 hours

Fluid 1 hour

2- clarithromycin

Git upset bid

3- erythromycin

6 hourly 10 mg / kg

problems of macrolides :

- Gastric outlet obstruction
- Pyloric stenosis

Prevention :-

Vaccination

nephrotic syndrome

-neophrotic syndrome related mainly to the proteins.
-in normal individual: there is minimal secretion of protein in urine and this is not more than 150mg/ 24hrs.
-whenever the proteinuria exceed 150mg this called abnormal proteinuria.

-there are non-pathological causesbcan lead to proteinuria:-
1.exercise
2.fever
3.emotional stress

-other causes (pathological) of proteinuria are glomerular disease which are either :
-primary glomerular disease.
Or-secondary glomerular disease
*nephrotic syndrome is a syndrome that characterized by:-
1.massive(heavy) proteinuria
2.hupoalbuminaemia (<2.5g/dl)
3.generalized oedema.
+_ 4.hyperlipidemia.

-massive proteinuria is called nephrotic range proteinuria.
-generalized edema is due to ed oncotic pressure caused by hypoalbuminaemia.

*pathogenesis:-
-the pathogenesis of nephrotic syndrome is related to cytokines theory.
*abnormal cytokines lead to dysfunction of glomeruli to refiltrate albumin.
-nephrotic syndrome common in males.

-age: 2-6 yrs

*pathophysiology:-
-proteinuria:
-there is increase in glomerular capillary wall permeability and due to negative charge of these protein so protein is losing in the urine
-mainly selective proteinuria (85%).
-degree of proteinuria: mild -moderate -or severe.

-generalized body swelling:-
-due to oncotic pressure caused by hypoalbuminaemia, while edema in nephritis is due to fluid retention.
-hyperlipidaemia:
-due to hypoalbuminaemia :the liver start to increase synthesis of protein and one of these proteins is lipoprotein.
-due to defect of lipoprotein lipase enzyme there is in cholesterol and triglycerides.

*clinical presentation:-
1.generalized body swelling: (main ……..)
-in :face, upper limbs, abdomen(ascites), lower limbs, pleural effusion, scrotal swelling in males and sometimes sacral edema.
*anthrax= generalized edema
2. change in urine color: urine is frothy urine due to proteinuria.

*causes:-
1.primary causes: due to glomerular disease:
-minimal change: account for more than 85% in children
-focal segmental.
15% - mesangio capillary.
-membraneous.
-membrano-proliferative.

2.secondary causes: due to disease(systemic):-
-DM.
-SLE.
-sickle cell disease.
-parasitic infection: malaria or schistosomiasis.
-viral infection: hepatitis B, C and HIV.
-certain drugs: pencillamine and gold therapy(use in RA).
-amyloidosis.

***investigations:-**

1.urine analysis: (mainly for proteinuria).
-frothy urine.
-heavy proteinuria: +++ and more.
-and you may find evidence of UTI (pus cell).
-no RBCs and cast.

*how to assess heavy proteinuria??
a.24 hrs urine collection(……..).
- significant: >3grams.
b.urine for albumin /creatinine ratio
-significant: >2
c. by surface area:
-significant: >40/m2/hr
2.serumalbumin:
-less than 2.5 g/dl
3.lipid profile:
-cholestrol +triglycrides high
-you can do LDL / HDL.
(assist investigation)
4.investigation for secondary causes:-
-RFT.

-abdominal U/S.
-RBC in diabetic pt.
-sickling test in SCD.
-ANA for SLE.
-screening for viruses.
-schistosomasis screening.
-s .ca2+ for hypocalaemia.
(كتير ca2+ بطلع مع البروتين, فاذا طلع بروتين كتير بطلع ca2+لانو ال)
*there is relation between nephrotic syndrome and hypocalcemia.
5.renal biopsy:-
-in certain cases.
-indications:
a. high BP + microscopic haematuria.
b. ……. Haematuria in nephrotic (to know what type of histopathological changes IgA , minimal)
c.steroids resistant.

*complications:-
1. infection: there is secretion of immunoglobulin in the urine with protein lead to immunocomprimised state.
Also: edema is good environment for b.infection
The most important organisms are the encapsulated organism, e.g: H.influenza, E.coli , pneumococcus , and meningitis.
The most common infections are:-
a.pneumonia.
b.UTI
c. peritonitis.
d.cellulitis.
2. liability to thromboembolic phenomena:-
-this due to loss of anti-thromboplastin
- 3 in urine lead to hypercoaguble state (DVT and PE) .
3. hyperlipidaemia.

4.renal failure: in few cases and esp. if the pt is not treated well .
5. complications of drug therapy.

*management:- (admission
1. you have to encourage movement to prevent DVT ,but in acute nephritis bed rest.
2.dietary management : restrict salt intake ,but donot restrict proteins.
3. no need for antibiotic therapy in nephrotic syndrome unless there is an evidence of infection.
4. the use of diuretics in nephrotic syndrome concides to 30% of pts .(use small dose of diuretics).
Use diuretics if there is:-
a.respiratory distress.
b.massive scrotal edema.
c.oedema not responds to steroids
5. specific treatment of nephrotic syndrome in steroids.

* steroids:
-use prednisolone 5mg.
-regimen : use 6 month regimen.
-regimen of 6 monthes:-
-according to the surface are.
-use 60mg/m2 surface area daily divided doses.
-then you have to follow-up your pt for response.

*response:-
-clinically: edema start to be subsided.
-labrotary: proteinuria start to be corrected and s.albumin start to increase.
Whenever the pt achives this state (clinically and labrotary response): this called remission.
*remission: is state of clinical and labrotary response to steroid.

This memission occurs in the first 10 -14 days after starting steroid.

Remission is an indicator of prognosis:

-the early the remission, the good the prognosis.

-the late the remission , the poor the prognosis.

So, the regimen of 6 month is:-

Start by 60mg/m2 daiy divided doses for 4 weeks.
Then 40mg/m2 alternative divided doses for4 weeks.
Then 30mg/m2 alternative divided doses for4 weeks.
Then 20mg/m2 alternative divided doses for4 weeks
Then 10mg/m2 alternative divided doses for4 weeks.
Then 5mg/m2 alternative divided doses for4 weeks.
(tabering of steroids)

This regimen prevent the relapse.

*relapse: the presence of clinical symptoms and signs + labortary findings for next time.

The relapsing rate in nephrotic syndrome is unpredictable and high (85%).

*terminology:-

Steroid sensitive :there is response to steroid therapy.

Steroid resistant: you give adequate dpse of steroids for adequate time (full 4 weeks) but still there is clinical and labortary evidence of nephroyic syndrome.

Steroid depeendent: the pt response to steroids, but while you start to ……. Or to stop steroids symptoms and signs start to appear.

*steroid dependent needs steroid therapy for long time.

(مثلا بحيث60, 40, 30 والمريض ماش كويس ,لكن لما حصلت 20الاعراض بدت تجيو تاني ما حصل ليو dependent 30ال علي ال 20مل لكن ما علي)

#in steroids resistant pt:-

-renal biopsy is indicated.

-use cytotoxic drug: cyclosporineA or cyclophosphomide.

*indications for cytotoxic drugs:-

1.steroid resistant.

3.non-minimal causes.

#complication(or SLE) of steroids:-

1.DM.

2.HTN.

3.cataract.

4.obesity.

5.suspectabillity to develop infection because steroid interfere with immune system.

- pallor, jaundice -> haemolysis. Fatigability -> decreased amount of O2 - spleen enlarged but not palpable.

* the spleen become palpable when is 3 times of normal size.

* severe anemia detect by severe pallor.

* Any pt come to you unless there is HF don't do transfusion unless you complete investigations (CBC) .

* neutrophil are neucleated cell

* Reticulocyte is neucleated cell.

* the test read reticulocyte as neutrophil that why you may find very high WBCs.

- ↑ esinophil -> allergy/parasitic infections

lymphocytosis ->cold/ malaria neutrophil if its increased may be otitis media.

* microcytic hypochromic may be thalassemia or sidroplastic anemia.

Auto haemolysis test :

Any blood cell intubate 37° 48 hour,
5% of the cells will be destroyed.

If hemolysis > 50% of cells this means cell membrane defect
* management:
Removes the organ that destruct and leave the spherocytes circulating in the blood.
* vaccinate pt if not vaccinated when you do splenectomy.

* G6PD deficiency: -
G6PD give H2 which keep glutathione in the reduced form.

sex link recessive :
Females are carrier, males are affected pt.
there are many Types :

Diabetes mellitus

د.بشير الناعم

Fanconi syndrome: characterized by:

1. Hypophosphatemia

2. Glucosuria

3. Aminoaciduria/ proteinuria

4. Hyperphosphaturia

5. Normal Ca

6. Hypocalcemia

8. Dehydration

9. Increased PTH

10. Increased AKP

(Fanconi syndrome = renal tubular acidosis)

* definition of diabetes mellitus:

Metabolic syndrome characterized by:

- Hyperglycemia.

- Glucosuria.

- Symptoms in form of polyuria, polydipsia and polyphagia. Caused by problems in insulin lead to disturbance of metabolism of CHO, lipids and proteins

Problems of insulin is either:

- in secretion: either absolute deficiency (type1) or relative deficiency

- in action: resistance.

=> disturbance of metabolism:

If there's no insulin (anabolic hormone) the body catabolizeproteins and lipids (this increase gluconeogenesis) and even starts glycogenolysis.

* why the body do that?? Because there's no insulin which is anabolic hormone.

Anabolic hormone: a hormone that is preventing lipolysis and enhance protein synthesis and enhance glycogen formation.

Classification of diabetes mellitus:

1. Type 1
2. Type 2
3. Type 3
4. Other forms of DM

- TYPE 1 DM :-

There's nothing called insulin dependent DM خلوها

type 1 diabetes mellitus is due to ABSOLUTE insulin deficiency

It's an autoimmune disease and can be associated with other autoimmune diseases:

- Addison disease

- Hashimoto thyroiditis

- Vitiligo

- pernicious anemia

- Celiac disease

- Inflammatory bowl disease (more => UC)

- Autoimmune hepatitis or chronic active hepatitis.

* it is the common type in pediatric

Commonly in children their age more than 6 months, but it is ranging from 6months up to 40 years .

TYPE 2 DM :

It's either due to:

- Absolute insulin deficiency

- Or relative insulin deficiency + resistance .

It's not common in pediatric, it's a disease of adults

It can present as young as 8yrs

Now it's new syndrome in pediatrics and this is due to current changes in lifestyle which make children more prone to be obese => this obesity is the commonest cause of type2 DM.

TYPE 3 DM : Is gestational diabetes mellitus.

Other forms of diabetes mellitus :

- **Neonatal** diabetes mellitus

- **MODY** (maturity onset diabetes in youth)

- **Cystic** fibrosis related DM _ cystic fibrosis lead to exocrine and endocrine pancreatic dysfunction.

- **Chronic** pancreatitis related DM

- **surgery** related DM (surgery can cause trauma to the pancreas or surgery that removes the pancreas out)

- **Medication** related DM : (steroids_ cyclosporines _ asparaginase)

\In pediatric our issue is neonatal DM and MODY.

Neonatal DM: occur during the first 6month of life and do not occur after 6month (inherited as autosomal dominant + single gene mutation)

Has 2 types:

1. Syndromic : baby has other syndromic features

2. Non-syndromic : baby is normal

Again it has 2 types:

1. **Transient** : give them insulin and then they improve => remission and diabetes disappear but they have 50% Chance to develop diabetes again.

2. **Permanent** : they need insulin for life.

Presentation: polyuria and polydipsia

دي قصة حكاها الدكتور كيف تعرفهم في طفل عمرو اقل من ستة شهور؟؟

In 2007 the mother came to ER and the baby is dehydrated and she said : my child wet his pampers frequently and I find it wet very heavy => polyuria.

And my child frequently suck my breast => polydipsia

And may say I have a child I lost him by the same presentation.

Then the consultant request RBG and the result was 600.and till 2014 he comes for the follow-up and his age at that time 6-7 yrs.

MODY:

Inherited as autosomal dominant, so you find (father, mother, grand mother) has DM so +ve family history in MODY .

MODY is a single gene mutation but type1 and 2 are multifactorial and not due to single gene mutation (polygenic).

It has different types 1-2-3-4-5-6

The age of MODY is adolescence and could be mild or severe.

=> : is a mild form of MODY and do not need treatment only follow up .

* Etiology of diabetes mellitus :

Varies according to the type:

TYPE 1:

- Genetic predisposition.

- Environmental factors.

Genetic predisposition:

- if the mother has DM the risk is 3-4 %

- if the father has DM the risk is 9%

- if both the mother and the father have DM the risk is 12-13%

So it's not inherited but your risk to have DM is increased when one of your family have DM. Because the general population risk is 0.05%

Environmental factors:

All environmental factors known to be a cause, so you have to differentiate between a causation and association.

Causation:

Bacteria that present in tonsils => it's the cause of tonsillitis .

If you have TB and you develop chest problem it is TB because of that mycobacteruim TB.

Association (Theories) :

Theory is not a cause of DM.

1. **Cow** - Milk - Protein Allergy (Theory):

is not a cause of DM, but it increases the chance of developing DM.

Study:

- They compare children who breast feed from their mom till 6 months and then they recieve supplementary feeding after that with other group of children who strat food &formula from day one.

Result:

- They noticed that those who start formula before 6 months are more prone to develop DM, But at the same time children who recieve breast milk again they will develop DM ..

So, be having a breast milk for 6 months is not eliminate the risk of DM, but it reduces the risk to DM.

(Breast feeding for short time -> more prone to Dm).

2**. Vit. D** deficiency -

with change of lifestyle children are not exposed to the sun, so the kids come at age of 1year with Rickets

to vit. D is not important for 70 absorption of Ca 2+ &r development of bon only, but it has other main many functions called Non-skeletal functions of vitamin D.

Studies & Relation of :

- Infections and Vit D deficiency

- Cancer prevention of Vit.D

- BP and Vit. D deficiency.

- DM and vitD deficiency

Studies found that :

pt with Vit.D deficiency are more prone to develop immune of related diseases and one of them is DM.

=> Dx by Vit. D level which is a very low in these children.

In world: 67% of children with DM have Vit. D deficiency, but

in sudan -> 10%.

so, They supplement their children with Vit. D at time of birth to keep Vit.D In a normal level, but inspite of that the incidence of DM doesn't decrease. And so vitamin D deficiency is just association with DM and not a cause.

Vit D supplement :

2000 units during pregnancy.

40 drops till 800 units in children after birth

3. **Hygienic** theory:

They hypophize that children were very clean, not exposed to anything so any infection (esp. viral) can lead to DM in those children (these children more prone to DM due to infection) So, Infections are one of the risk factors to develop DM .

TYPE 2 DM:

- most important cause is obesity

- +ve family history

- not all obese children will develop diabetes and this depends on the quality of genes.

--> family with bad genes their obese children will develop DM

--> family with good genes their obese children will not develop DM

TYPE 3 DM : this is related to the gestation

MODY:

Neonatal DM: they're not something preventable and they're related to single gene mutation (monogenic).

DIAGNOSIS:

By history, examination and investigation .

=> our issue is type 1 DM.

-History:

=> type1 DM is very rare to discover accidentally because in type 1 you must have symptoms .

HPI: classical symptoms {polyuria, polydipsia, polyphagia and wt loss}

Nocturia: form of polyuria and may be the first symptom.

Fatiguability: is a very important symptom.

The three commonest presentation are:

Polyuria or may present as nocturia, Polydipsia and fatigability

Fatigability and nocturia may be the early symptoms => prefer classical symptoms . The pt may present with these symptoms or may present with symptoms of DKA which are:

- Nausea and vomiting

- Diarrhea
- Abdominal pain
- Abnormal breathing (kussmaul breathing)
- Confusion and loss of consciousness (coma), lethargy and fever.

All these symptoms may be said by the mother or you must ask about them.

Ask the mother how many days/wks having these symptoms .

You will get DM if 80-85% of Beta cells of pancreas destroyed .

If 70% are destroyed you will not get DM but recurrent infections (esp. tonsillitis and malaria)or any other stressful condition may aggravate the symptoms of DM and the patient may present with polyuria and polydipsia or even may present with symptoms of DKA in this condition. WHY does this occur?? BECAUSE in the stressful condition glucagon is secreted which is a hormone that counteract insulin.

Hormones that counteract insulin :

1. Glucagon
2. Growth hormones
3. Cortisol

DD of **polyuria** and **polydipsia** :

1. DM
2. Diabetes isipidus (DI)
3. Chronic renal failure- that caused by obstruction*

4. Hypokalemia

5. Hypercalcemia

* polyuria occur in CRF with obstruction because obstruction will affect the reabsorbtion of water, so after you relief the obstruction the ability of the renal tubules to reabsorb the water is affected => so the kidney can't reabsorb all the water => transient polyuria.

* hypokalemia and hypercalcemia , in the kidney there're two regulatory hormones for water reabsorbtion => ADH and aldosterone. Hypokalemia lead to decreased production of aldosterone (some sort of aldosterone deficiency) and aldosterone stimulates Na and water reabsorbtion . So when it's deficient there's decrease NA and water reabsorbtion so there's polyuria.

Hypercalcemia affect the receptors of aldosterone and ADH in the kidney => some sort of resistance of aldosterone and ADH => decreased reabsorbtion of water => polyuria.

Past medical history:

Ask about **surgery** (pancreatectomy)

Ask about **chronic pancreatitis**

Ask about **cystic fibrosis .**

You must ask about **pregnancy** especially ask about **TORCH** infections particularly **rubella** because it **increases the risk** to DM.

Family history:

Ask about **consaguinity**

Similar disease

Autoimmune diseases (**very important point**)

Nutritional history:

It's concerned to type 2 DM.

In type 1 DM you can ask about:

1- **breast feeding** : because breast feeding for short period increase the risk of DM.

2- **vitamin D** due to its association.

3- **cow milk** and formula milk due to its association.

Drug history:

Ask about **previous** drugs (**steroids**).

Examination:

- general examination: if the patient not presented with **DKA** usually the examination is normal .

If the patient presented with DKA :

1. Patient may be **confused, lethargic** or **comatose**

2. **Vital signs are very important**.

You should measure the weight because all the treatment depends on the weight.

Vital signs in patient with DKA ;

- Pulse: to know the patient is in **shock or not**

- Temp: increased or normal

- Capillary refill: **hemodynamically stable or not!**

- RR: to know **pattern of breathing**

- BP: **decreased** in patients in **shock**

Pulse: **normal** in pts not in shock ,

rapid small volume of difficult to palpate => **shock**

RR: **kussmaul breathing**: rapid deep breathing => **DKA**

rapid shallow breathing => **pneumonia**

BP: **hypotension indicates shock** and the patient needs **fluid**

Temp: if it is **increased** may indicates **infection** which is a precipitating factor for DKA

If there's fever examine (ear, tonsils, throat and the chest) to know the source of the infection.

If the temperature is normal don't give antibiotics (it is not indication for antibiotics).

Examination for association:

General examination :

Pallor : to know if DM is associated with pernicious anemia or celiac

Jaundice: association of DM with autoimmune hepatitis .

Skin examination: for

1. **Acanthosis nigricans** (esp. in type 1 DM)
2. **Hyperpigmentation** for association with Addison disease
3. **Hypopigmentation** for association with vitiligo

*acanthosis nigrican seen in: **axilla, back** of the **neck** and **under** the **breast**.

DD for acanthosis nigrican:

1. Obesity
2. DM esp. type 1
3. Polycystic ovary disease
4. **GIT malignancies** , Ca stomach causes severe form of acanthosis Nigricans.

Hypopigmentation: may be generalized and this occur in severe types of Addisons disease

May be localized hyperpigmentation and this may occur in mild types of Addisons disease.

Area of localized hyperpigmentation:

Palms and soles

Buccal mucosa (lips)

Teeth

Crackle

Nipple

Umbilicus

Any scar

Thyroid examination:

For **goiter** => association with Hashimoto thyroiditis

Examination for complication:

GIT examination:

Abdominal distention as a result of paralytic ilues that caused by hypokalemia.

Or upper abdominal distention due to gastric dilataion.

Abdominal abscess=> one of the complication of DM is abcesses in a funny place.

Skin examination:

For fungal infections

Genitalia examination:

Vaginal infection esp Candida

{Candidal infections may be an early sign of DM}.

The rest of the examination depends on the history (clinical situation), if patient present with cough you should examine the chest and so on.

(pneumonia is one of the risk factors to DKA).

Investigation:

For DM:

1. RBG

2. FBS (both FBS and RBG used for diagnosis of DM)

3. Urine analysis (for glucose, acetone, pus cells).

4.HA1c: it's the percent of glucose binding to hemoglobin .

Normally there's percent of glucose binding to hemoglobin and this percent depends on the level of glucose in the blood.

So when glucose in the blood increases => increase the percent binding to hemoglobin => **HA1c**

HA1c is not a diagnostic criteria for type 1 DM. (Diagnostic for type 2).

Normal HA1c = 5.7%

Borderline = 5.7-6.44%

Level of DM more than or equal 6.5%

5. Antibodies: not done for every patient .

Antibodies that can cause type 1 DM:

1. Glutamic acid decarboxylase antibody (GAD) the most important one

2. Insulin antibody

3. Islet cells antibody

*note: we don't need to do GTT in type 1 DM.

<u>Other investigations:</u>

1. Investigations for association

TFT

Investigation for celiac disease (duodenal biopsy)

2. CBC: don't do CBC unless the patient is febrile

Hematocrit is increased due to dehydration, use it for follow-up

ولي شنو CBC دا لو في واحد عاوز يتفلسف ليك قال ليك إلا تعمل

3. RFT: very rare to develop AKI in diabetics in pediatric

4. Electrolyte: very important

Na and K , K may be normal or high (acidosis) , low .

Even you find normal or increased K the total body K decreased.

Na is low caused by:

1. Hyperglycemia causes pseudohyponatremia

2. Hyperlipidemia

* Criteria for diagnosis of DM:

1. RBG more than or equal 200 with the presence of one symptom at least OR

2. FBS more than or equal 126 with presence of one symptom at least

3. If the child present with FBS more than or equal 126 without symptoms repeat it the next day if also more than or equal 126 then it is DM

If RBG more than or equal 200 mg without symptoms search for stress factor that cause hyperglycemia. Why it is not DM? Because in DM when RBG reach 200 or more symptoms must be there.

So type 1 is not like type 2:

RBG =300mg

In type 1 symptoms must be there

In type 2 may be asymptomatic even at this level.

If the patient present with febrile illness and RBG =300mg it's not DM it's stress hyperglycemia

So absence of symptoms in presence of triggering factor is not DM.

* stress hyperglycemia can be caused by:

1. Infection (fever) esp tonsillitis

2. Convulsions

3. Any stressful condition.

Complication of DM:

The most common and most important in type 1 DM is DKA.

Diabetic ketoacidosis DKA

Can be either first presentation or complication.

First presentation in 15-75% of diabetics => depends on the community awareness about DM (in sudan 36%).

Definition:

Is a clinical syndrome characterized by:

1. Hyperglycemia
2. Glucosuria
3. Ketonuria

In addition to fluid, electrolytes and acid base disturbance.

Fluid disturbance = dehydration

Electrolytes disturbance = hypokalemia and hyponatremia

Acid base disturbance = acidosis

* Presentation

1/ abdominal pain

2/ vomiting

3/ level of consciousness may be decreased but even if it is not criteria for DKA

4/ breathing: rapid deep breaths- rapid shallow breathing indicates pneumonia (bronchiolitis)

5/ symptoms of DM

Abdominal pain is due to:

A/ acidosis

B/ dehydration (mesentric vascular ischemia)

C/ gastric dilataion

* any patient with DKA having a degree of dehydration.

By **examination**:

1. GCS: normal GCS doesn't exclude DKA because level of consciousness isn't one of criteria for DKA.

2. Pattern of breathing + odor of breathing

3. Signs of dehydration (ranging Signs)

4. Vital signs: most important

5. Weight: Rx depends on the weight.

Classification:

According to hemodynamic status :

1. **Shock** : either compensated or uncompensated shock (patient is weak and tachycardic).

A. **Compensated shock**: Normal BP + prolonged capillary refill = more than 3 seconds.

B. **Uncompensated shock**: hypotension + prolonged capillary refill

2. Severe dehydration: without shock

لو جاك البيشنت بي شوك اول شي تبدا تعالجها بعد تعالجها بيبقى البيشنت في سيفير ديهايدريشن بعداك تعالج الدي كي ايه

Which lab results help you to confirm suspicious DKA:

1. RBG more than or equal 200 mg

2. Urine analysis: acetone, pus cells => infection

3. ABG, VBG

PH less than 7.3

Bicarbonate less than 15 .

Ketonometer:

-in the blood measures β-hydroxybutyric acid which appear first and disappears first.

-in the urine measures acetone which may disappear after 24hrs- عشان كدا ما بعتمد عليو

If the patient is a known case of DM and present with DKA:

Ask about : dose and duration, site of injection :

Lipohypertrophy lead to decreased absorption of insulin

Source of Infection lead to abcess and then DKA.

Ask about fever (infection esp tonsillitis) examine the chest and site of injection this called sick day management الام بتتصرف مع اليوم البتجيو فيو الحمى

Treatment of DKA

1. Correct dehydration
2. Correct hyperglycemia
3. Correct electrolytes disturbance
4. Correct acidosis
5. Treat underlying infection or any other precipitating factor.
6. Anticipate prevent and treat complication of management
7. Treat the cause of DKA to prevent recurrence.

1) Correct dehydration: what type of fluid are you going to give?

How much? On how many hours?

When I change?

When I stop?

Type of fluid ; normal saline

Amount. A) shocked:

- uncompensated shock => 20ml/kg IV as quick as possible then assess vital signs if still shocked repeat till out of shock (up to three times)

- compensated shock => 10ml/kg over 1hr then asses vital signs after 1hr if still shocked repeat till out of shock (up to 3-4 times)

The patient is either presented with shock or with severe dehydration.

- if shocked treat as mentioned above, if you treat the shock => patient shifted to category of severe dehydration and then treat dehydration.

- if severe dehydration treat it and then proceed to the steps of DKA treatment.

يعني لمن تعالج الشوك كدا يا دوب البيشنت بكون بقى في اول خطوة لي علاج الDKA

B) severe dehydration:

1. Mild:

PH 7.3-7.2

Bicarb 15-10

Deficit is 5%

2. Moderate:

PH 7.2-7.1

Bicarb 10-5

Deficit 6-7%

3. Severe:

PH less than 7.1

Bicarb less than 5

Deficit 8-10%

(But we consider deficit 10% for any patient because of lack of facilities) .

e.g wt= 35 kg => deficit = wt ×100 (if deficit is 10%) = 35×100= 3500ml.

If you want to calculate the total amount of fluid you should calculate the maintenance

Maintenance: 1st 10 kg×100

2nd 10 kg×50

Then each kg×20

e.g wt = 35 => 10×100 + 10×50 + 15×20 = 1000+500+300= 1800ml

We will correct dehydration over 48 hrs so maintenance of two days 1800×2=3600ml.

Amount of fluid over 1hr :

= (deficit + maintenance) ÷ total hrs .

e.g weight = 35kg

Deficit = 3500

Maintenance of two days = 3600

Total hrs = 48

=> (3500+3600)÷48 = 7100÷ 48 = 147.9 = 150 ml.

=> this patient need 150 ml of normal saline every 1hr .

Drops calculations: how many drops of NS per hour

= (amount × 15) ÷ (60×total hrs) = drops/hr

e.g wt = 35 , amount = 7100 , total hrs = 48 hrs

=> Drops = (7100×15)÷ (60×48)= 106500÷ 2440 = 36.9 = 37 drops/hr

* When I change NS? Change NS to dextrose NS (DNS) .

I will change if:

1. If blood glucose drop rapidly => more than 100/1hr (normally it drops from 50-90 per hr)

(if glucose drops rapidly and you don't change to DNS this will change the osmolarity and this is a risk factor for cerebral edema)

2. If RBG less than 300 with normal NA change to 1/2 NS + D5% But if the NA isn't normal keep on NS .

If the NA is normal and you didn't change to 1/2 NS + D5% this will cause hyperchloremic metabolic acidosis => this is due to rapid dilution of plasma bicarbonate with saline.

(Decreased plasma bicarbonate lead to increased plasma chloride)

* When I stop fluid? Continue till the patient is out of DKA and can tolerate oral intake.

Don't stop abruptly but over 2hrs (decreased 50%).

2) Correct hyperglycemia:

You give insulin:

Type: regular

Dose: below

Frequency: every hour

When to change: if the patient is out of DKA .

Dose: 0.05-0.1 unit/kg/hr , you give 0.05 in some patients to avoid hypoglycemia especially:

1. If age is less than 3yrs .

2. If the patient is known case of DM وجينا الجرعة اديناو ليك بتقول امو

When I give insulin? 1-2 after starting fluids and this to: 1. Avoid cerebral edema

2. To avoid hypokalemia.

Route of insulin:

The best route is IV عندنا ما ودي

Either infusion pumb, syringe pumb or subcutaneously .

Infusion pumb:

توصل الجهاز مع سير الدرب وتدي الجهاز اوردرانو انا عاوز كدا في الساعة مثلا خمسة مل او عشرة

syringe pumb: العادية الحقنة زي ودي

Two types: 50 cc , 100 cc

بتخت ١٠٠ مل من النورمال سالاين و ١٠٠ وحدة انسولين سو كل واحد مل فيو وحدة واحدة انسولين

e.g weight = 30 and the dose is 0.1 = 30×0.1= 3 ml = 3 units.

يلا ال ٥٠ سي سي الخط بتاعها يساوي ١ مل (وحدة) وال ١٠٠ سي سي ٢ مل (٢ وحدة) طيب لو البيشنت وزنو ١٥ كج => ١٥× ٠,١ = ١,٥ ، طيب دي اديها كيف؟ ح اخت ١ مل من ال ٥٠ سي

سي و ١ مل من ال ١٠٠ سي سي دي كدا ٣ وحدات وبدل اديها في ساعة واحدة بديها في ساعتين، عادي ماف مشكلة

subcutaneously if there's no pumb <=

Because the patient dehydrated so decreased absorption you start by leading dose 0.3 then continue by maintenance dose 0.1/hr

When I stop insulin? I stop insulin when the patient out of DKA, and I shift for other regimens of insulin therapy . *

first ولو ,known case of DM الكان بياخدها لو كان اصلا dose ح ارجعو لل presentation تحدد ليو دوز

3) Correct electrolytes disturbance :

The problem is Na and K, Na treated by NS

K: either normal , increased or decreased

When start K? At the time of insulin

Dose: 40 mmol/litre or 20 mmol (ampulla) per drip

How long? As far as the patient on IV fluids .

When stop K? With stopping of fluid as the patient is out of DKA unless the patient is hypokalemic.

Contraindications of K?

1. If the patient is not passing urine (no urine output AKI which is rare in DKA in pediatric

2. K more than 6.5

3. If there's ECG changes of hyperkalemia (tall tented T-wave)

If the patient is not passing urine you should examine the bladder because:

If the patient is unconscious may develop urine retention

Patients may also develop Neurogenic bladder (long-term DM) .

- increased may be just a compensation for acidosis and it retains to normal after 1-2 hrs so you need electrolytes.

4) Correct acidosis: it's corrected spontaneously with correction of dehydration and hyperglycemia.

There's no benefit for giving bicarbonate to treat acidosis in diabetic child . (Increased risk for cerebral edema) .

Bicarb is only given if there's persistent hyperkalemia.

5) Treat underlying infection or any other precipitating factor:

Infections: febrile patient with increased TWBC => give antibiotics.

Afebrile patient with increased TWBC (not infection so don't give antibiotics) differential increased neutrophils , search for non infectious cases for increases TWBC which may be one of these:

1. Stress

2. Hepatic failure

3. Corticosteroids

4. Adrenaline

Other precipitating factors => treat accordingly.

6) Anticipate, prevent and treat complications of management:

The most two important complications are hypoglycemia and cerebral edema.

1. Hypoglycemia : treated by shifting fluids from NS to D5% .

If the patient still hypoglycemic => D7% or 7.5%

If the patient still hypoglycemic => D10%

If the patient still hypoglycemic => D12%

If the patient still hypoglycemic you decrease dose of insulin by 20%

Don't stop insulin never ever because your target is not glucose but your target is to correct acidosis.

And when you stop insulin you will never correct acidosis so don't stop insulin.

Diagnosis of hypoglycemia:

In DKA: RBG less than 90

In DM: RBG less than 70

2. Cerebral edema: risk factors :

- Severe acidosis

- DKA at home for Long time

- Age less than 5yrs

- Increased urea

- Very large amounts of fluid

- Early insulin

=> usually occurs in the 1st 4-12 hrs

Dx: symptoms:

1. Severe headache

2. The patient from the start may present with vomiting and then stop and then come back and this is called recurrence of vomiting which is a symptom of cerebral edema.

3. The patient from the start don't present with vomiting and vomiting develop during treatment and this is also a symptom of cerebral edema.

Signs :

1. Irritability

2. Increased BP, especially diastolic BP

Cushing triad : decreased pulse, increased BP, widening pulse pressure.and this triad is associated with increased ICP.

3. Incontinence

4. Decorticate and decerebrate of position

5. Kussmaul breathing change to chimy stock breathing

6. Cranial nerves palsy (convergent)

Investigations :

No investigations

We do CT scan but not to confirm diagnosis because normal CT scan doesn't exclude cerebral edema , we do CT scan to look for associated complications : hemorrhage, thrombosis and infarction.

=> Treatment of cerebral edema:

1. Exclude hypoglycemia
2. Elevate the bed at the head site
3. Decrease the rate of fluid by 50%
4. Give the patient oxygen
5. Give manitol and if not available give hypertonic saline (3% saline) .

** When the patient is out of DKA? **

Clinically:

1. Conscious
2. No dehydration
3. No acidotic breath

Laboratory: ABG, VBG:

1. PH more than 7.3
2. Bicarb more than 15

Treatment of DM

Treatment of DM depends on the presentation of the patient:

1. Patient presented with DKA=> diseased

2. Patient presented without DKA:

- Insulin therapy

- Education and counseling

Insulin therapy: Types of insulin:

1. Long acting insulin

2. Intermediate acting insulin

3. Short acting insulin

4. Rapid (ultra rapid) acting insulin

5. Mixed insulin.

1. Long acting insulin :

e.g Lantos (acts for 24 hrs)

Levimir, Deltomir (daily)

Leglodet (can act up to 72 hrs)

So you can give it every 24 or 36or 72 hrs

2. Intermediate acting insulin :

e.g NPH reaches the peak after 6 hrs , start after 3hrs and can act from 12-18 hrs

3. Short acting insulin: sometimes called rapid :

=> soluble (regular) insulin => start after 1/2 an hour and can act for 6-8 hrs and reaches peak after 3hrs

4. **Rapid acting insulin** (if you called short acting rapid then you call this ultra rapid) :

e.g Novorapid, apidra , lespro

Peaked after 1.5 hrs and cleared after 4 hrs .Don't give it alone give it with other type of insulin.

5. **Mixed insulin**: NPH + any other type of insulin

(soluble regular insulin + NPH)

Methods of delivery of insulin:

1. Syringe _ most available

لازم تعرف الخط الواحد فيو كم وحدة

2. Pen

3. Insulin pumb

دي عبارة عن جهاز صغير فيو حتة للانسولين وفي سير بتركب في اليد أو ال abdomen او ال thigh بكون SQ وفي منو different types:

Fully automated: this is called CGM continuous glucose monitoring <= device, it's device connected to insulin pumb , it measures blood glucose every 5 minutes and sends informations to insulin pumb => .calculate the dose accordingly

not automated <=:

دي بحتاج تشتغل عليها تقيس السكر بعد كل وجبة وتقول ليها انا اكلت كدا جرام كاربوهيدريت وسكر كدا وتحسب الجراعة براها وتضغط اوكي تبدا تشتغل.

<=Another one is with CGM

لكن انت بتحسب ليها السكر بتاعك كم وتدخل الرقم بتحسب براها

<=Another one is with CGM

دي بتحتاج بس توصل ليها عاوز تاكل كم من الكارب وبتشتغل براها تاني بل بتوقف لو الجلوكوز نقص

<=Another one is with glucagon

دي برضو بتشتغل براها ولو السكر نقص بتقيف ولما السكر ينقص بتفرز جلوكاغون عشان ترجع الجلوكوز للنورمال.

*Regimens of insulin:

1. Basal bolus regimens (most common worldwide)

2. Twice daily regimens (2 choice regimen, most common here)

3. Pumb (the best one) - discussed

The regimen depends on the available insulin and the socioeconomic status of the family.

1. Basal bolus regimens: To mimic the normal physiological insulin secretion .

--> at the day because any food is absorbed after 2hrs=> you need insulin to store glucose in the liver and muscles .

--> between meals : you need insulin to regulate amount of glucose that comes out of the liver between meals.

--> after meal you need insulin to cover glucose that comes from those so :

Basal => at the day + between meals

Basal => after meals (each meal)

- for basal use Lantos.

- for bolus use Novorapid

* you give 30 => Lantos 15 once per day and Novorapid 15 three times per day .5 after each meal .

2. Twice daily regimens:

You give mixed insulin. 2/3 the dose in the morning and 1/3 in the evening

In the adolescence (more than 8 8yrs) you give extra dose after dinner .

Don't give mixed insulin for less than 3yrs because they may develop severe hypoglycemia so you avoid this by giving NPH only .

* How to store insulin??

If you store the insulin improperly it will be not functioning.

الام تقول ليك اديناو الجرعة لكن السكري ابى ينزل نهائي

تحفظ الانسولين في تلاجة ممكن يقعد تلاتة شهور يتخزن في الرف

او في نقاع الزير تخت الانسولين في كيس اول حتى يتخت في الموية عشان الموية ما تخش الفتيل بالاسموزية.

ممكن تخزنو في حفاظة تخت التلج وتكب موية عشان الانسولين ما يجمد

ممكن برضو تخزنو في قربة

*How to prepare fthe dose?

طلع الفتيل و رجو براحة او ما ترج اصلا

<=take units of insulin accordingly to the dose and then inject .

لو تمنية وحدات شيلهن وما بتشتغل بالهوا والموضوع دا

*where to inject Insulin?

1. Arm : away from the joint
2. Abdomen: around the umbilicus
3. Thigh: lateral aspect
4. You can inject at the buttocks

Don't inject at one site but rotate between these sites to avoid lipodystrophy

Complications (S/E) of insulin:

1. Hypoglycemia
2. Lipodystrophy

Hypoglycemia: by definition: glucose less than 70.

Symptoms and signs: fatigue,tremor , palpitations and late patient may develop convulsions.

معاهن جوع وزهج.

Rx :

Measure glucose by glucometer

Juice, if not present give 100 ml of water with sugar

كباية شاي موية فيها ملعقتين سكر

After 20 minutes measure the glucose:

- if improve give him meal with juice at the same time because there's delay in absorption.

- if not improved repeat again .

If the patient become unconscious give him a tablespoon of heavy jam , repeat after 20 minutes if not improved.

After all this if patient not improved give him glucagon IM.

* Hypoglycemia caused by:

1. Take the dose but not take the meal

2.increased dose

3. Exercise

* But DKA caused by:

1. Missed dose

2.decreased dose

3. Stressful condition

Lipodystrophy:-

May be lipohypertrophy or lipohypotrophy

Lipohypotrophy is very rare because the insulin used is human insulin.

Lipohypertrophy usually happens and lead to delayed absorption and infection (abcsess).

The patient prefer to inject Insulin at the area of lipohypertrophy because it not painful and you have to prevent him from this. The area of lipohypertrophy is at increased risk for infection due to decreased blood supply to this area and due to this there's no pain.

So, it's very important to ask about the site of injection in known case of DM because it can be the cause of DKA by:

1. Being a source of infection

2. Decreasing insulin absorption .

=> to avoid all these problems of area of injection you must rotate the site of injection.

Dose of insulin for newly discovered patient:

1. Patient less than 3yrs: give NPH because it takes long time to work 3hrs and this to avoid hypoglycemia.

2. Patient 3-5 yrs give mixed insulin 0.05 unit/kg

3. 5yrs up to puberty give mixed insulin 0.75-0.1 unit/kg

- during puberty 1-2 units of mixed insulin.

Education and counseling:

1. About the disease

2. About the treatment ; insulin

- How to store insulin

- How to prepare dose

- whereto inject Insulin

- complication and side effects of insulin

3. About glucometer:

- How to use glucometer

- what time of the day we need to check blood glucose.

4. About diet:

- what to give what not to give

5. Hypoglycemia

- definition, symptoms and signs , glucometer, how to treat .

6. About the importance of follow up and date the 2nd(next) visit.

* Diet: according to:

1. Socioeconomic status of the family, if poor eat any thing

2. Age of the child : less than 3yrs, no specific feeding pattern

ياكل مع اخوانو اي حاجة ياكلوها

3. More than 3yrs start to restrict.

Restrictions:

Avoid high sugar food and drinks and soft drinks (cola, 7up..etc)

Can eat one banana or one apple or one orange or even ice cream but only at special occasion or not every day .

ما تحرم الطفل بس كمان ما كل يوم

ياكل اي حاجة في الصينية بس كمان ما يركز على حاجة واحدة

باختصار الاكل

حاجات يأكلها زي ما عايز مثلا البروتينات

حاجات نص نص زي الرغيف والفواكه والكسرة والبسكويت

حاجات ممنوع منها نهائي زي البساطة بالشوكولاتة إلا يكون عايز يلعب كورة

The objectives of treating child with DM are:

1. To prevent complications (acute and chronic)

2. To all patients to grow normally(normal growth and puberty)

e.g height:

<=in the 1st yr increases by 25 cm

<=in the 2nd yr increases by 12.5 cm

<=in the 3rd yr increases by 7.5 cm

<=from the 4th yr till puberty then each year increases by 5-6 cm.

Follow up:-

1. Measure weight and height of the patient.

2. Check the records of the mother at home

3. Look for infections: for candiasis in mouth, in nails, in vagina for vaginal candiasis, Examine the chest for chest infection, exami e the tonsils, look to wounds if present.

4. Ask about school performance

5. Ask about social problems or economic problems

6. Ask about insulin : storage, injection and so on, to check if they know or not

7. Do HA1c to check control of DM the last three months.

طلع عالي معناها بتكضب عليك وما بتديو اما مشاكل HA1cلو الام قالت ليك بتديو الانسولين و مالية أو اجتماعية

Other complications of DM:

Acute: DKA and hypoglycemia

Chronic:

<=macro: absolutely rare in children

<=micro:

1. Retinopathy

2. Nephropathy : nephropathy and retinopathy are more common in pediatric than neuropathy

3. Neuropathy => need uncontrolled DM for longtime

Screening for complication in type 1 DM:

screen if:

1. The duration of DM is 3-5 yrs

2. Child entering puberty

3. Age reaching 10 yrs

لو السكري ليو خمسة سنة اعمل سكرين لكن لو ليو تلاتة سنة شوف الكونترول كويس ولا كعب لو كويس خلي السكرين بعد خمسة سنة ولو كعب اعمل ليو سكرين طوالي

(Good control <= HA1c less than 7.5)

لو واحد عمرو حداشر سنة وشخصت السكري قبل ستة شهور وهو دخل البيوبرتي طوالي بعمل ليو سكرين

ولو واحد عمرو تسعة سنوات وشخصتو بي سكري بعد سنة طوالي بعمل ليو سكرين عشان بكون حصل عمر عشرة سنة

#Sick euthyroid syndrome:

=> In stressful condition (any illness) such as DKA: T4 isn't converted to T3 but it's converted to reverse T3 which is an inactive form of T3.

=> so if you do TFT in patient with DKA you'll find increased reverse T3, decreased T3 , decreased T4, decreased TSH (TSH is normal in the beginning but decreased later)

=> why T4 converted to reverse T3 (inactive form)??Because thyroid hormone lead to increaseBMR and if patient in stressful condition we don't need to increase BMR so we don't need conversion of T4 to T3. After recovery T3,T4 and then TSH return to the normal.

So don't take TFT if the patient in DKA because it will give you false abnormal result due to sick euthyroid syndrome.

=> so check TFT and celiac disease after the patient is out of DKA

For thyroid : if TSH is normal i=> repeat every year

If TSH is increased and normal T4 => repeat every 6months

For celiac => repeat every year.

Hyperglycemia without ketones:

- Very common, patient present with hyperglycemia 400-450 but no ketones in urine, no acidosis and no dehydration.

=> **Rx**:

1. If :

- known case of DM => increase the dose by 10-20% of his primary dose , and you repeat 2-3 times and don't start treatment for DKA.

- If newly discovered => we start the dose from 0.7 unit/kg in children more than 5yrs and calculate the dose and start immediately, 2/3 in the morning 1/3 in the evening.

لو جاني نص النهار ح احسب ليو الجرعة بتاعتو واشيل عشرة لي عشرين في المية منها واديها ليو والمساء ابدا ليو جرعتو عادي

e.g pt wt = 30

30×0.7= 21 unit , 14 units in the morning and 7 in the evening .

In mid day: 20% of 21 is almost 4 units so give 4 units with meal and in the evening start the dose 7 units and continue.

1. Identify the cause and treat it:

- Didn't take the dose - Infection. - Eat too much .

Honeymoon period:

=> period of time shortly following the diagnosis of DM (wks- months especially 1st to 6th months and up to one year).

Criteria:

1. Insulin less than 0.5 units

2. Unexplained hypoglycemia

3. HA1c less than 6.5

=> **caused** by:

1. Hyperglycemia

2. Hyperlipidemia - hyperglycemia => block beta cells

=> Never stop insulin but give the smallest dose of insulin.

لو وقفت الانسولين نهائي الاهل بقولو الولد بقى كويس وبنسو انو عندو سكري ودا ممكن يدخل الطفل في كوما

Type 2 DM in children:

Risk factors:

1. Obesity.

2. Family history

Diagnosis:

By the same criteria for type 1 DM. HA1c is one of the diagnostics and this is the difference from type 1. HA1c more than 6.5 => diabetic

=> borderline:

RBG: 140-200

FBS: 100-126

HA1c: 5.7-6.5

=> if the patient is borderline with no symptoms do GTT.

GTT in pediatric: you don't give 75 this will lead to hyperglycemia طوالي

You give 1.75g/kg and then you measure at 0,1,2 hrs .

=> patient with nocturia, RBG =150 and HA1c more than 6.8 this is diabetic. Because HA1c more than 6.5 is significant.

Rx:

By tabs: metformin 500/850

Dietary control

Lifestyle modifications

(ONLY indicated tabs in pediatrics is metformin).

You should start insulin in type 2 DM in children if:

1. Severe DKA.

2. RBG more than 250

3. HA1c more than 9

* symptoms of DKA or symptoms of hyperglycemia indicates insulin *

In these conditions you start insulin till the patient is out of DKA or return to the normal (no symptoms of hyperglycemia) then introduce metformin tabs and withdrawal of insulin.

/// 12yrs old boy presented with polyuria and polydipsia with +ve family history, RBG =220, HA1c =8% , this patient is for metformin tabs and not insulin.///

Compilations: same as type 1 .

Acute: hypoglycemia and DKA

Chronic:

1. **Micro**: absolutely rare in children

2. **Macro** :

a) retinopathy . b) nephropathy. c) neuropathy. a+b are more common.

Screening for complication:

Not like type 1, you screen for complications in type 2 DM at the time of diagnosis.

Diabetes mellitus

د.الفاضل عيسى

What diabetes type 1 means?

History?

Examination?

Investigation?

Management and follow up.

* Type 1 DM:

Due to problem in production of insulin.

* History of DM:

* Main complain: patient come with three presentations :

1. Not dehydrated : patient had sugar and acetone in urine => management => don't give insulin don't manage as DKA don't give fluid.

2. Classical type (severe dehydrated) => manage as DKA

3. Shock => manage as DKA.

DKA:-

Acetonuria, glucosuria and full dehydration

History of presenting illness: patient has:

- polyuria

- polydipsia

- polyphagia

- weight loss

* polyuria: increased amount and frequency, if just increased frequency it could be (UTI, stone, cystitis, urethritis).

* you ask about enuresis which called bed wetting.

Types of enuresis:-

1. Primary

2. Secondary

الطفل لمن يتوقف عن التبول في السرير في عمر سنتين ونص مثلا لو تبول مرة أخرى بعد أقل من ستة شهور دي برايمري ولو تبول مرة أخرى بعد اكتر من ستة شهور مثلا بقى عمرو تلاتة ونص دي سكندري

Ask about polydipsia, polyphagia (appetite) and weight loss.

* Systemic review:

CNS:

- ask about loss of consciousness (DKA, hyperglycemia).

- ask about convulsions

- ask about visual abnormalities, visual acuity, blurred vision)

Very rare to develop complications (cardiomyopathy, renal problems, microvascular)need 5-10 yrs to develop complications.

Cardiopulmonary: infection

GIT: abdominal pain, abdominal distention, diarrhea, abdominal swelling.

GU: urine: color, amount and frequency, burning micturition => UTI

vaginal discharge in female is a common presenting feature.

Skin: changes , hypopigmentation and hyperpigmentation.

* Past history: ask about condition which preceded or lead to DM such as viral infection

Types of viral infection :-

- mumps which cause pancreatitis in children and distruct the beta cells.

- rubella

- cocxaci virus

Ask about past history of hospitalization if he's a known case of DM to know PH of DKA.

هل قبل كدا رقد مستشفى ورقد بشنو وأدوهو شنو

*Family history: DM type 1 isn't familial but it could be increased incidence.

- FH of viral infection (TB)

- Family pedigree.

* Social history : you ask to know the impact of the disease on the family.

DM occur in poor and rich people no variation.

DM has seasonal variations , increased in winter more than summer because of viral infection.

Social history is important in the follow up of patients , nutrition, drugs, complications and diagnosis.

* Drug history:

* Developmental history

* Nutritional history: very important due to exclusive breastfeeding for 6months and continue for 2yrs this decrease the incidence of DM.and the Atopy is decreased.

And you ask about nutritional history to know the diet controlled or not.

We don't restrict diet in the children because they're growing, but we don't encourage diet containing modified glucose. البيبسي بالشوكولاتة والباسطة:

*Developmental history: if the patient is 8yrs old we don't need to take details about milestone.

يا أمو الطفل دا زي اخوانو ولا أتأخر في المشي والجري

لكن لو الكيس كانت جهاز عصبي بسال من التفاصيل بدا يحبا في كم شهر صرخ طوالي ولا لا اخد اوكسجين ولا لا صنقر في كم ومشى في كم

Also in case of malnutrition most look for regression.

Regression مثلا طفل عمرو تلاتة سنة اسي ما بمشي لكن كان يمشي دا

Delay طفل عمرو تلاتة سنة لكن اصلا ما مشى دا

* in diabetic child more than 10yrs must be ask about schooling.

Delay schooling لو قالت لي بقرا في سنة أولى وهو عمرو عشرة سنة

Diabetes is one of the chronic causes that delay puberty.

Examination:

1. Higher function. Conscious or not, commonest presentation of

DM is either bed wetting or DKA(comatose), so level of consciousness is important.

Is patient lethargic or not, lethargy => dehydration

Irritability: irritable or not.

2. Examination of the eye:

- visual acquity: wearing glass or not? Contact lenses?

- look for squint and cataract

- look for anemia

- signs of dehydration => sunken eye , loss of tears

- visual field

*why anemia in DM? Because it's an autoimmune disease, cause pernicious anemia.

- fundoscopy: for optic atrophy

عشان كدا في ال systemic reviewبسال بشوف كويس ولا لا

3 .Examination of the ear:

Ask about discharge, deafness

DIDMOD syndrome; diabetes insipidus diabetes mellitus optic atrophy and deafness

DM and deafness come together .

4. Mouth:-

- look for signs of severe dehydration

- can't take orally

- Drug coated tongue

- look for oral hygiene (fungal infection)

- look for pernicious anemia

- hyperpigmentation of the tongue indicate Addison disease which associated with DM.

- smell of acetone in mouth

5. Neck:

- look for thyroid (Hashimoto thyroiditis) associated with DM.

6. Chest:

- kussmaul breathing, acidotic deep breathing

7. Abdomen: abdominal distention due celiac disease => associated with DM

hepatomegaly => ....syndrome? , in which the patient is short and have hepatomegaly.

8. Trunk: anteriorly and posteriorly for hypopigmentation(vitiligo) which is autoimmune disease

9. Genitalia: DM cause delay in puberty so breast in female and pubic hair in male.

10. Thigh: site of injection of insulin look for atrophy and hypertrophy.

11. Legs: look for operations, neglected feet.

12. Hands:

- for pallor => anemia

- there's limitationson movement of the joint => joint become stiffed like rheumatoid arthritis,

this is called primer sign due to deposition of cholesterol and lipids in the interphalangeal joints.

- look for hyperpigmentation => Addison disease

- fungal infection

- pulse: small thread pulse.

- blood pressure:

- reflexes: delay relaxation which is called autonomic neuropathy

* Investigations:

Don't depend on GTT and FBS, just urine and random blood sugar.

- if urine contain sugar and acetone => DM, if dehydrated => DKA , not dehydrated! This is called diabetes with ketonuria .

- RBG more than 200=> DM

- RBG more than 250=> DKA غالبا

-if you find just acetone <= starvation.

Dx of DM depends on:

- History (symptoms)

- Examination (signs)

- Urine and RBG .

If sugar in urine + or ++ this is normal, if more than +++ this is DM.

Diabetic patients with acetone => starvation hypoglycemia

** diabetic patient presentedto you comatose what do you think? And what you do? Either hypoglycemic coma or hyperglycemic coma (DKA).

Examine and take blood sample for random , give glucose 10% if hypoglycemic will improve,

If hyperglycemic coma he doesn't affected.

If you give him insulin, you cause destruction of the brain , it's fatal.

* management of diabetic who has acetone and glucose in urine and not dehydrated => give him daily requirements 0.5/kg .

e.g wt 40kg => 40×0.5= 20

العشرين دي بقسمها ليو خمسة كل ستة ساعات اذا كان ما مرقد في المستشفى وإذا كان مرقد بقسمها ليو تلتين الصباح وتلت المساء وافحص ليو كل ستة ساعات .

If he's dehydrated => treated as DKA .

Signs of DKA:

- abdominal pain
- vomiting
- fever
- lethargic
- may be comatose

In DKA there're 4 things to treat:

1. Dehydration:
2. Hyperglycemia
3. Electrolyte: K most important
4. Acidosis

- classification of dehydration according to loss body weight:

1. Mild => loss of weight < 5%
2. Moderate => loss of weight 5%-9%
3. Severe => loss of weight > 10%

Dehydration ودا تقسيم صعب بحتاج نوزن قبل ال

If wt before dehydration 10kg and after 9.6 kg then he loss0.4 = 0.4÷10×100= 4% => mild dehydration.

- classification of dehydration according to osmolarity:

1. Hypertonic: Na > 145

2. Hypotonic: Na <145

3. Isotonic : Na 135-145 (which is normal range of Na).

ودا برضو تقسيم صعب :).

1 .Dehydration : treatment of dehydration is treat the deficit + maintenance for 2 days.

10× percentage of dehydration × wt of the child

Wt =40kg = 10×7.5×40= 3000

In diabetes you care for cerebral edema in severe dehydration so between 5 and 10 is 7.5 => deficit => 3000

Maintenance = 1st 10kg ×100

2nd 10 kg×50

rest of kgs×20

For patient wt 40 => 10×100+10×50+20×20= 1900 in one day, so maintenance of two days 1900×2= 3800.

Total of this child = deficit +maintenance = 3000+3800= 6800ml.

** deficit = 10× percentage of dehydration + wt of the child **

1. Call for help when you need to treat DKA.

2. Wt of the patient must be measured

3. Calculate the dose of fluid, insulin, K

4. IV cannula

5. Take blood sample

6. Introduce catheter

7. Cardiac monitoring by ECG.

- blood sample for RBG, electrolytes, PH, ABG, ketones, lipids profile, urea, TH.

- catheter to monitor urine (glucose and acetone).

- ECG => signs of hyperkalemia due to acidosis In emerges K at side which cause of death.

After calculate fluid give in first hours as quick as possible. ،

20×40(wt)=800ml , give 800 ml with one hour.

Drops per minute = amount ×15÷ 60× hrs =
15×800÷60×1=200drops/min

Total - 800 = 6800-800= 6000 => must be finished in the other 47 hrs.

6000×15÷60×47= 70 drop/min .

- every hour I take blood sample and urine and monitoring by ECG.

- every hour ask from RBG and acetone and K .

When RBG < 300 , the acetone disappear, even if there was acetone I take by the RBG because this acetone is just residual.

لو الممرضة قالت ليك الراندوم بقى اقل من تلتمية قول ليها اديتيو كم درب لو قالت ليك خمسة قول ليها السبعة الباقيين جيبي بدلهم جلكوز .

*convert from normal saline to glucose when RBG <300 or no acetone in urine. After this monitor every 6hrs instead of 1hour .

* continue for two days with glucose

2. Hyperglycemia: immediately give 0.25 unit/kg soluble insulin IV and IM (half IV and half IM).

- every hour must take blood sample for RBG.

After hour if RBG still high and HCO3 and K high , give 4units IM and after hour again check RBG if high and there's acetone in urine and hyperkalemia give again 4 units IM and so on till RBG is less than 300 and no acetone then give insulin as follow:

ادي أربعة وحدات لكل صليب سكر في البول ما عدا الصليب الأول

يعني لو لقيت تلاتة صلايب حتدي تمانية وحدات انسولين تحت الجلد ولو لقيت صليب واحد ما بتدي انسولين

واصل في الانسولين تحت الجلد لمدة يوم كل ستة ساعات

*when RBG <300

- convert from normal saline to glucose

- convert from IM insulin to SC insulin.

3. Treatment of electrolytes: K is very high in patients of DKA due to extracellular mobilization.

- daily requirements of K is 1.5 mmol/day.

- give 2.3 mmol in hypokalemia with fluid

- when patient recover from DKA that's mean all K shifting to intracellular, usually that's occur after 1st hour (after 800ml of fluid) patient develop hypokalemia. usually give K after 1hr but don't give it unless you make sure by ECG changes that there's signs of hypokalemia.

* when you give K?

When patient passing urine, that's usually sfter 1hr , afraid in severe dehydration from renal failure , that's either hemolytic due to acute tubular necrosis or uremic syndrome (HUS).

- if severe dehydration cause acute renal failure that mean high K due to renal failure.so don't give K.

- so give K if patient pass urine because that mean kidney is intact and there's normal renal function.

لو كنت في مكان ما فيو فحوصات ادي بوتاسيوم اول ما العيان يبول لانو كدا الكلية بتكون سليمة .

ما بحسب انا بدي بوتاسيوم كم لكن كل درب انا بديو بدي معاو بوتاسيوم بمعنى كل درب بعد الساعة دي يكون فيو بوتاسيوم.

4 .Treatment of acidosis:

- usually you don't need to give diet bicarb, you don't need to correct acidosis because it correct itself spontaneously, insulin is anabolic hormone so stop catabolic effect.

- if PH <7.2 , bicarb <12 this mean severe acidosis which cause hyperkalemia and interfere with contractility of the heart => must be treated by give one third of the deficit , don't do full correction.

Normal bicarb = 22-28

- if you find the bicarb 10 and PH 7 do correction.

Correction = 1/deficit × weight

= 1/3×12×40=160

Don't do full correction because insulin do the correction and if you do correction the patient will enter in alkalosis which cause expansion of of blood pressure and cerebral edema and hypocalcemia (tetani) so correct one third of the deficit.

Summary to manage patient with DKA:

-shout for help

-take blood sample

-IV line

-ECG to cardiac monitoring

Honeymoon period:

Occurs in newly discovered diabetics there's antibodies destruction beta cells the child have hyperglycemia so give insulin after short period.

New beta cells forms because the child is growing and produce insulin. This insulin and exogenous insulin cause hypoglycemia , don't stop exogenous insulin but minimize it .

Tell the patient that DM is for life but this called honeymoon period and then it'll come back after a month or two.

* Semogyi phenomena:

Caused by doctor (iatrogenic), if patient weight 40 and you give him 2/3 in the morning and 1/3 in the evening and next morning you find +++ sugar , that means evening dose isn't enough , you increase it and check the sugar next morning وتزيدها تاني لو ظهرت ليك صلايب لحدي ما تزبط

In phenomena you give the patient large dose => at mid night there's severe hypoglycemia which stimulates the countercurrent hormone (GH, glucagon, epinephrine and cortisol) these high hormones stop the action of insulin.

And cause glocogenesis causes high random in the morning you have to decrease the dose not increase it.

* normal person at midnight have hypoglycemia but patient with Semogyi phenomena have hyperglycemia at midnight.

We give up to 1.5 unit/kg for diabeticpatient and more than 1.5unit/kg if we expect Semogyi phenomena.

After treatment of DKA give the patient.0.5 units/kg 2/3 before breakfast and 1/3 before dinner.

If diabetes occur during illness you can increase the dose by 10% .

- decrease dose by 10% in الرياضيين.

*follow up:

1. History

2. Examination

4. Investigations : every three months.

* 1. History:

- ask about bedwetting

- ask about polyuria

- ask about appetite and diet

* 2. Examination:

- anthropometric measurements: wt / ht

- systemic examination

* 3. Investigations:

1- urine general which detect control within last 24hrs.

2- RBG which detect control within last 24hrs

3. HA1c: normally saturated with glucose, should be < 7% .

- if HA1c < 7% => excellent control

- 7-9% => mild control

- 9-12% => poor control

- >12% => very poor control دمو مشبع بالجلكوز

HA1c detect control in previous three months

4. Baseline urea and electrolytes

5. Baseline lipid profile

6. Baseline thyroid (antithyroglobulin antibody- antimicrosomal antibody)this is firstindicator that this patient had Hashimoto disease.

- urine => first change in urine is microalbuminuria > 20 mmol , this means this patient will develop renal problems in few years.

6. Baseline for celiac disease

7. Baseline every year

8. Follow-up every 3 months.

* normal HA1c <6% .

DM and DKA

د.هشام العمدة

The Common presentation of DM in children

is with DKA Younger specially at but adult children present with classical presentation "increase urine frequency, polydepsia polyphagia, wt loss "

In your history you should mention the +ve things firstly, then the important - ve.

* In the FH first ask about similar condition then chronic diseases to then the parents and if there is consanguinity

- You should mention their numbers the age of both the elder one and the youngest ones then mention if there is a simillar conolition in his siblings and if there is adeath and it cause.

You ask about viral infection and Mumps specifically.

- Parotitis is destruction of B-cells in pancrease .

You ask about skin changes vitiligo and other autoimmune diseases associated with DM.

as Hashimoto's thyroiditis, Addison disease, Pernicious anemia, celiac disease.

On examination: "Focus exam : * In the chest: RR, type, patteren of breath

acidotic breathing => rapid shallow breathing, rapid deep breathing => chyne stok breath.

Wt=> Age x 2 +8

expect to find in 7-12 yrs wet loss.

* Pallor = pernicious anaemia, signs of dehydration , deafness "diabetes insipidus", optic atophy.

- Thyroid enlargement"Hashimoto's thyroiditis" skin pigmentations "Addison disease " abd distension "celiac " hepatomegaly " Moreac syndrome" => hepatomegaly + DM+ short stature + infantile

genitalia" due to poor control. .

those are the acute signs that you should look for.

* Now some acut cases are presenting with neuropathy.

DM in children is Type 1, due to insulin

deficiency or relatively due to destruction of the receptors

We diagnose DM clinically and Laboratory :

The clinical presentation in DKA is: .

1. Acute abdomen
2. Vomitting
3. Alternation of consciousness > Coma
4. Dehydration
5. Acidotic breathing

* Pathophysiology of DKA:

* To diagnose as diabetis the glucose level

should be : + Strong FH + Change in lifestyle

- Random blood glucose > 200 mg/dl.

- Fasting blood sugar > 126 mg/dl.

*in DKA the RBG should be > 300

* Other Laboratory finding in DKA:

- low pH <7.3

- HCO3 < 12 mmol.

Glucosuria - Proteinuria.

Other changes with DKA:

- Electrolytes

- Hypokalemia

* How to manage DKA:

1. ICU admission + monitor

2. Give fluids => Normal Saline "isotonic solution"

Amount: first you should calculate the deficit and the maintainance.

Severe dehydration deficit = Wt of pt x 100- 10% deficit 100 ml/kg .

Maintainance 1st 10kg x 100

2nd 10kg × 50. Residual Kg x 20

To calculate for severity of dehydration = wt x100

* We give the maintainance + 1/2 deficit within 24hrs. "Guid line" " infusion pump"

* If you want to give the full maintainance +full deficit should be within 36-48 hrs.

- Give insulin , at least 1/2 hrs after fluids.

- in the 1st hour we give 0.25 IU/kg. 1/2 IV and 1/2 IM

- Then 0.1 IU/kg IM /hr

And Give 0.3 IU/kg IM "Guideline"

Then 0.1 IU/kg subcutaneously, hourly till the

RBG is 250 mg/dl

* When RBG becomes below 250 mg/dl we Should change the Normal saline to Glucose 5% .

Because giving insulin will stimulate B cells => insulin

* After you achieve a good control RBG <200mg.

you should change the insulin.

If you didn't change the NS=> Hypercholremic metabolic Acidosis.

Potassium replacements:

- If there is ECG changes.

-If the child not passing wrine.

- if there is signs of hypokalemia. ..

- Serum K is low

=> 20mmol = 0.5-1 mmol/kg/day

* In newly discovered patient 0.5-1.5 IU /kg /day - - Mixed insulin containing soluable 1/3 + long

acting 2/3.

If the child > 3yrs - we give 2/3 of the dose & in morning and the 1/3 in the evening.

If the child <3yrs we use long acting insulin only. 50% before breakfast and 50% before dinner.

In diabetic Ketosis: no acidosis . hyperglycemia acetonuria, Just to adjust the dose.

* In children we are not going to restrict Carbohydrate from diet => important for growth.

Counselling :

- Sign's of hypo/hyperglycemia

- How to preserve insulin

- How to inject insulin

- How to draw adequate close of insulin

- How to use stick at home to investigate for proteinuria, glucosuria or using touch for RBG.

* How to inject:

vertically in the arm , abdomen, thigh with changing the site of injection.

How to preserve:

- Not to freeze insulin and not to put in a hot weather.

* Problems of injection:

- Lipotrophy _ decrease the absorption of insulin.

- Lipohypotrophysl

* Doing HA1C reflect the control of blood glucose for the last three months *supportive investigation *

Sign's of hypoglycemia:

- Hunger

- Sweating

- Headache

- Blurred vission was

- Convulsion.

Immediately you should give highly sweatest thing.

Give the patient the fixed dose (2/3,1/3) and measure RBG if > 300 give 10% soluble of the total dose.

e.g if he's taking 30 IU/day, measure blood sugar every 6 hrs and give 3 IU insulin. If RBG > 300.

Long term complication of DM :

- Nephropathy & predict by looking for:

1. Microalbuminuria

- Neuropathy

- Retinopathy

Complications of DKA:

1. Cerebral oedema

2. Pulm edema

3. Aspiration Pneumonia

4. venous thrombosis

5. Acute pancreatitis

6. Rhabdomyolysis

7.Cardiac arrhythmia

GTT is not indicated to diagnose DM type 2.

- HbAK is just supportive and not diagnostic in type 1. but diagnostic in Type II.

_* Important investigations in diagnosis:

1. Serum electrolytes.

2. Acid base ese status

3. HA1c

4. Glutamic acid decarboxylase

5. Anti-islet cells antibody.

6. Anti-insulin antibodies => presents in 85-90% in newly discovered patients.

For autoimmune type of DM do:

1. TFT

2. Thyroid antibodies

3. Anti-gliadin

4. Anti endomysial + tissue transglutaminose antibodies => for Celiac disease.

* CBC => Leukocytosis

DM and DKA

د.جورج ظريف

We deal with DKA as severe dehydration so deficit = 100ml/kg.

This patient with DKA is lethargic and has decreased input and increased output in form of vomiting so he need to maintenance in form of IV fluid.

* maintenance =

1st 10 kg = 100 ml/kg

- 2nd 10 kg = 50ml/Kg

The rest = 20 ml/Kg

* the pt must be correct from DKA within 48 hours.

Correction of fluid :

deficit + 2 maintenance within 48 hour

* correction of fluid :

1/2 deficiet + maintenance within 24hrs

* DKA with shock :

- managment of shock :-

- bolus IV fluid 20 ml/kg to be given rapidly within 15 minutes.

- reassessment of The pt.

If the patient still in shock => put for plan for dehydration.

(sure amount running equally for each hour)

* signs of shock hypovolemic shock:

- rapid weak pulse

- comatose

- Lethargic

- cold extremities

- delay cappillaries refilling

- hypotention

* in DKA pt start with fluid, Then give

insulin after half an hour.

insulin Dose => 0.3 IU/kg subcutaneously then bolus every hour

0.1 IU/kg sc till The pt get out of DKA.

How to know the patient is out of DKA??

– absence of metabolic acidosis which detect clinicaly by absence of Kussmaul breathing or laboratory by Normal PH and bicarb become Normal.

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- presence or absence of acetone in urine doesn't confirm or exclude DKA. Some residual of acetone present in the bladder.

- Don't catheterise pt with DKA unless he is comatose. In this situation catheterization is part of DKA managment.

* when pt get out from DKA shift from

Normal saline to glucose with normal saline.

* when Random reach 250 ml/dl I do the shifting for fluids from Normal saline to glucose with Normal saline.

I continues on regular hourly insulin till pt get out from DKA.

* pt may have RBG less than 300 but

still have Kusnamal breathing at This point you give more Glucose with

regular hourly insulin.

* Don't stop regular hourly insulin acc to random Blood sugar but stop it

Acc to metabolic acidosis. shift from regular hourly insulin to fixed dose of insulin.

* acidosis responsible for continue or stop regular hour insulin. After stop it you convert patient to mixed insulin , fixed dose if chronic case.

- if newly discovered: dose= 0.5 IU/kg (mixed insulin) 2/3 in the morning and 1/3 in the evening.

Rx according to age:

- < 3yrs give zink insulin

- > 3yrs give mixed insulin.

- some schools give 0.1 IU/kg IV but this is difficult here.

* clinical indicates of absence of ketoacidosis:

- RR and character of respiration becoming different (from kussmaul to normal breathing).

- he returned his consciousness

- want to eat (insulin consuming glucose, that's enhance starvation.

When you give patient insulin and normal saline you push patient for hypokalemia and hypoglycemia.

There're 2 types of patients:

1. Newly discovered

2. Patient for follow-up.

* newly discovered patient come with classical presentation (polyuria, polydipsia, and weight loss). Usually patient doesn't know he have diabetes till there's infection and present with DKA.

- in history if suspect diabetes find out the etiology, patient may have pancreatic tumor affect beta secretions and cause diabetes.

* ask about alteration of the bowl habits due to endocrine disorder such as hypothyroidism, and autoimmune disorder which associated with multiendocrine disorder such as Addison disease.

* GIT problems as cause => pancreatic problem

* endocrine causes: cushing, hypothyroidism, acromegaly and pheochromocytoma.

Ask About chronic medication in drug history : precipitate DM.

Alpha adrenergic or beta adrenergic as long acting cause diabetes, thiazide diuretics.

Ass about FH of similar condition => genetic cause of DM.

Examination of newly discovered diabetes :-

- search about etiology :

- 4Feature of cushing

- High BP => pheochromocytoma

- multi endocrine disorders.

There are causes of DM, we find this causes rough dismorphism such as Turner, down, Kelinfilter Syndrome.

* After Hx and Ex you must treat underlying cause of DM.

2. In pt who come with follow up I search about complication in History an examination

Patient with DM type1 very important to know awareness of the patients and their parents about:

- hypoglycemia

- how to store insulin

- type of food

- how to give insulin

* Compilations:

1. Acute: DKA

2. Chronic:

- microvascular : such as retinopathy, nephropathy (which end by albuminuria and renal failure) and arthropathy (cheira arthropathy in which patient has stiffed tendon)

- macrovascular : such as MI

Patient with chronic DM and had decreased sensation , had trauma to joint and it become chronic arthritis till it become destructive joint disease.

- CNS may be affected and give distance sensory neuropathy- stock and gloves.

proximal motor neuropathy affected the proximal muscles of upper limb and thigh.

autonomic disturbance : such as maintenance.

Search about site of injection of insulin due to if there is lipotrophy. the absorbtion of insulin from this site is impaired and there is risk to develop

DKA.

* check site of infection

- oral thrush / between the tooth

Diagnosis of DKA:

1. Clinically: metabolic acidosis with Kussmaulbreathing

2. Laboratory

RBG >300

PH <7.3

HCO3 <15mmol/litre

* present of Acetonuria doesn't indicate DKA.

* fatigability is important symptoms in DKA

- must ask about insulin :

- from where you buy it, how much it cost?

- where you reserve it.

- compliance or not.

- who inject the child.

* what is the intervention of mother toward hyperglycemia :

- glucometer

- dipstick

* Complication of therapy :

- chosity

- hypokalemia

3. Hypoglycemia:

How you Treat hypoglycemia at

home ?

- give him sweet , 2-3 spoons of sugar in water, give him a sandwich.

lipohypertrophy :

- injection of insulin in one same place.

How you diagnose it ?

- Palpation of injection site.

4. Lipohypertrophy give poor control DM due to insulin doesn't absorbed from subcutaneous.

How you manage it ?

change site of injection at least three months.

5. Lipoatrophy :-

How you diagnose it ?

- inspection and palpation

- How you manage it ?

inject around the area.

6. Infection

7. Poor Compliance : worst complication of insulin is poor compliance.

8. Cerebral edema:

- convert of fluid from compartment to another compartment

- No insulin=> ↑ Extra cellular glucose => increase osmolarity => polyuria, if you correct This hyperosmolarity quickly there is sudden drop in hyperosmolarity and become hypoemolarity => take fluid to the cell=> oedema.

Increase osmolarity take fluid from the cell produce intra cellular counter molecule when glucose > 300 the doctor push pt. to take much fluid and insulin to decrease osmolarity because rapid correction cause edema.

* what is diagnosis of cerebral oedema?

- signs of cerebral edema ?

1. headache

2. Irritability

3. Loss of conciousness.

4. Convulsions.

5. Increased ICP.

DKA daginosis is made by :

1. Blood glucose is 300 mg/dl or more 2. PH < 7-3

3. HCO3 < 15 mmol/L

4. Urine +ve for acetone.

If no lab facilities you can relay on clinical signs :

1. High blood glucose

2. Ketonuria.

Management :

if patient is in coma :

1. Coma care

2. air way

3. O2

4. Gastric aspiration

5. Treat shock with 20ml/kg of of NS or ringer lactate.

#steps of managment -

1. Fluid =

- start with NS - as if it sever dehydration => weight × 100 = Deficit

- maintenance

- then give 1/2 deficit + maintenance in 24hrs

- given by Infusion Pumb hrly.

- start with NS and then when RBG reach 250 change to D5% + NS to avoid hypoglycemia.

2. K+

- maximum is up to 25 mmol/ kg/Day

- ampulla with every drip

give it when the pt start to urinate to avoid AKI.

3. Insulin

- give 0.1 unit/kg/ hour as continous effusions till the pt recover from DKA by infusion pumb.

- If no Infusion pumb => 0.3 ml/kg SC (ballous dose) and then 0.1 mg/kg/hour.

- Insulin is given after 2hrs from the Fluid because it need time to be Absorbed so pt can benefit from it.

- So the correction is by fluid not the Insulin

- continue given Insulin till the patient is recovered from DKA:

1. No acetone in the urine

2. PH > 7 3

3. start pass urine

4. HCO3 >15

5. Clinically

when patient recover from DKA we return him to to the previous fix dose 0. 75 ml/kg/hr or the dose that control him If newly discovered.

- Clinically to HA1c less than 7.

4. Brcarbonate not recomended. $NaHCO_3 + H_2O$ in the blood=> H_2CO_3=> H_2O+CO_2. CO_2 is diffuse rapid at the level of the brain barrier,

- cross BBB and cause cerebral edema.

- If you correct dehydration give Insulin+ K.

treat the underlying cause then bicarbonate will be corrected.

5. Correct the underlying cause:

#Primary and secondary enuresis

also there is loss of phosphate so we can give it as K+ phosphate.

For hypoglycemia -> give D5%

DkA heated with soluble insulin

Intermediate for young infant (NPH)